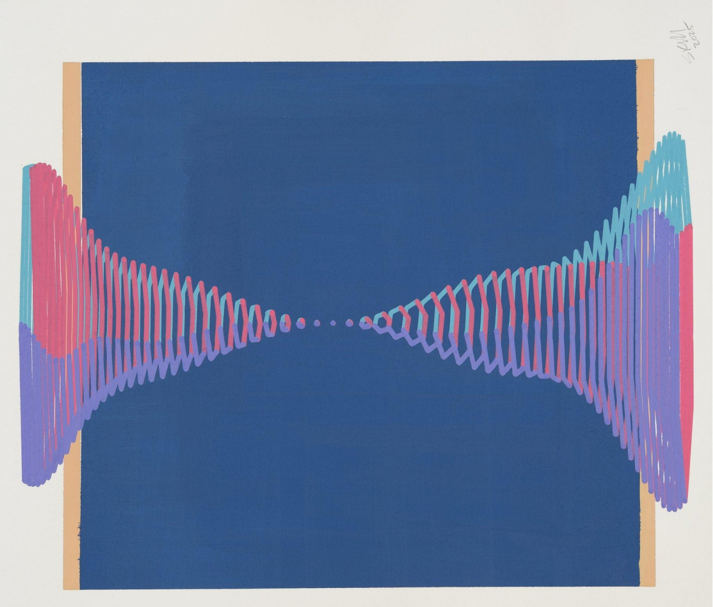
Optimizing the extraction of entanglement from quantum fields

Marcos Morote Balboa

T. Rick Perche



Credit: Sohan Murthy

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The Nordic Institute for Theoretical Physics

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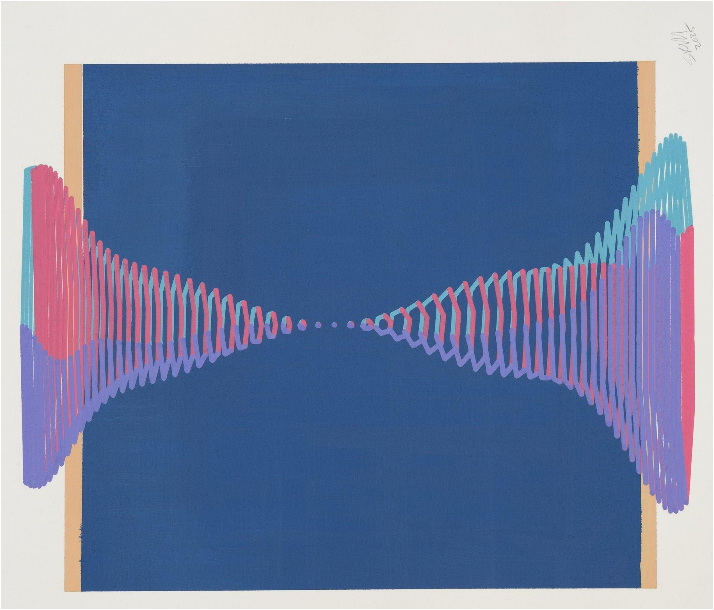
Outline



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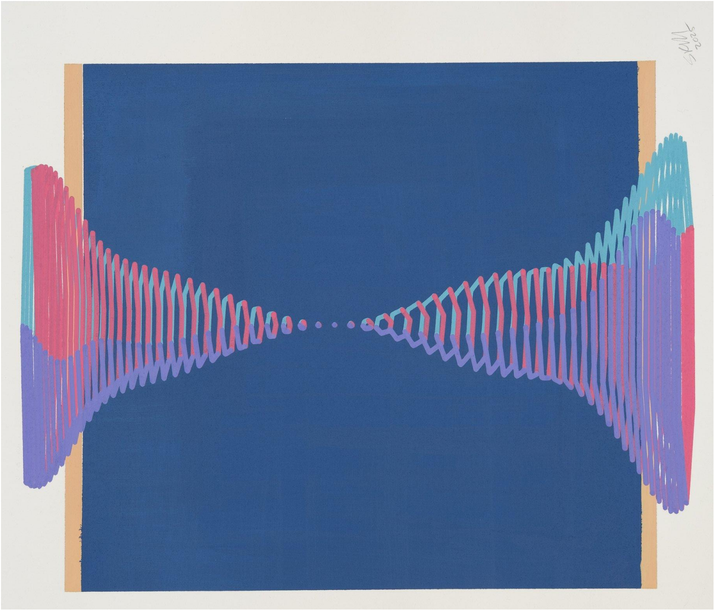
Outline

1. Entanglement



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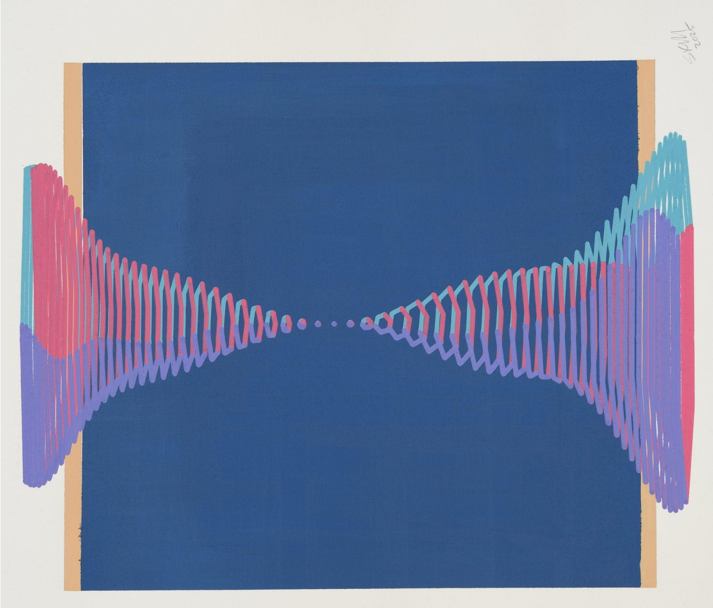
1. Entanglement

2. Entanglement Harvesting (EH)



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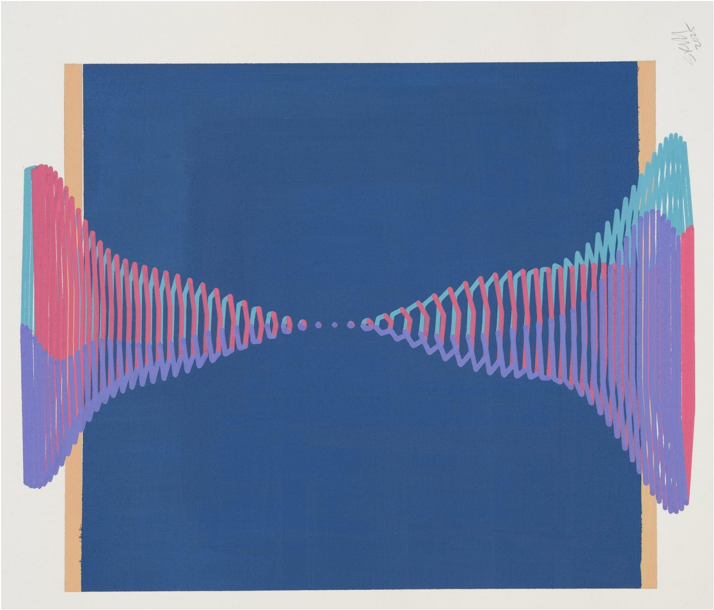
3. EH in Bose-Einstein condensates



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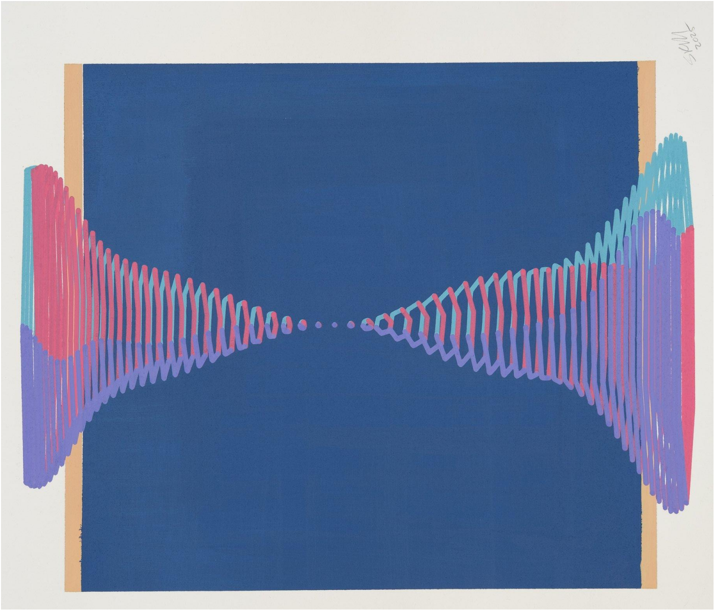
4. Optimization of EH



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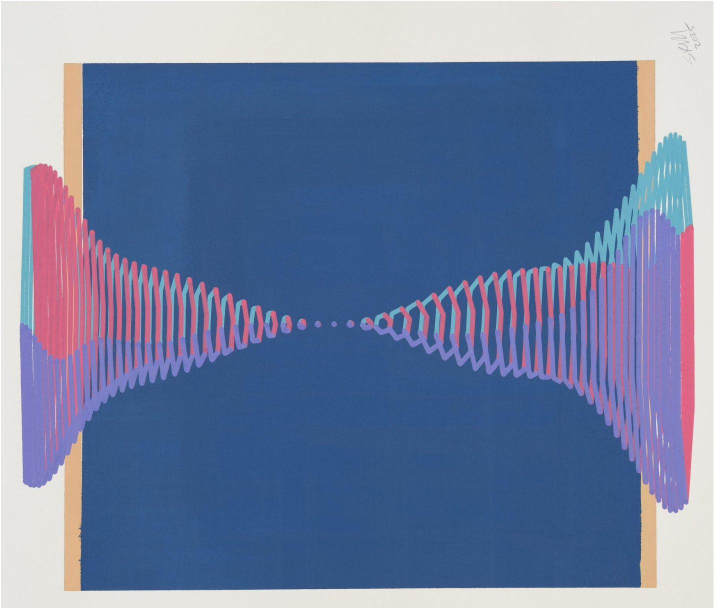

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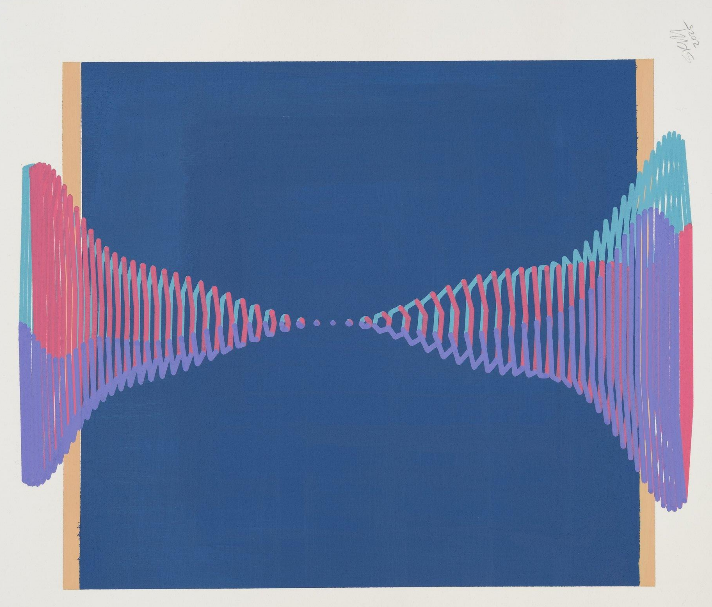

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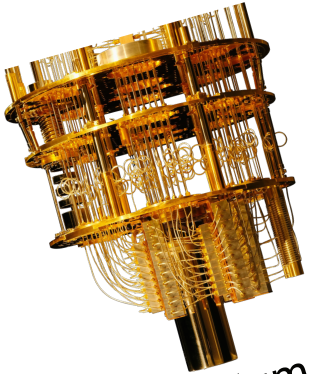
Entanglement

Entanglement

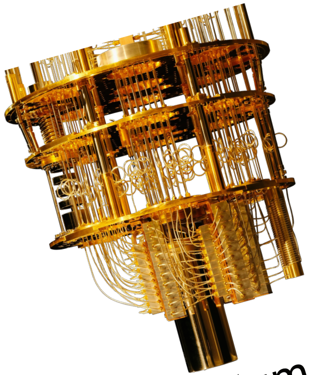
Credit: John Richardson

Entanglement



Quantum computing

Entanglement



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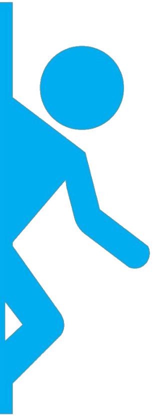
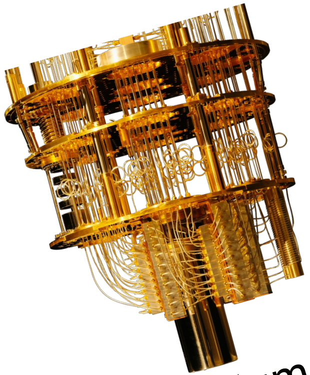


-tation

Quantum telepor-

Entanglement



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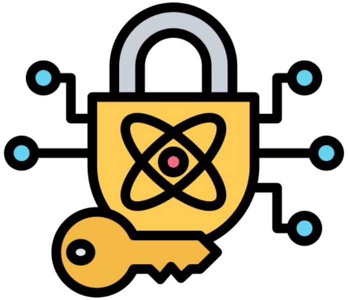
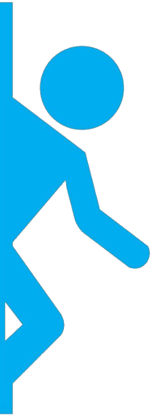


Quantum cryptography



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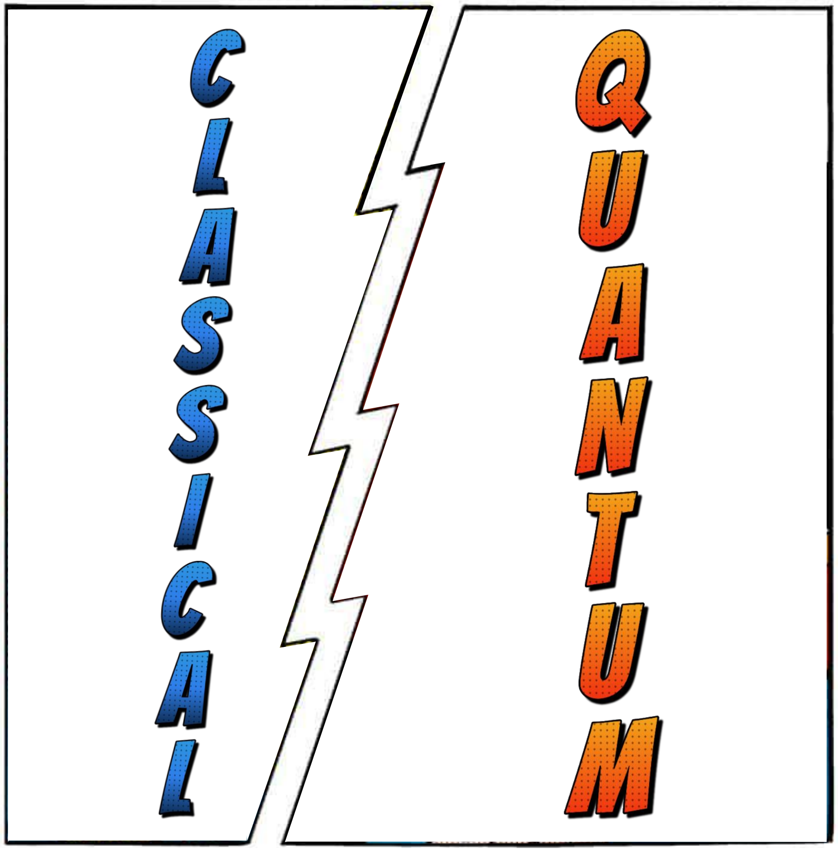

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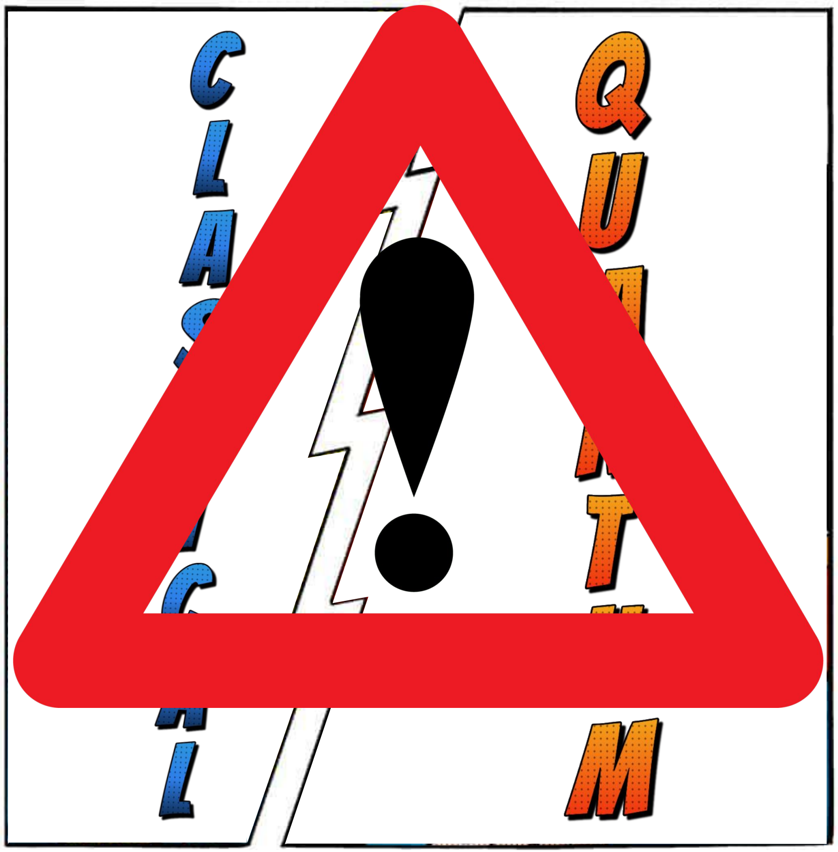

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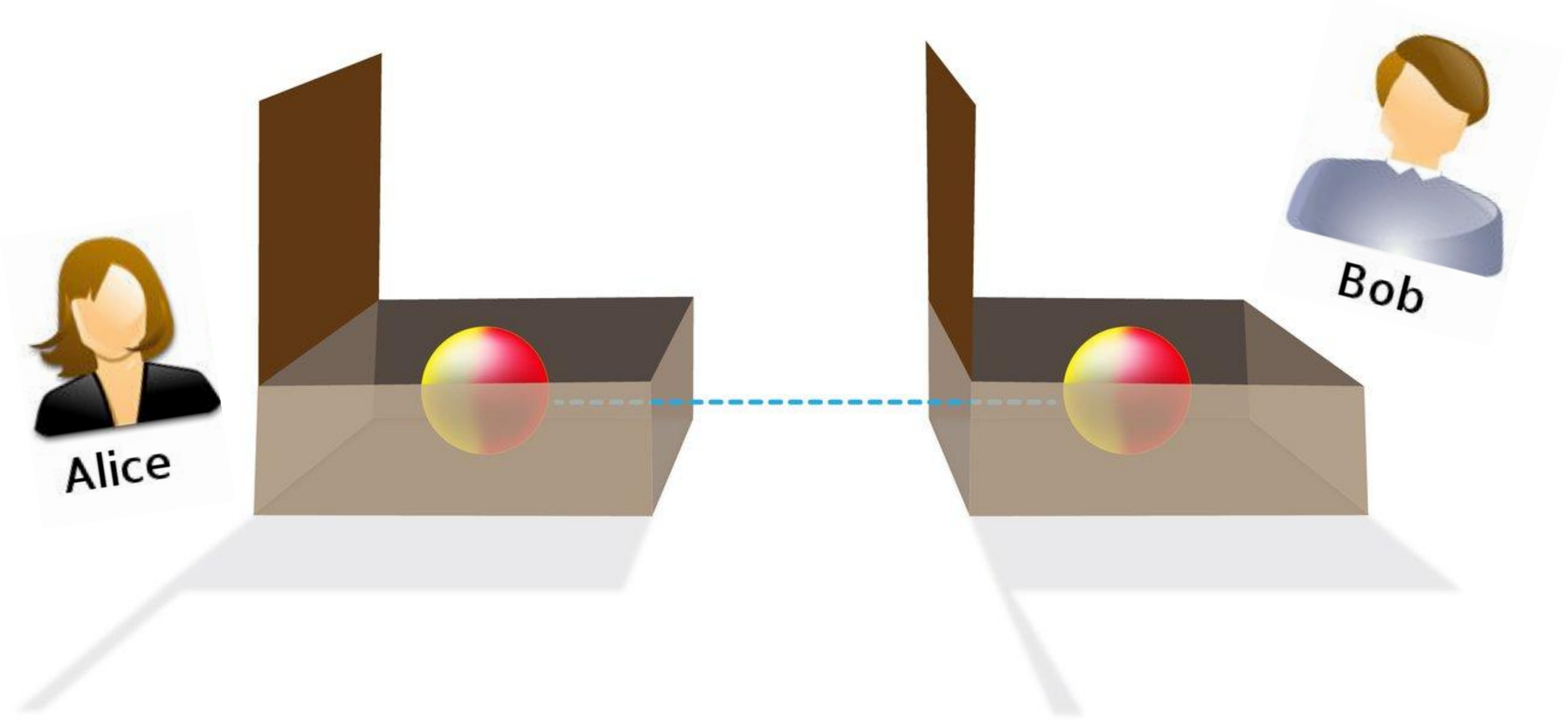

Credit: Lance Hayshida for Caltech Science Exchange

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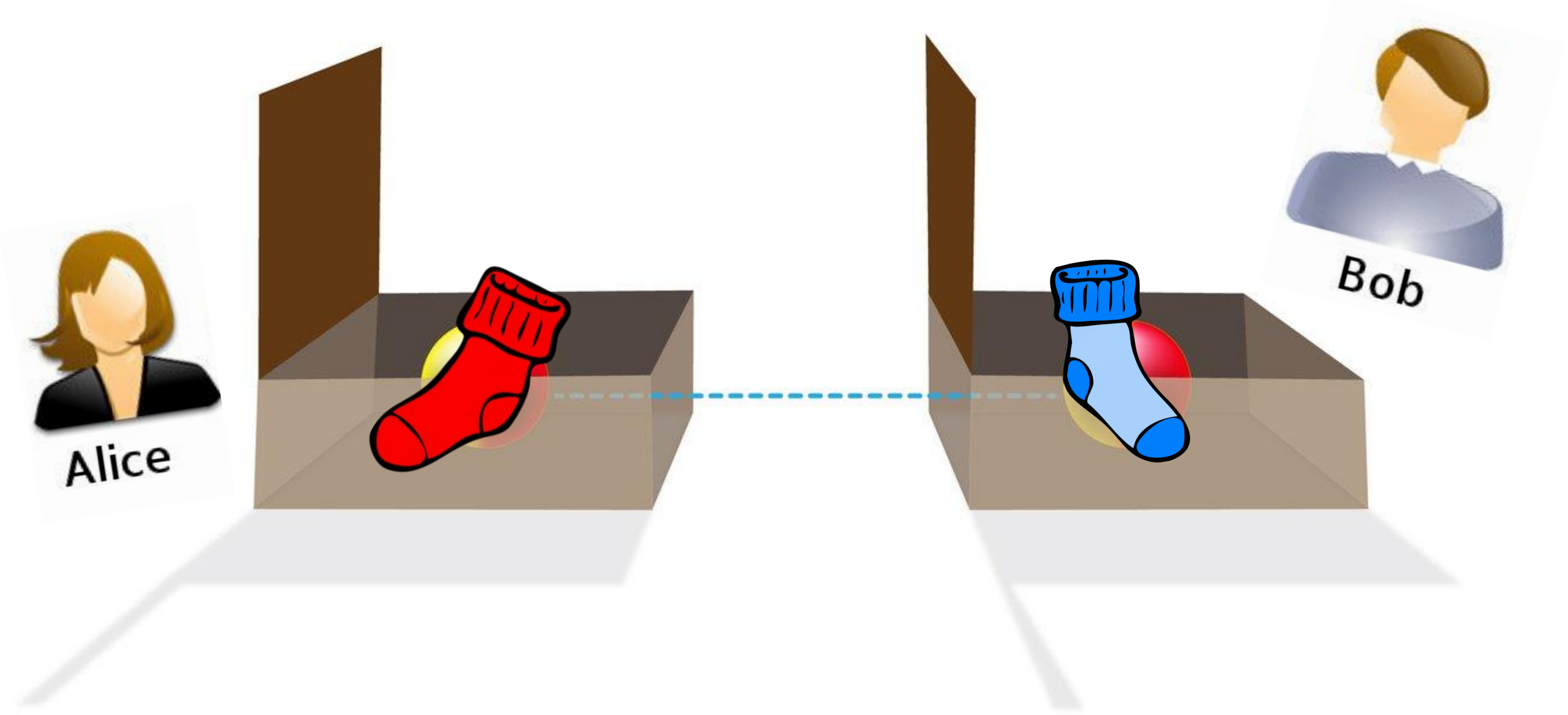

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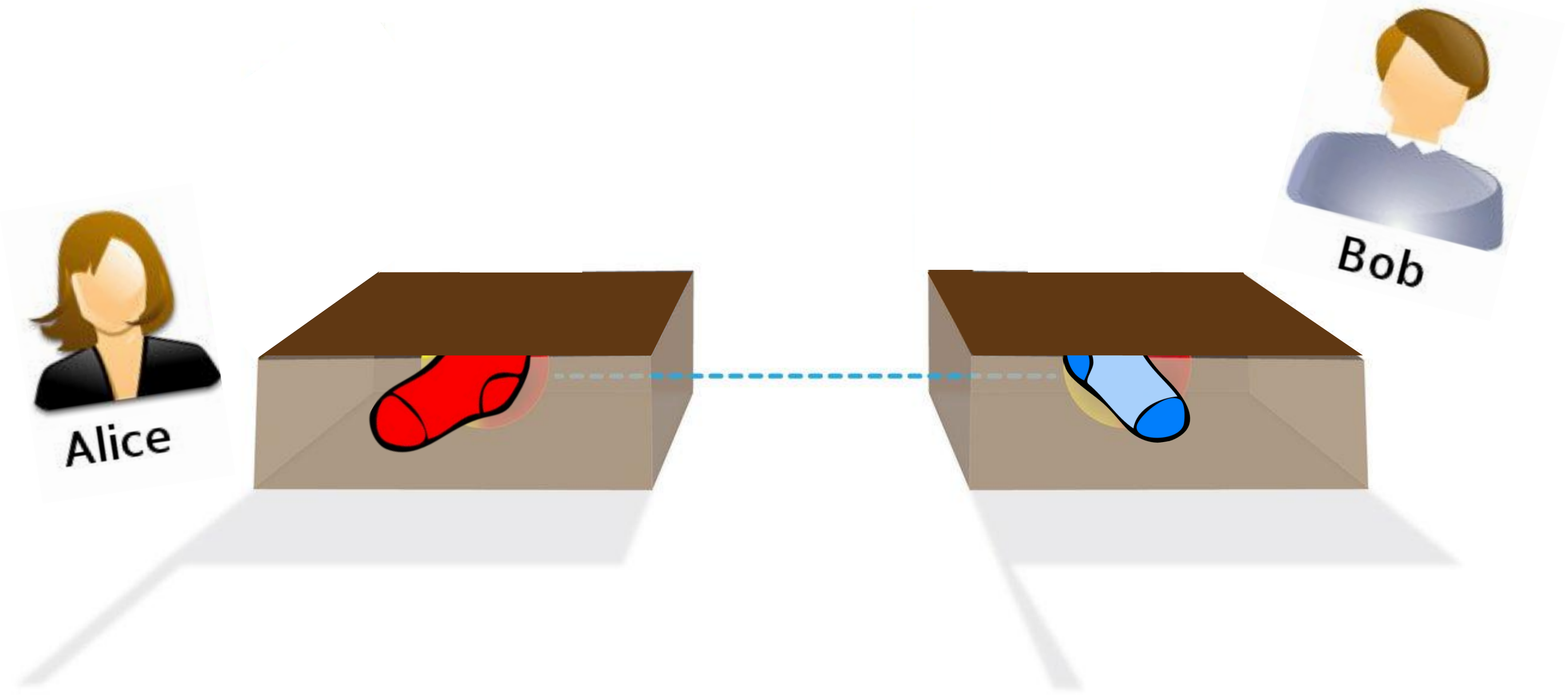

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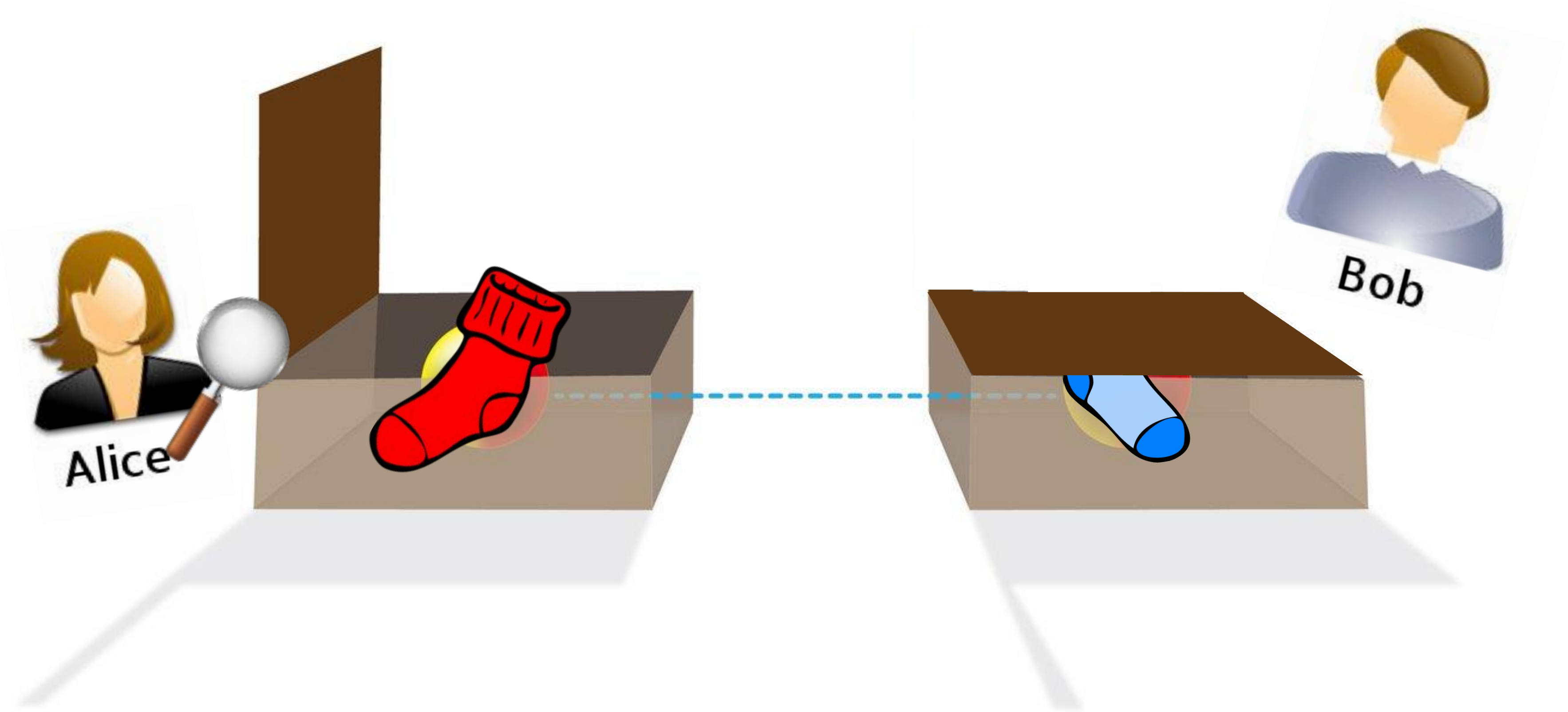

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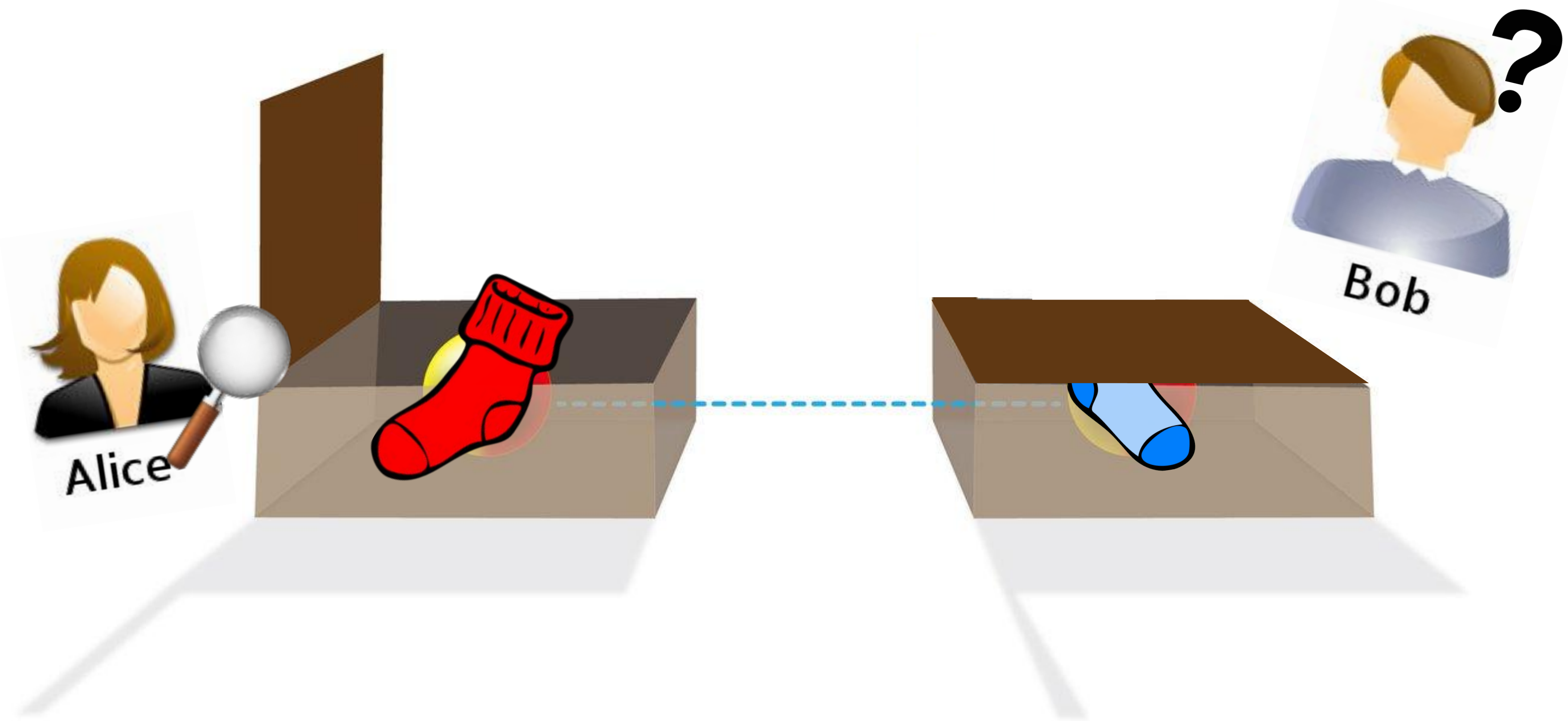

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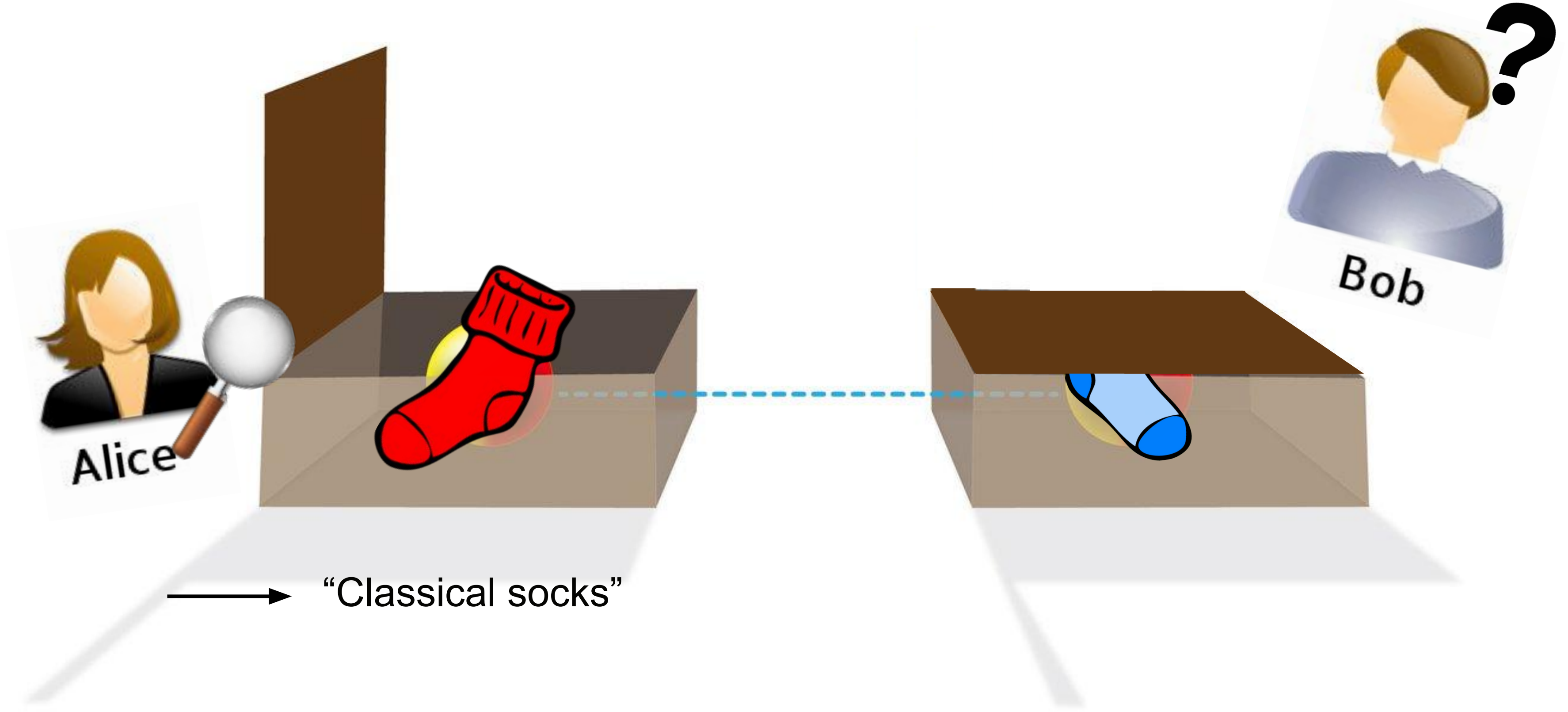

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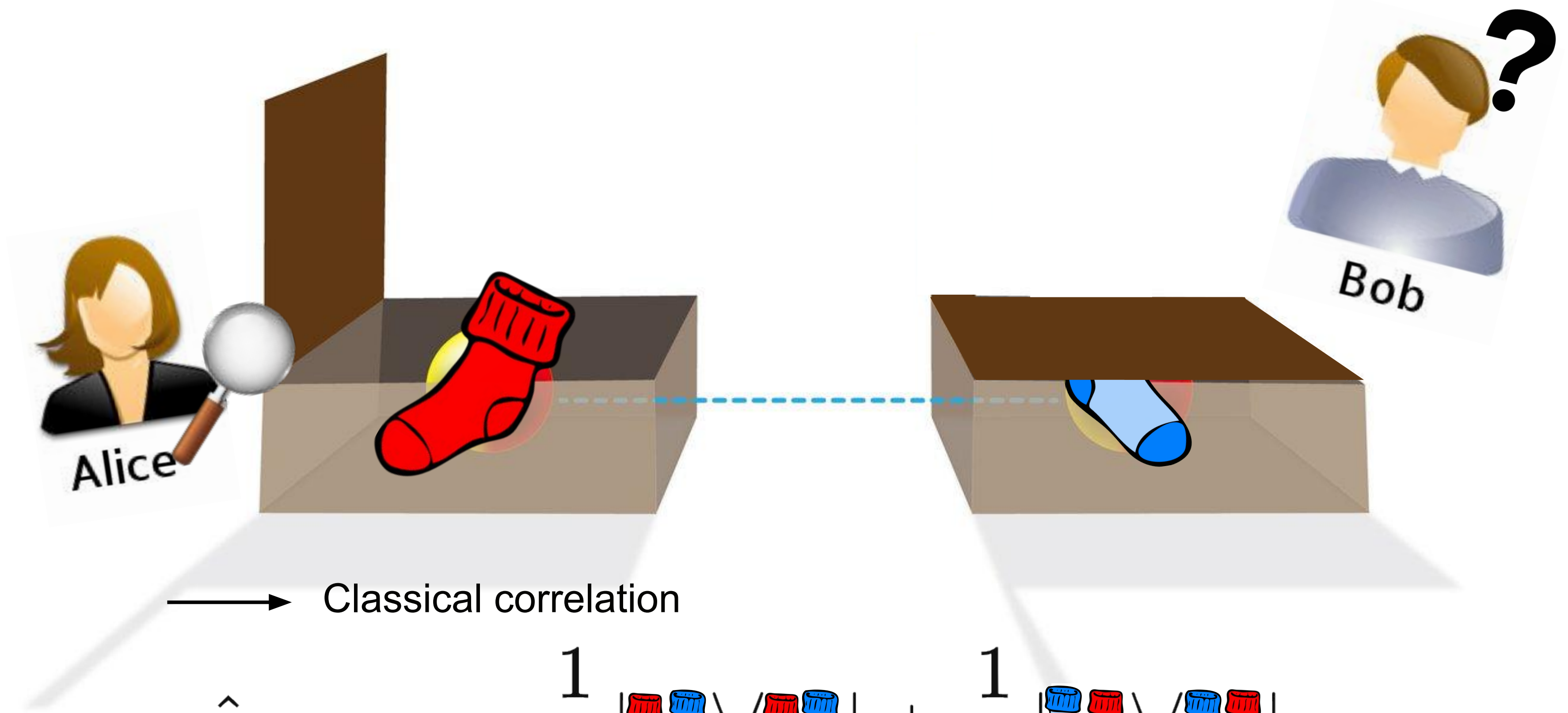
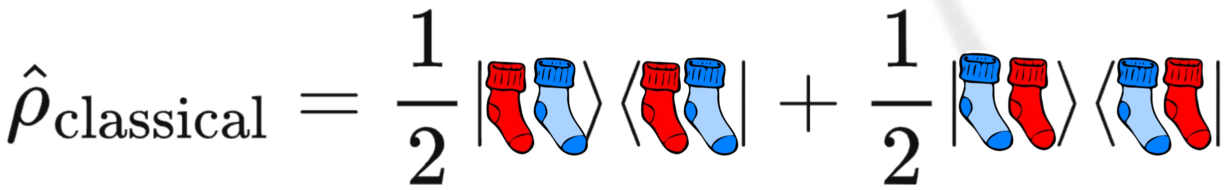


Classical correlation



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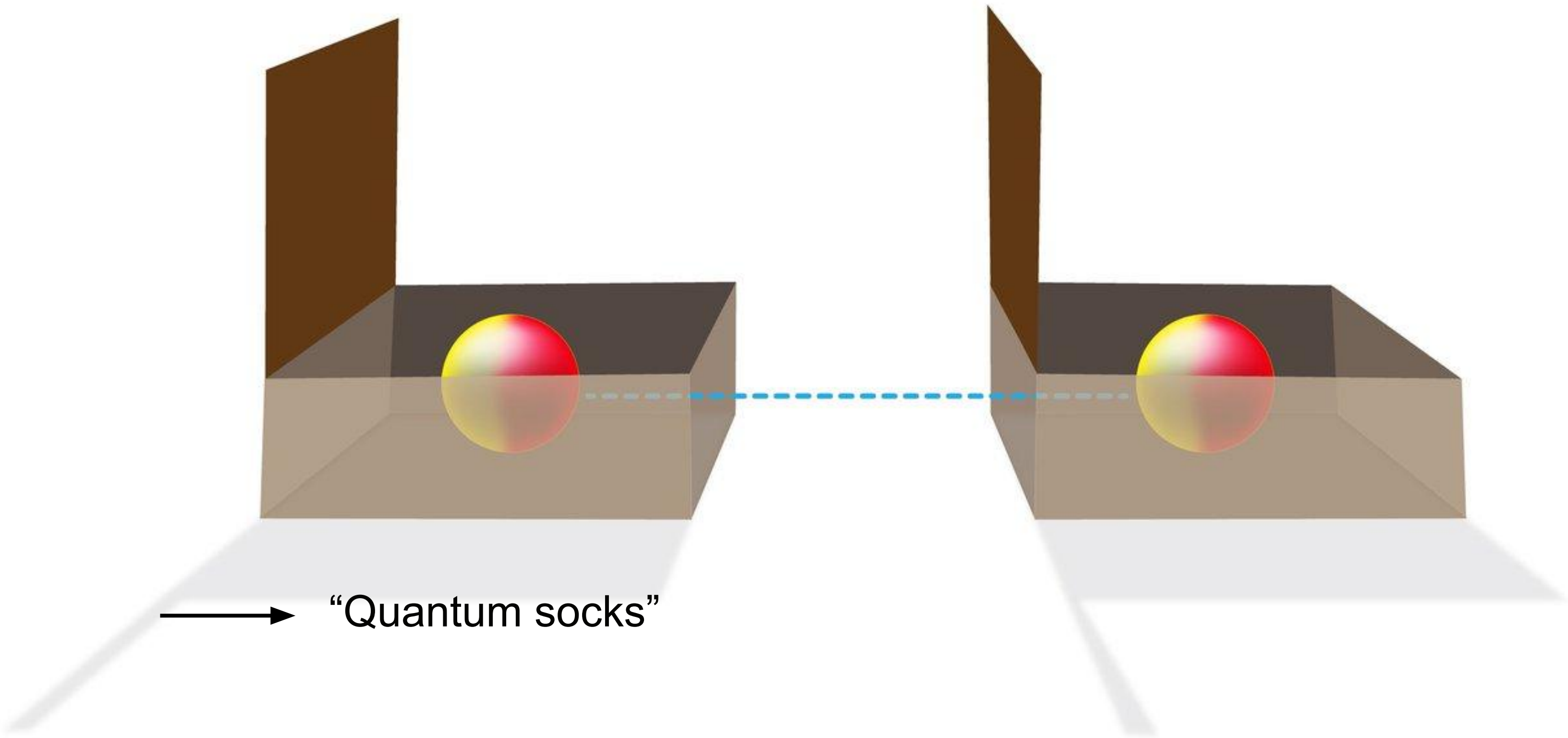

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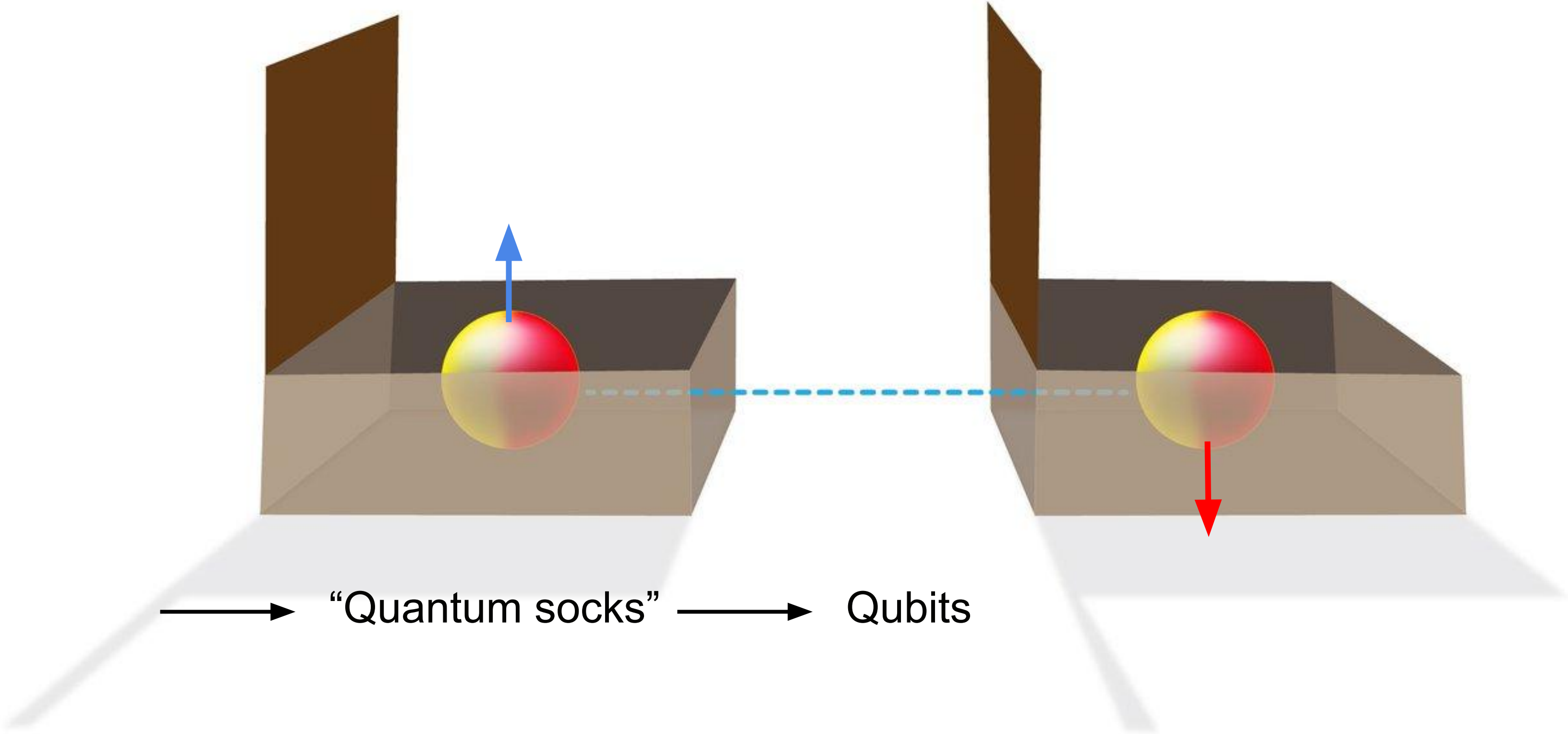

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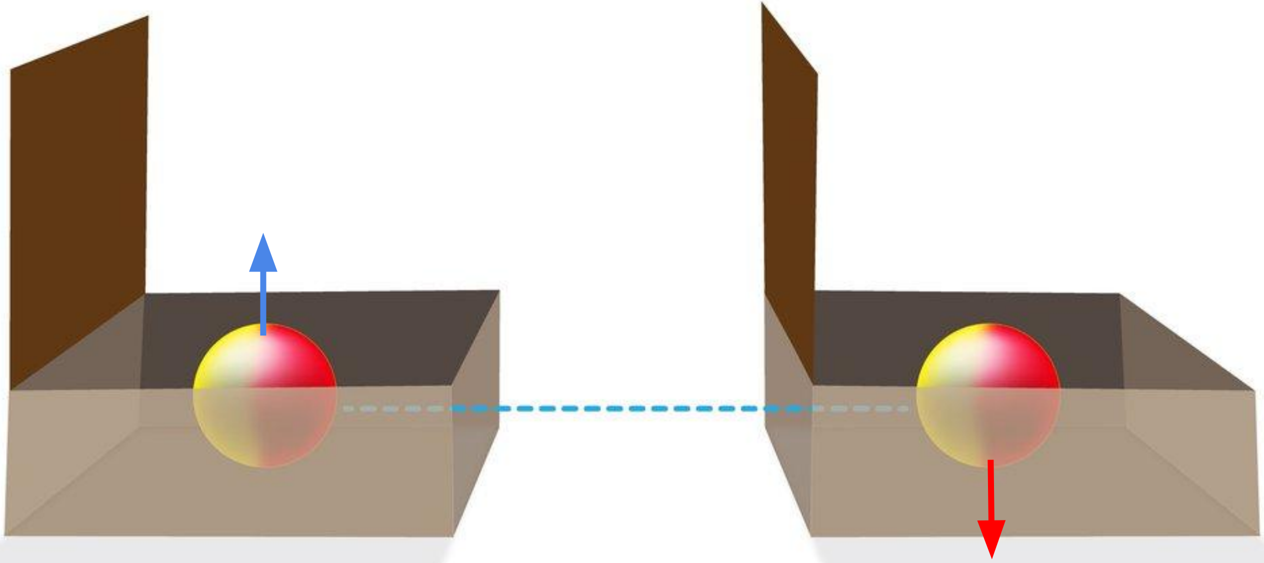

→ “Quantum socks” → Qubits

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

Entanglement

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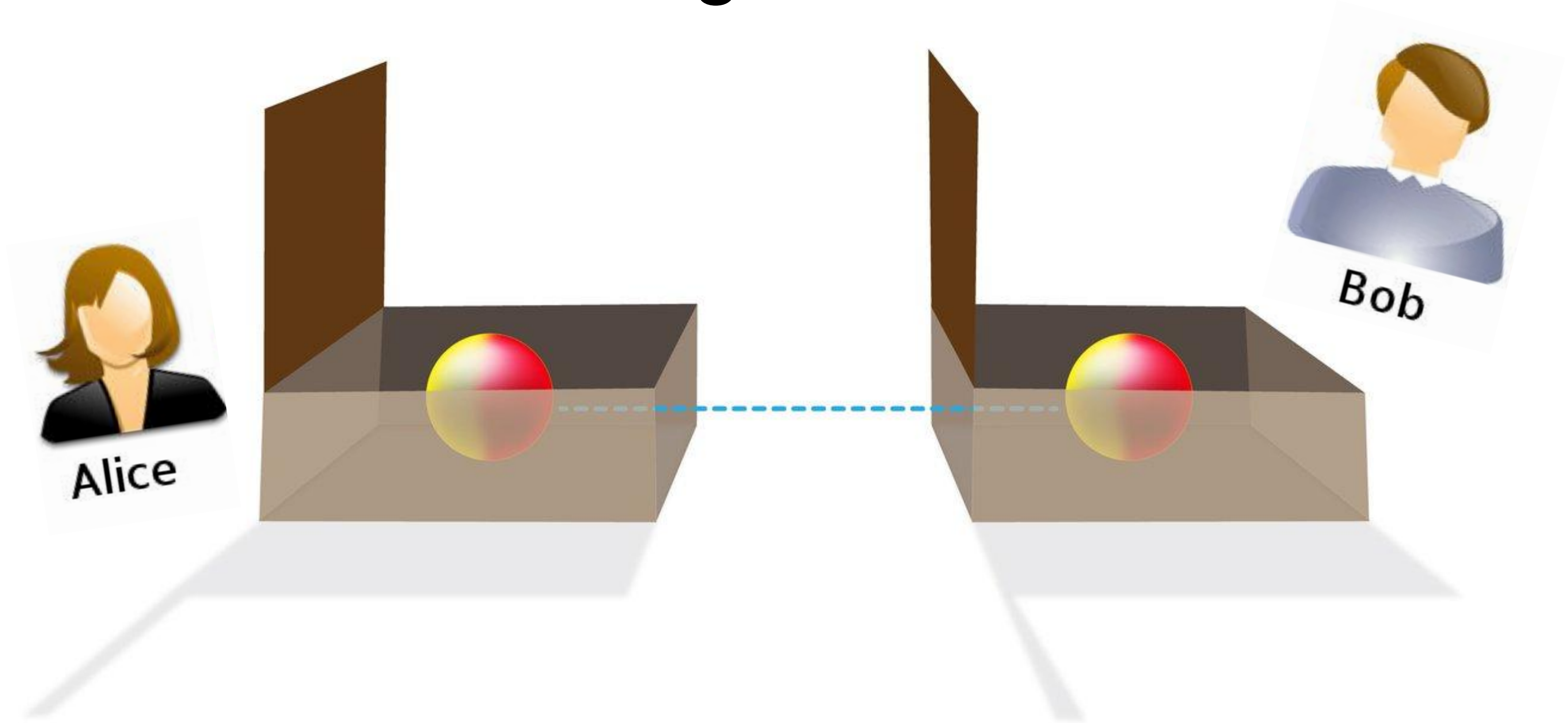
Entanglement



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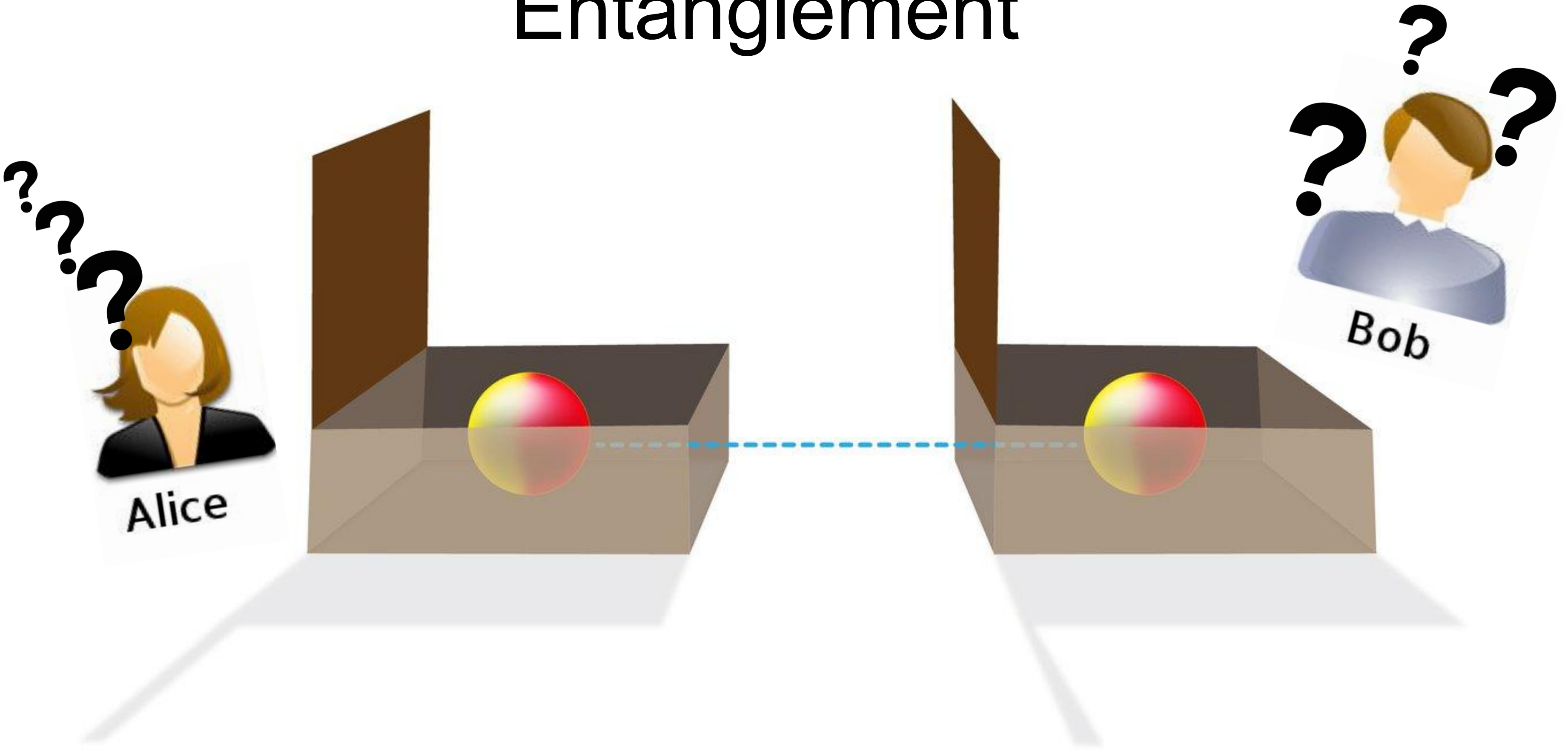

→ It can be very hard to quantify (or even witness) entanglement in general!

Quantifying Entanglement

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→ **Pure states**

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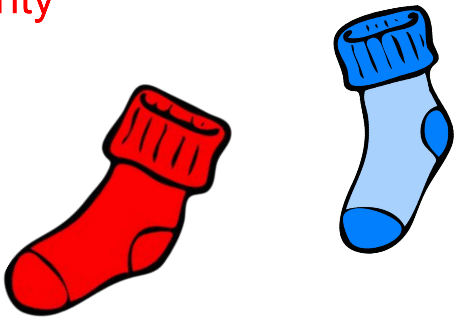

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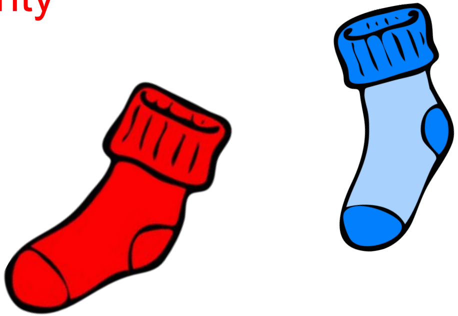

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→ **Negativity** $\mathcal{N}(\hat{\rho}_{AB})$ λ_i

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Summary of entanglement

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- Property of quantum systems
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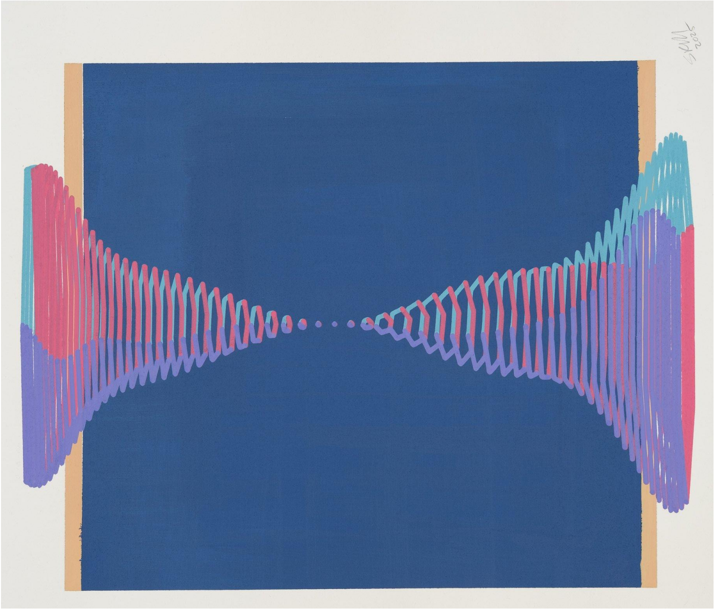
- Very hard to quantify or even witness
- One way of quantifying entanglement: Negativity

Entanglement harvesting



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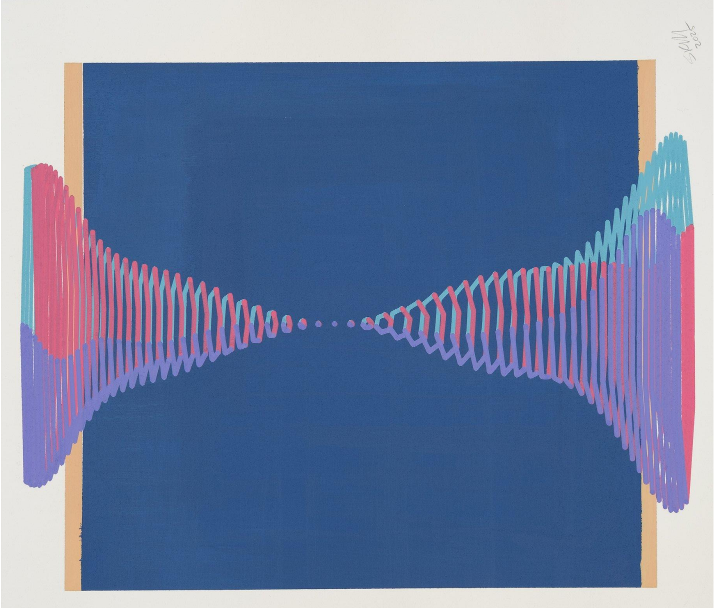
Entanglement harvesting

→ How much entanglement between regions of a quantum field?



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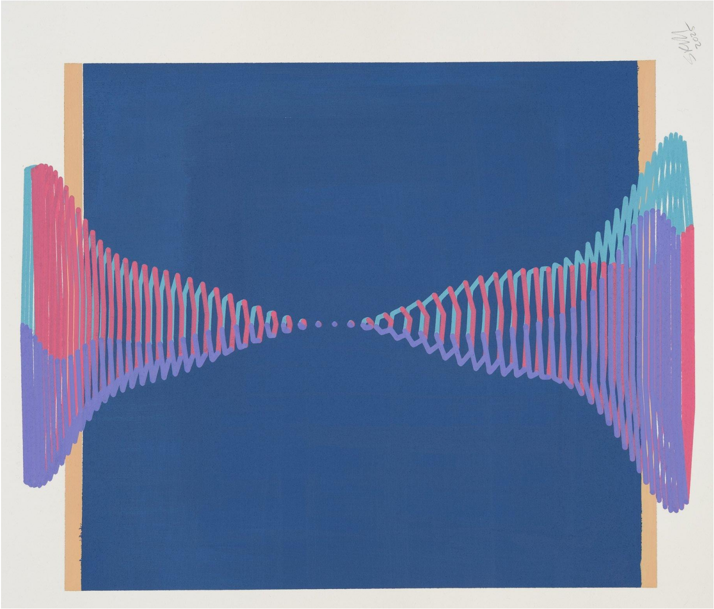

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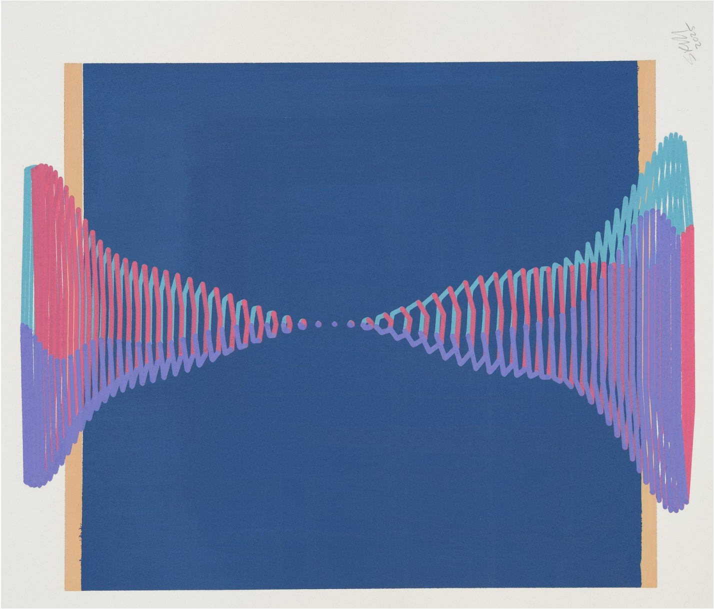

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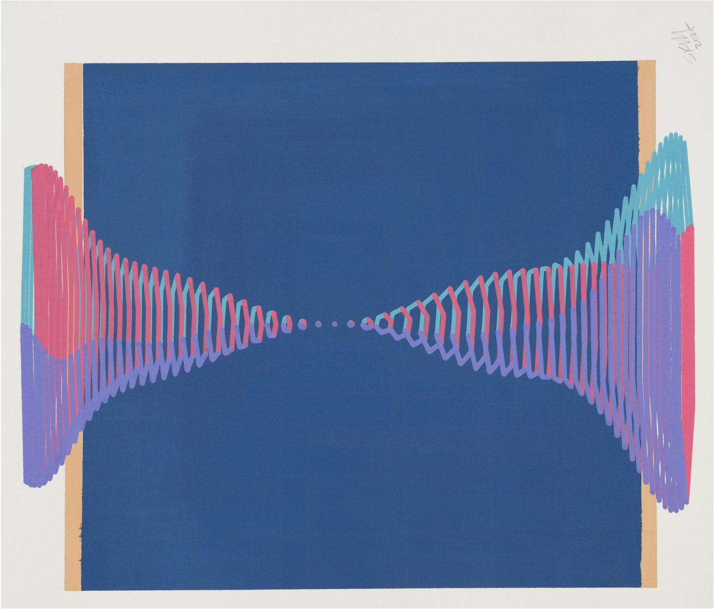

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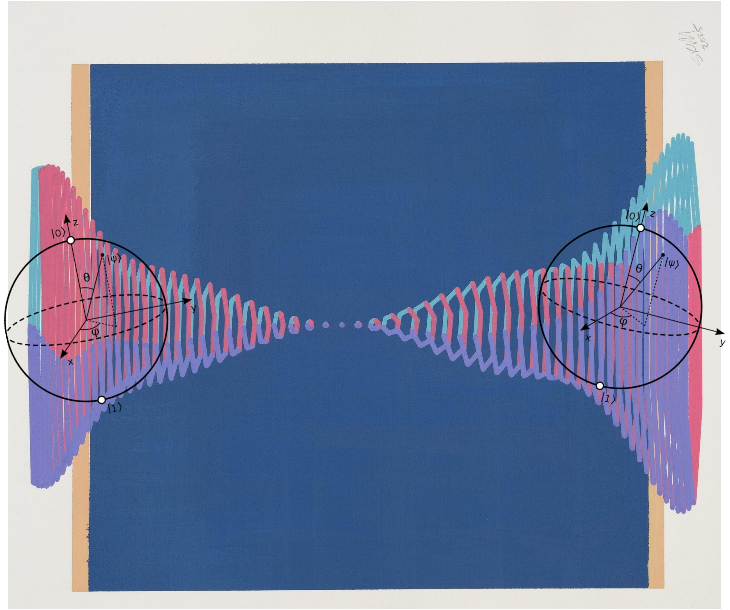

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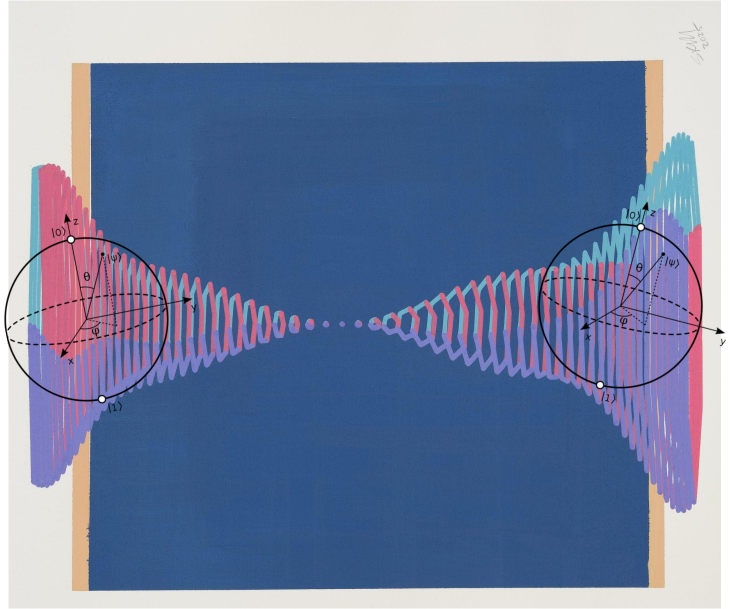

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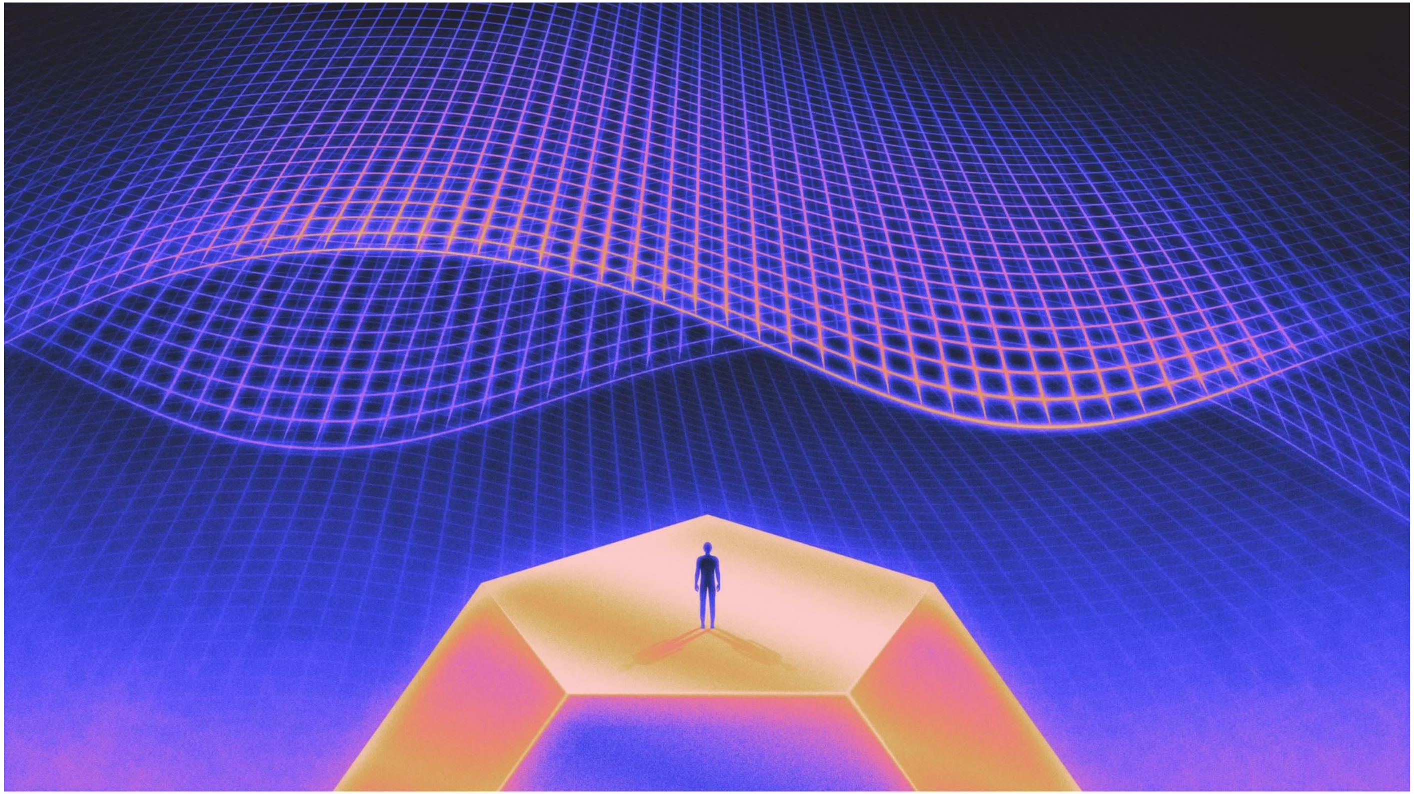
Quantum Field Theory



Quantum Field Theory

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→ Excitations of the field ~ particles

Quantum Field Theory

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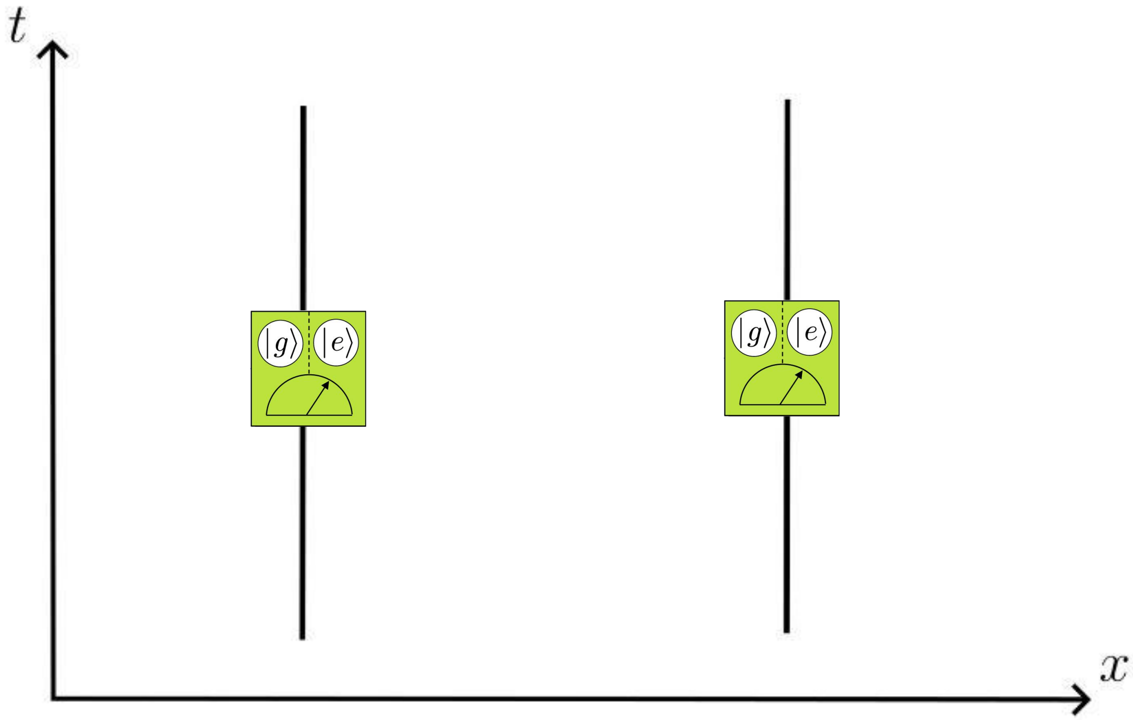
- Excitations of the field ~ particles

- Particles follow dispersion equation $E(p) = cp$

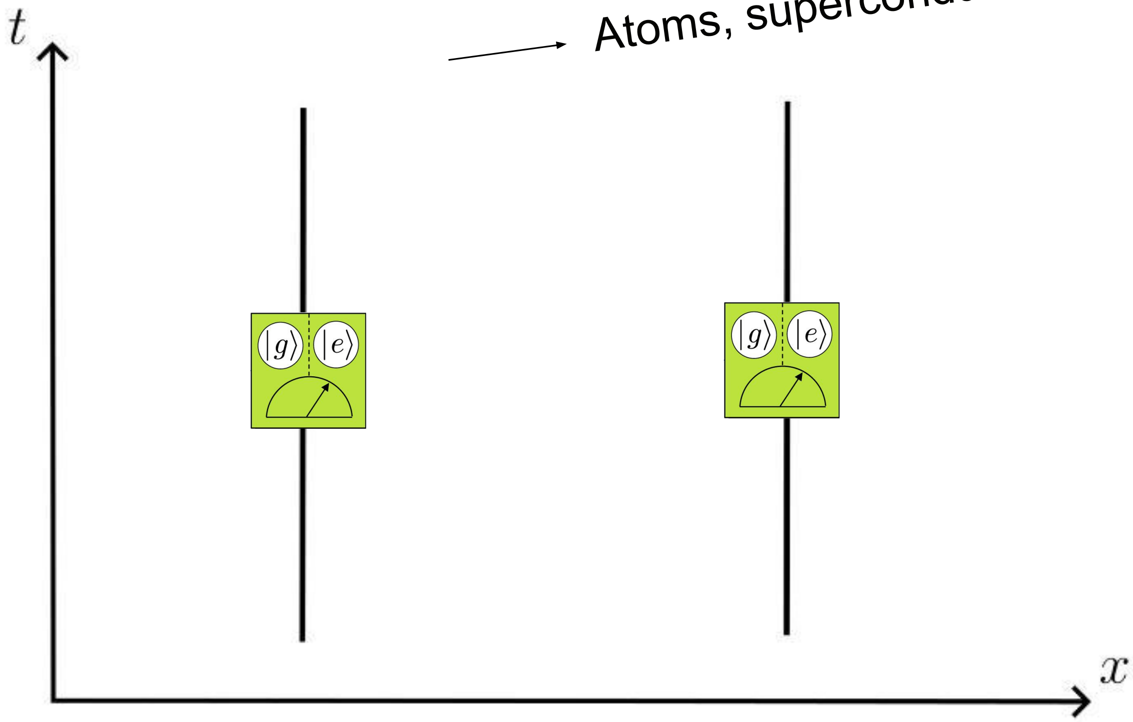
Entanglement Harvesting

Entanglement Harvesting



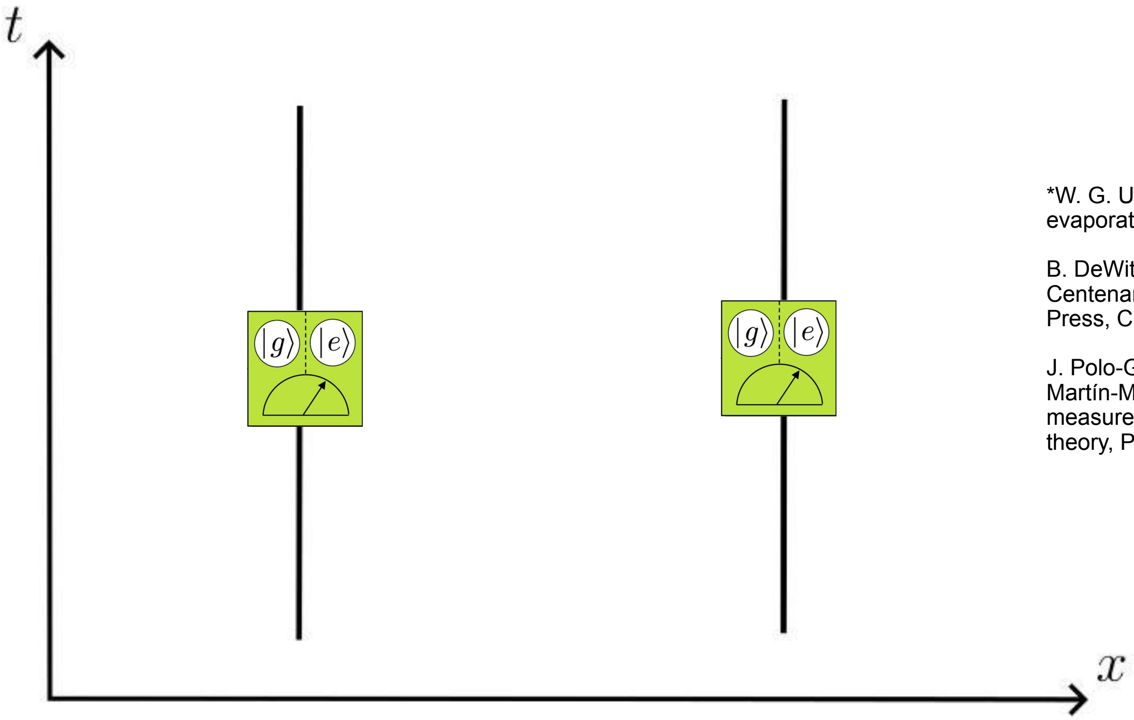
Entanglement Harvesting

Atoms, superconducting qubits



Entanglement Harvesting

→ Unruh-DeWitt (UDW)*



*W. G. Unruh, Notes on black-hole evaporation, Phys. Rev. D 14, 870 (1976).

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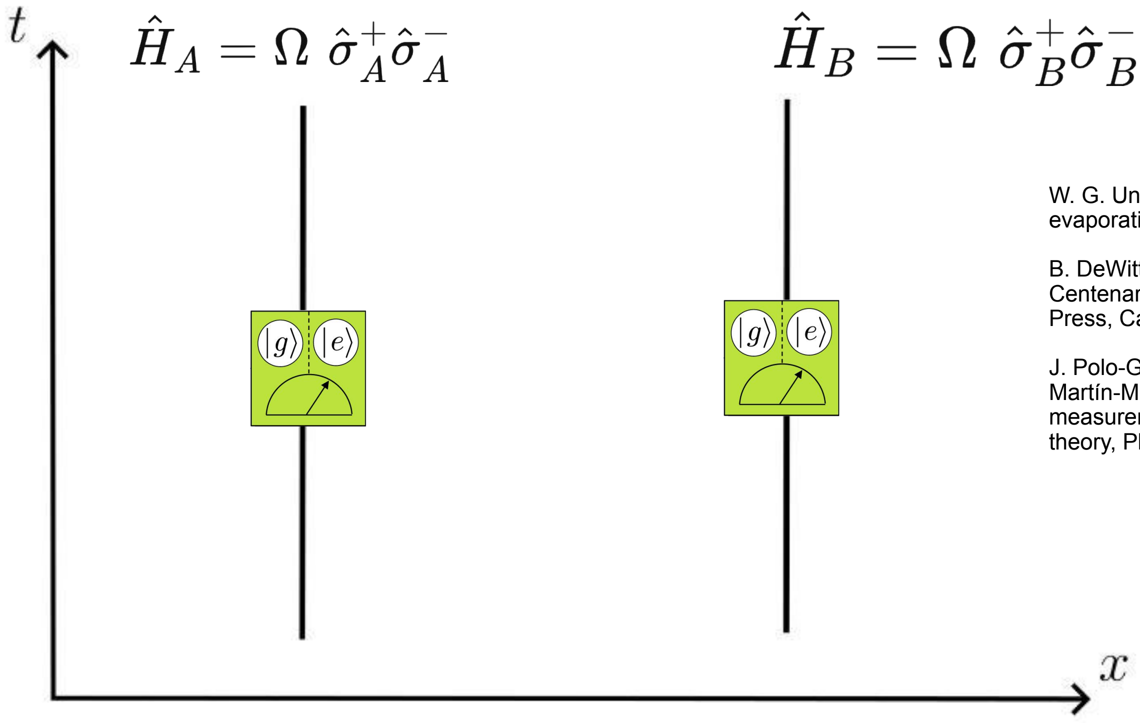

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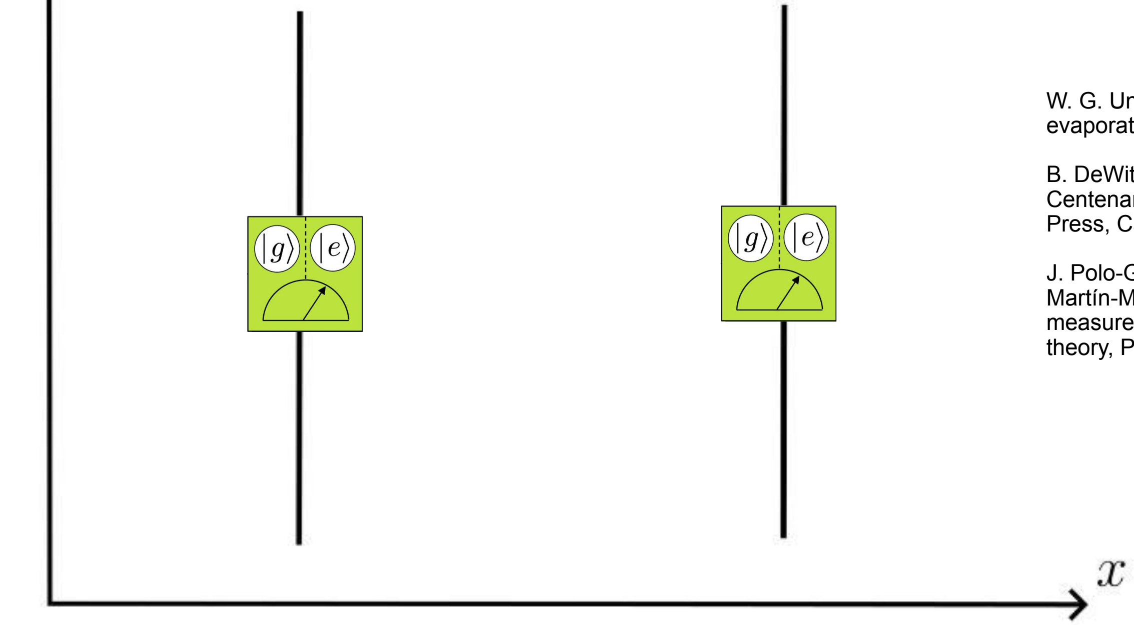
Entanglement Harvesting

Energy gap

$$\hat{H}_A = \Omega\ \hat{\sigma}^+_A \hat{\sigma}^-_A \qquad \hat{H}_B = \Omega\ \hat{\sigma}^+_B \hat{\sigma}^-_B$$



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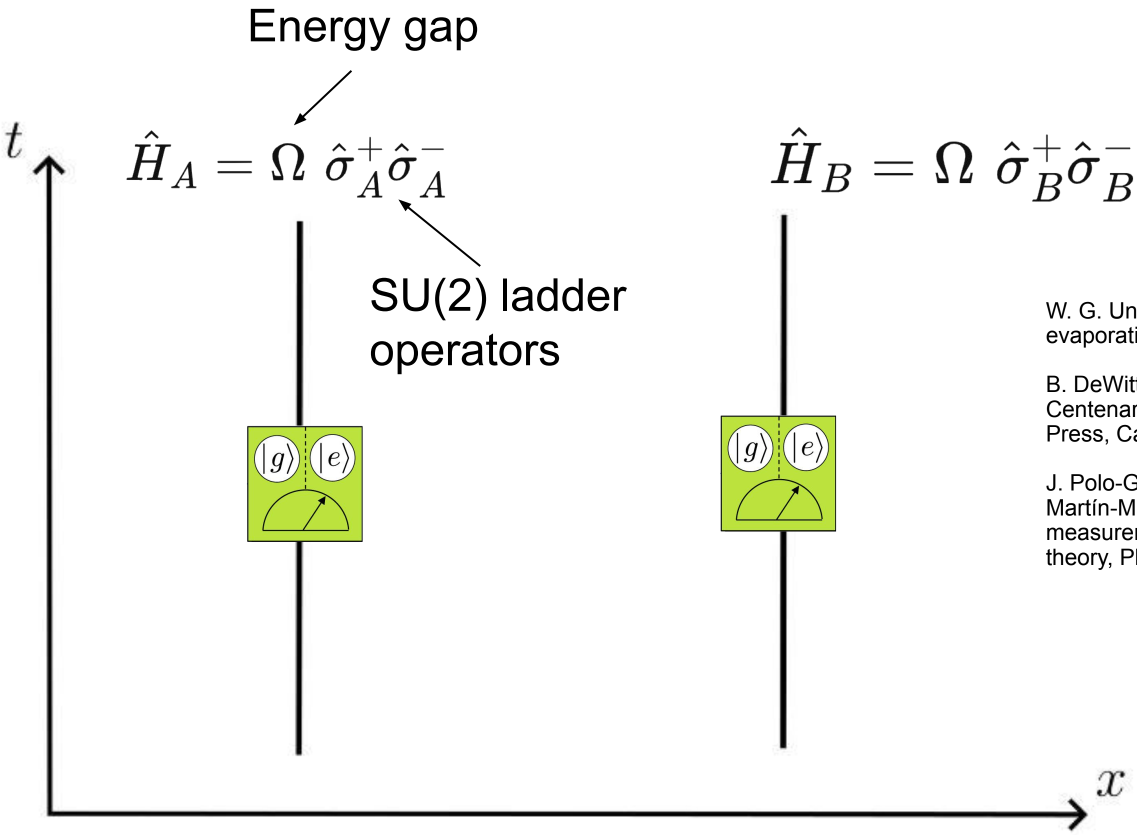

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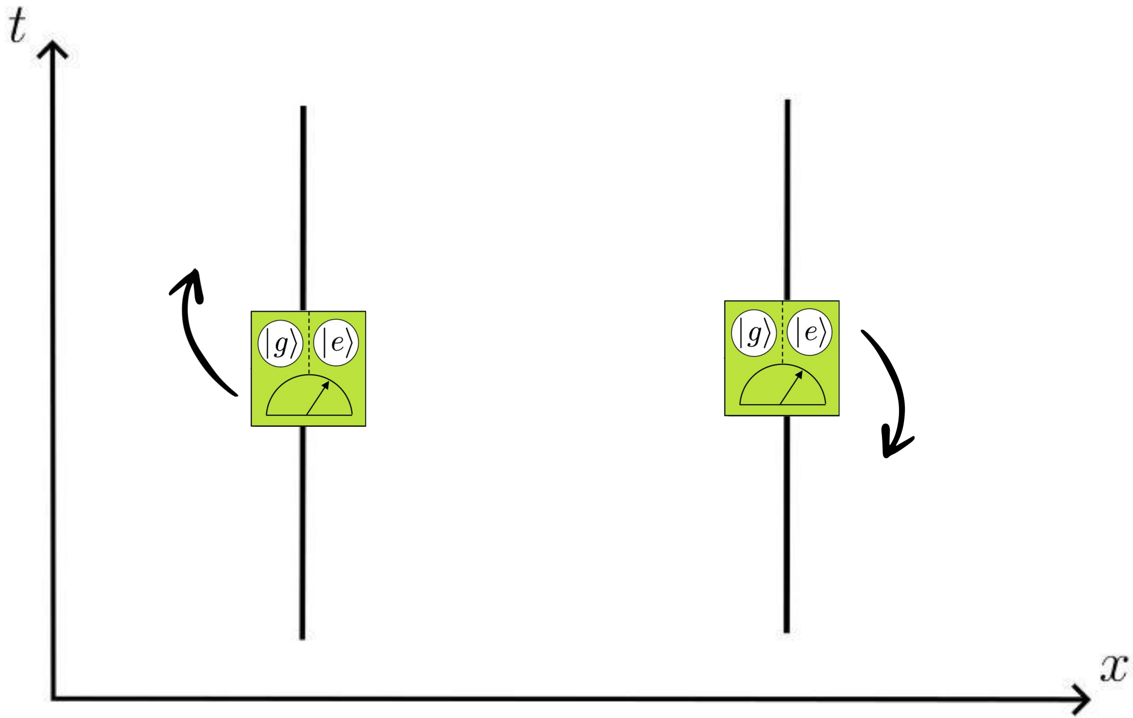

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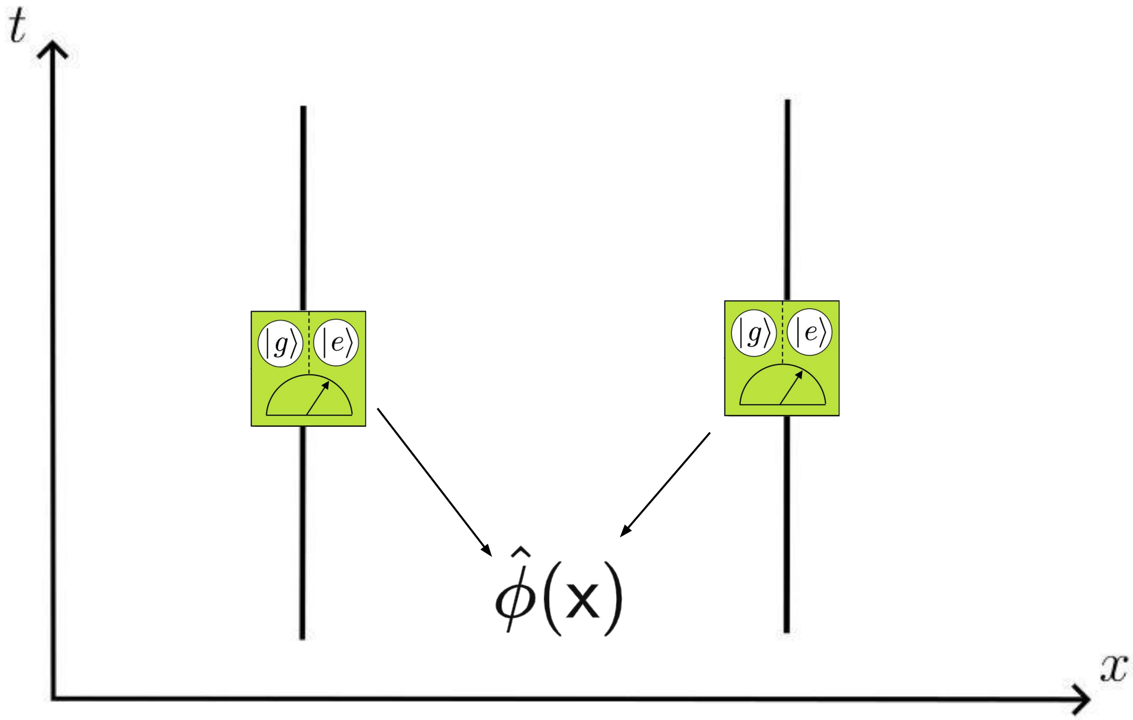

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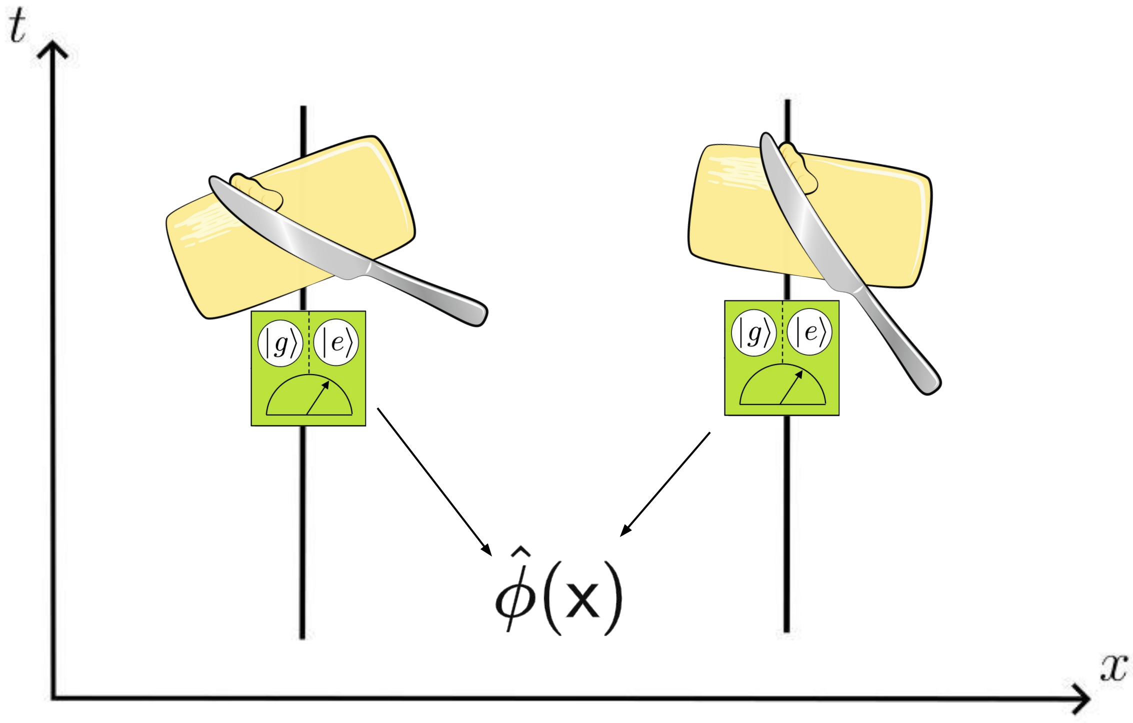
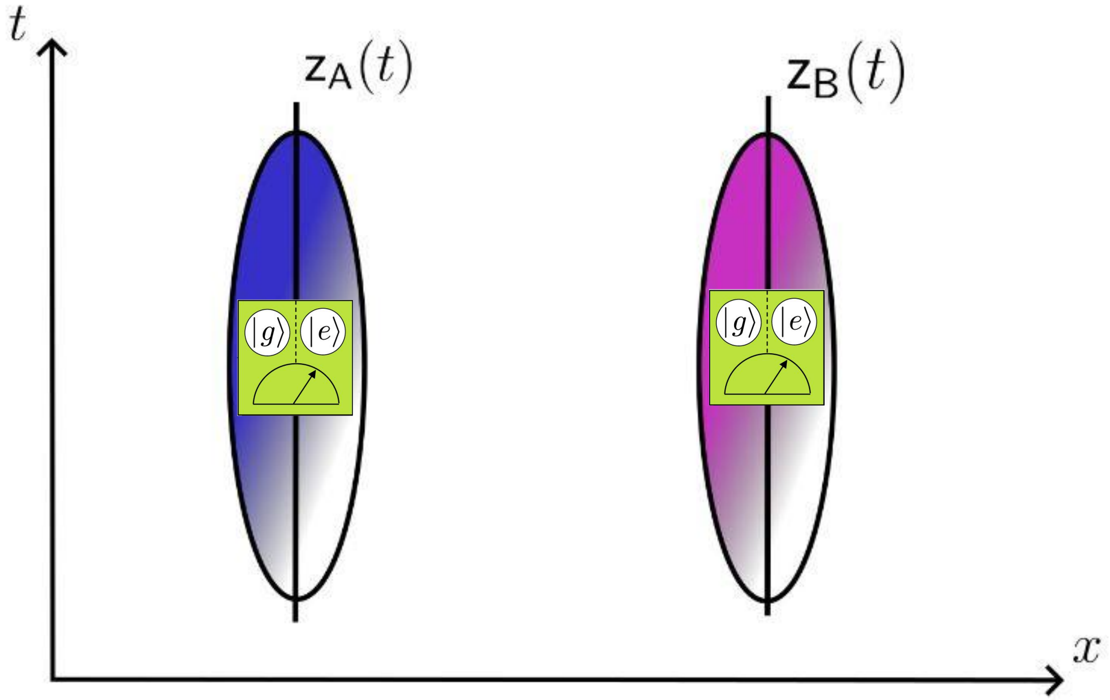


Entanglement Harvesting

→ Physically, the detectors have finite size



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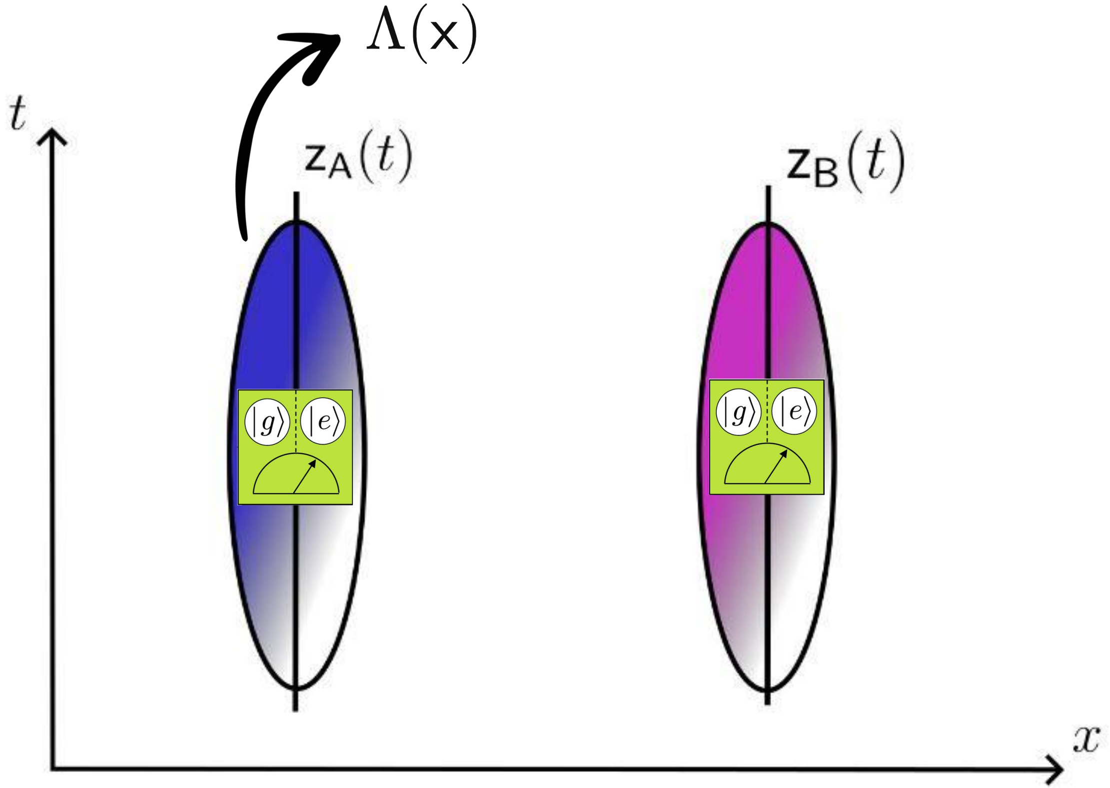

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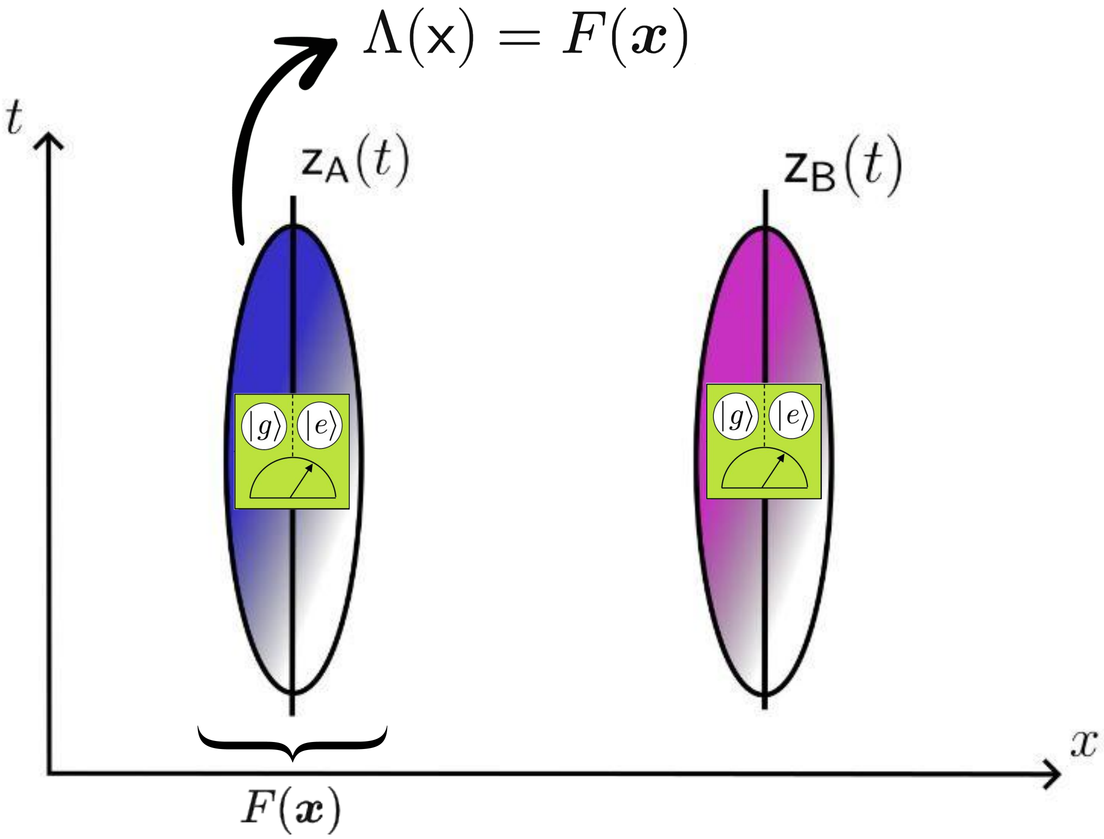

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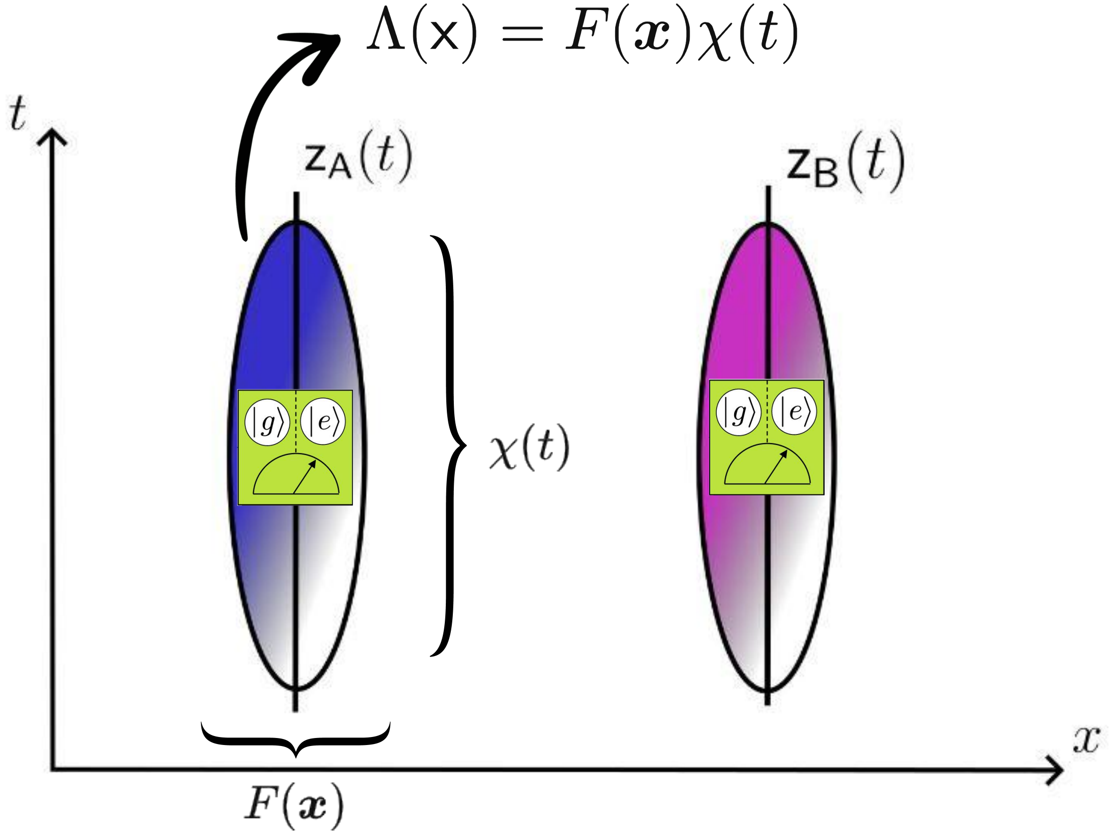
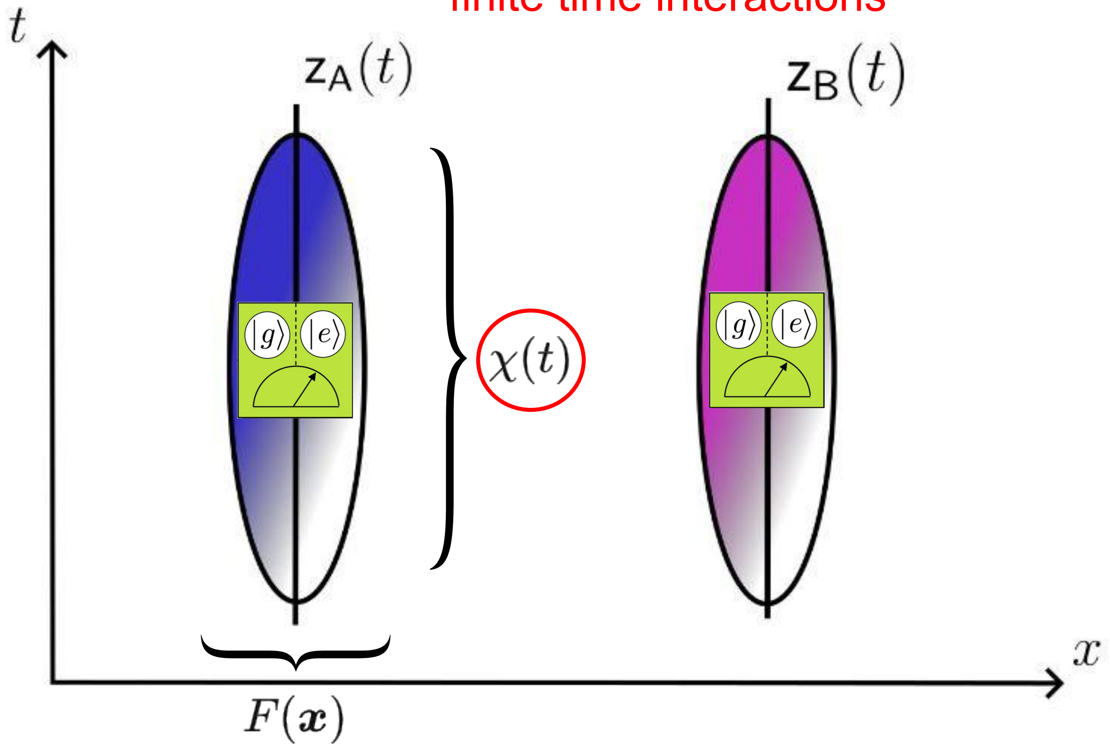


Entanglement Harvesting

We can only see vacuum effects in finite time interactions



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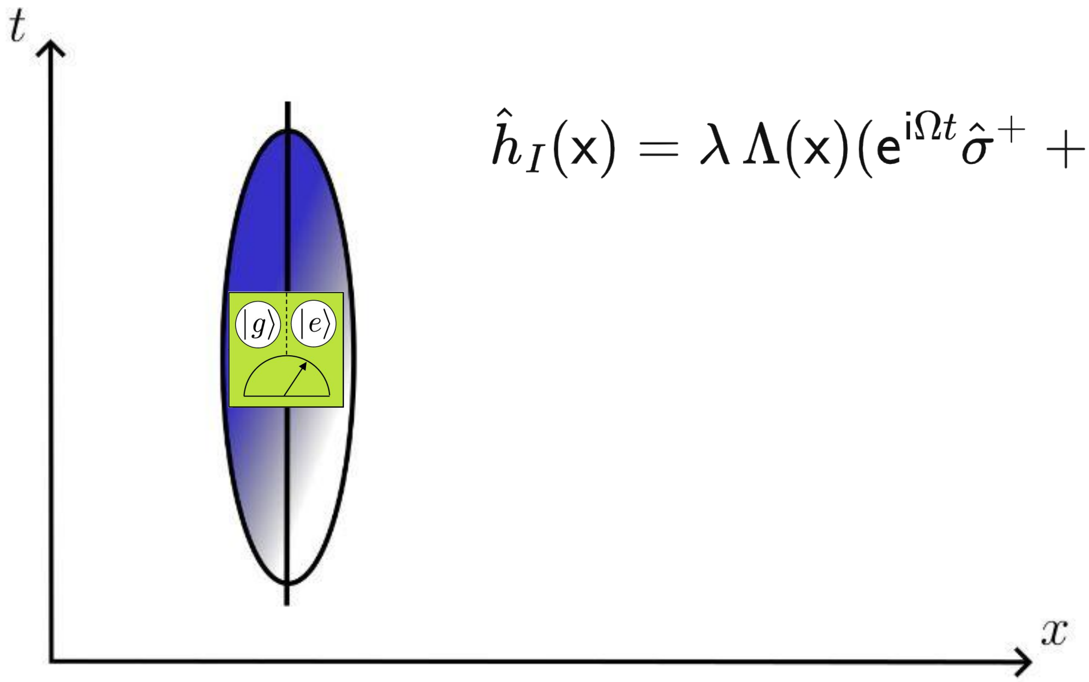

$$\hat{h}_I(\mathsf{x}) = \lambda\,\Lambda(\mathsf{x})\big(\mathrm{e}^{\mathrm{i}\Omega t}\hat{\sigma}^+ + \mathrm{e}^{-\mathrm{i}\Omega t}\hat{\sigma}^-\big)\,\hat{\phi}(\mathsf{x})$$

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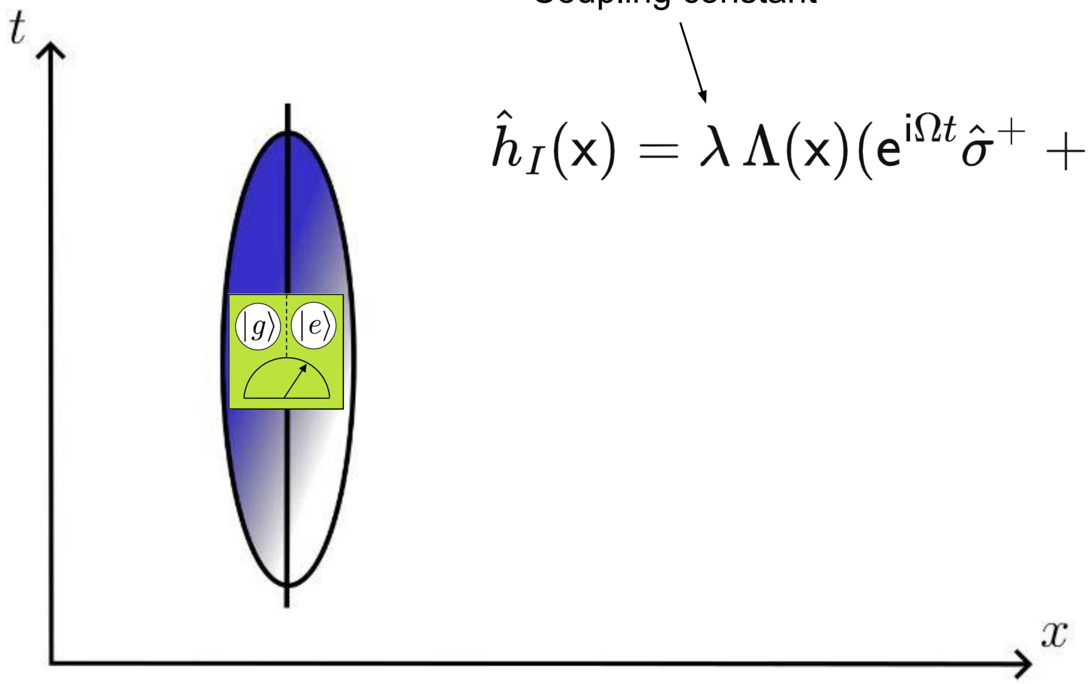

Coupling constant

$$\hat{h}_I(\mathsf{x}) = \underset{\uparrow}{\lambda}\,\Lambda(\mathsf{x})\left(\mathrm{e}^{\mathrm{i}\Omega t}\hat{\sigma}^+ + \mathrm{e}^{-\mathrm{i}\Omega t}\hat{\sigma}^-\right)\hat{\phi}(\mathsf{x})$$

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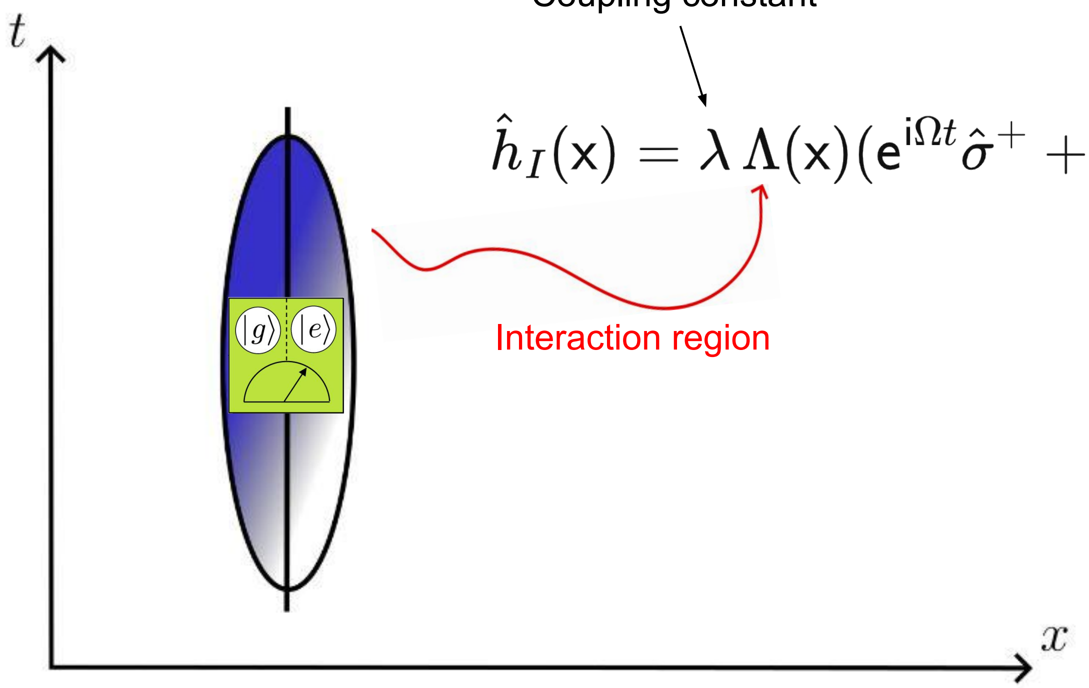

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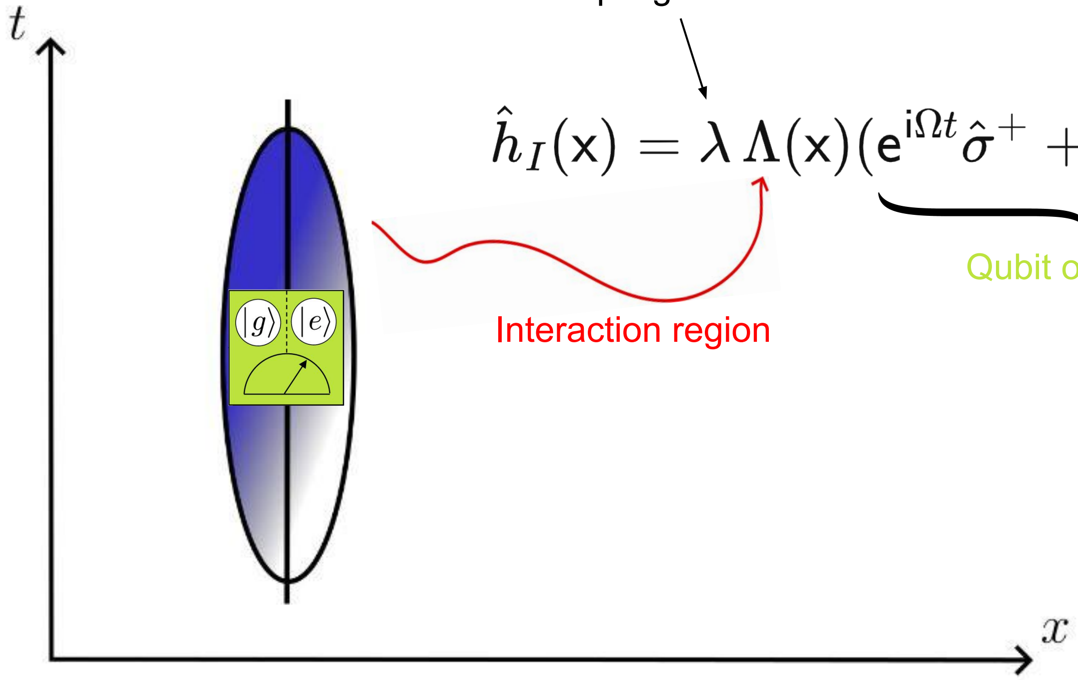
Interaction region

Entanglement Harvesting



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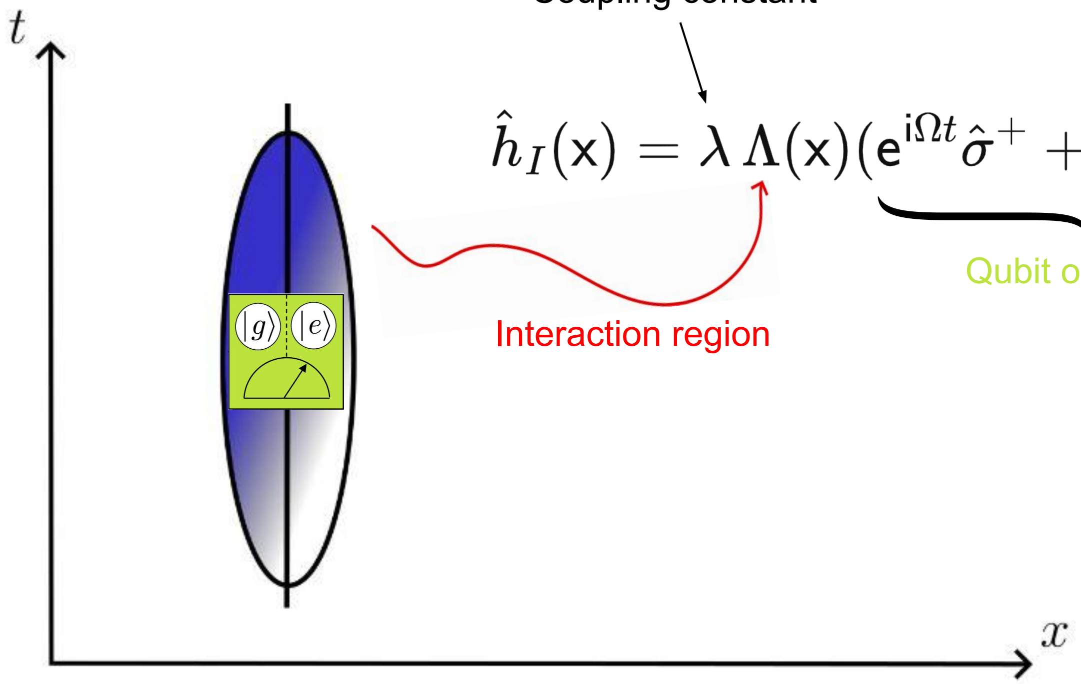

Coupling constant

Quantum field

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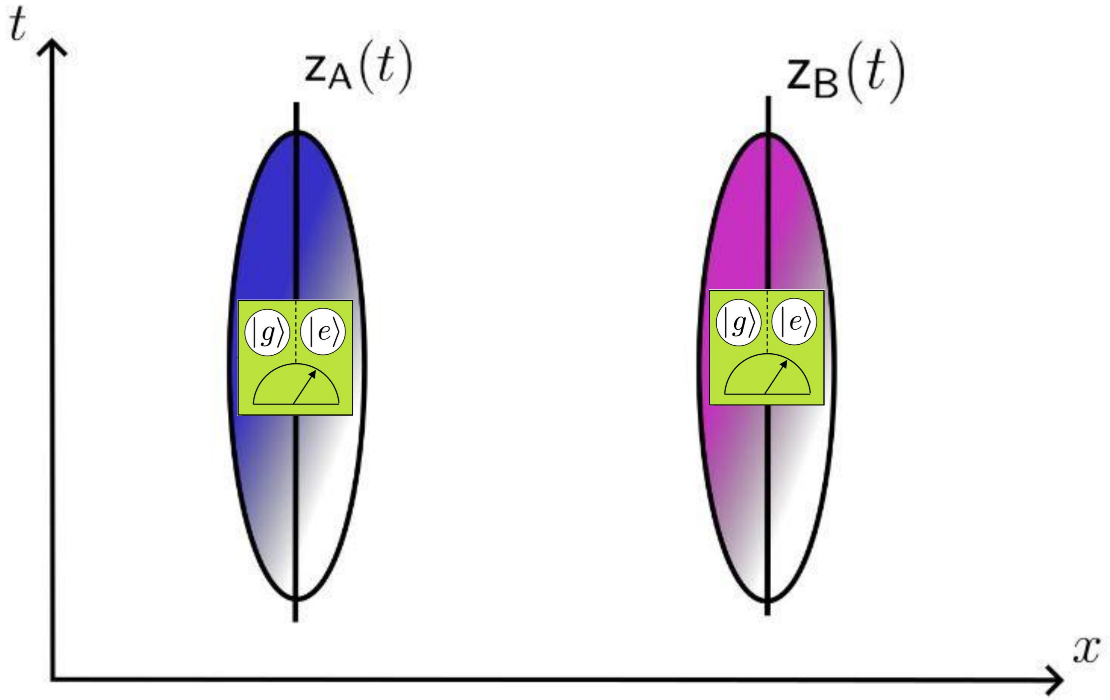

Entanglement Harvesting

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→ Final state of the detectors depend on 2 things:

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$$\boxed{\lambda_{\text{neg}} = -|G_{AB}^{++}| + W^{-+}}$$

R. T. Perche, “Closed-form expressions for smeared bi-distributions of a massless scalar field: non-perturbative and asymptotic results in relativistic quantum information”, 20 Dec 2023, https://doi.org/10.48550/arXiv.2312.13343

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$$\boxed{\lambda_{\text{neg}} = -|G_{AB}^{++}| + W^{-+}}$$

→ $G_{AB}^{++} = \lambda^2 G_F(\Lambda_A^+, \Lambda_B^+)$

→ $W^{-+} = \lambda^2 W(\Lambda_A^-, \Lambda_A^+) = \lambda^2 W(\Lambda_B^-, \Lambda_B^+)$

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Negativity

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- $\longrightarrow$ Feynman propagator
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Negativity

→ $G_F(\Lambda_A^+, \Lambda_B^+)$

- → Feynman propagator
- → Non local
- → Encodes quantum correlations → Entanglement

Negativity

$\longrightarrow G_F(\Lambda_A^+, \Lambda_B^+)$

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- $\longrightarrow$ Encodes quantum correlations $\longrightarrow$ Entanglement
- $\longrightarrow$ Related to **probability amplitude** that **both** detectors become excited mutually **due** to the quantum correlations

Negativity

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$\longrightarrow W(\Lambda_A^-, \Lambda_A^+) = W(\Lambda_B^-, \Lambda_B^+)$

Negativity

$\longrightarrow$ $G_F(\Lambda_A^+, \Lambda_B^+)$

- $\longrightarrow$ Feynman propagator
- $\longrightarrow$ Non local
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- $\longrightarrow$ Related to **probability amplitude** that **both** detectors become excited mutually **due** to the quantum correlations

$\longrightarrow$ $W(\Lambda_A^-, \Lambda_A^+) = W(\Lambda_B^-, \Lambda_B^+)$

- $\longrightarrow$ Wightman function

Negativity

→ $G_F(\Lambda_A^+, \Lambda_B^+)$

- → Feynman propagator
- → Non local
- → Encodes quantum correlations → Entanglement
- → Related to **probability amplitude** that **both** detectors become excited mutually **due** to the quantum correlations

→ $W(\Lambda_A^-, \Lambda_A^+) = W(\Lambda_B^-, \Lambda_B^+)$

- → Wightman function
- → Local

Negativity

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- $\longrightarrow$ **Probability** that **a** detectors become excited

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$$\mathcal{N} = \max\{0, |G_{AB}^{++}| - W^{-+}\}$$

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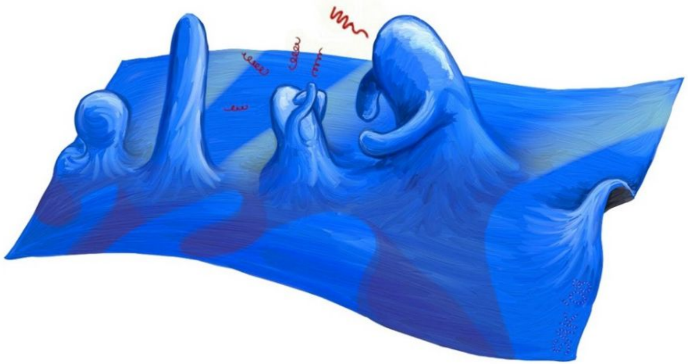

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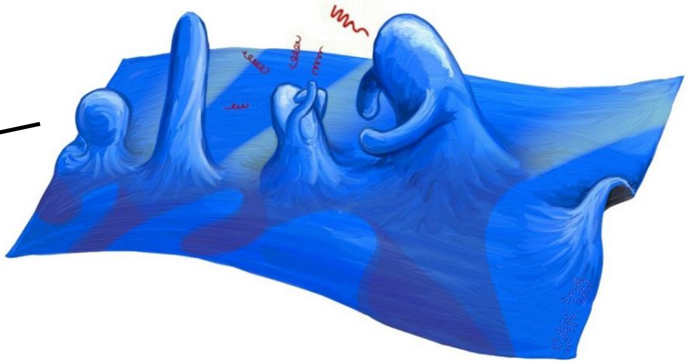

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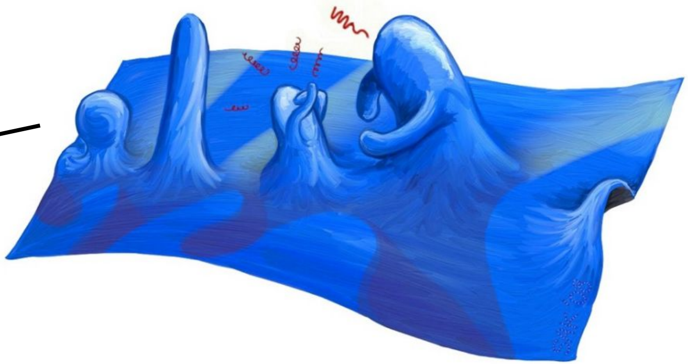

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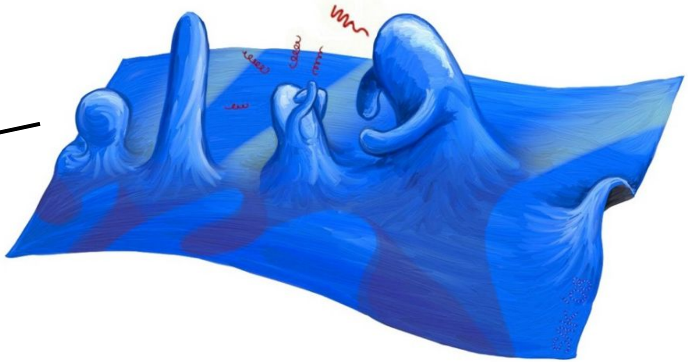

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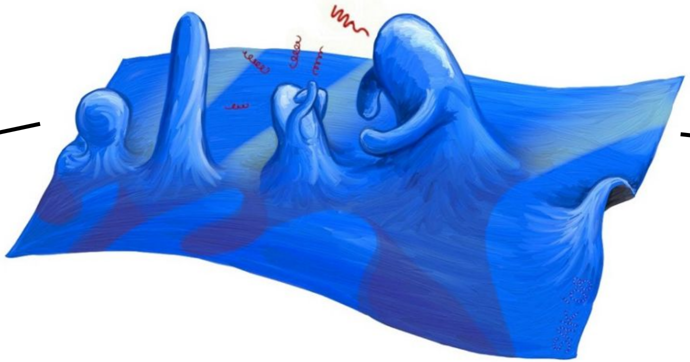

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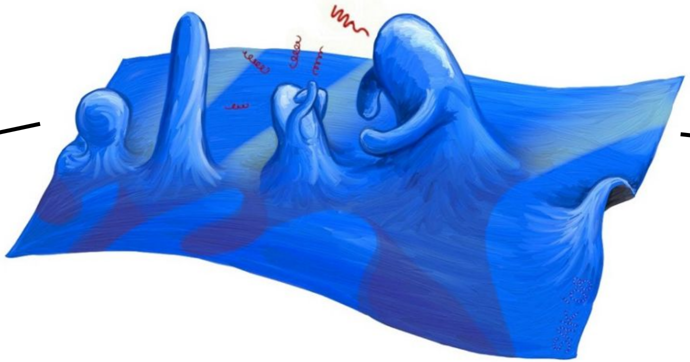

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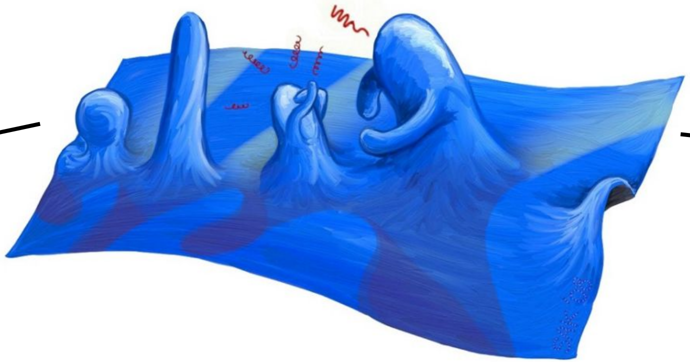

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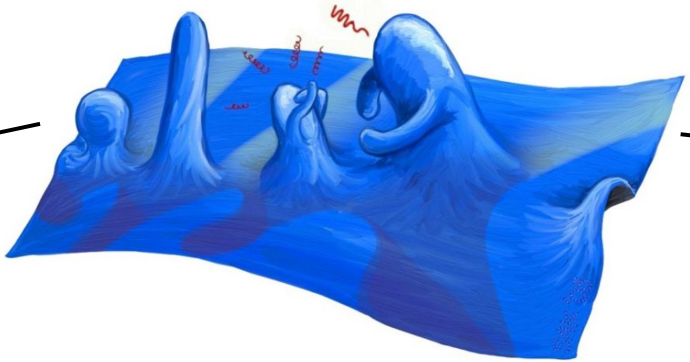

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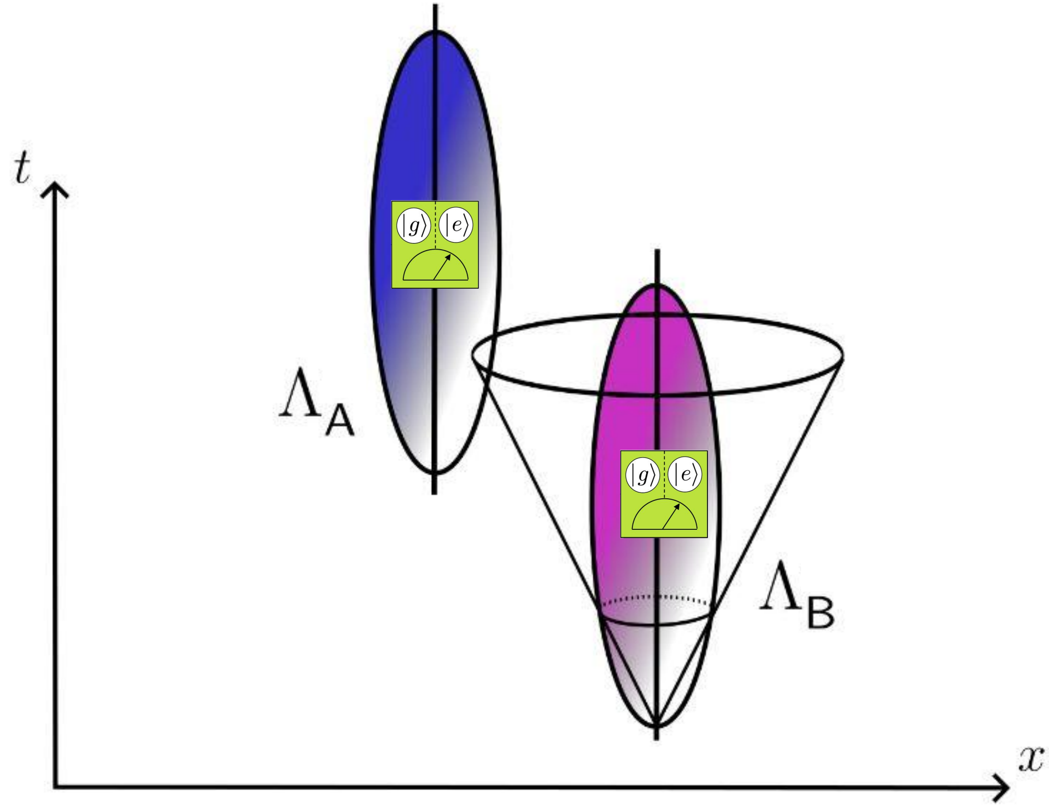
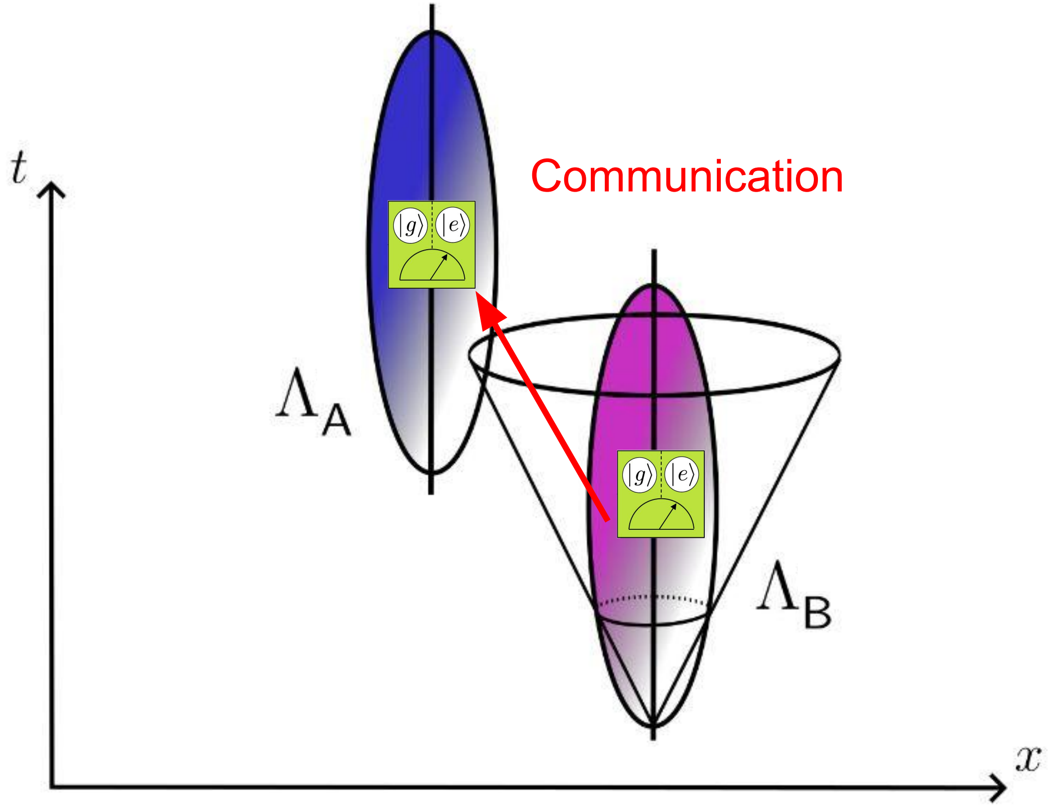
Entanglement Harvesting



Entanglement Harvesting



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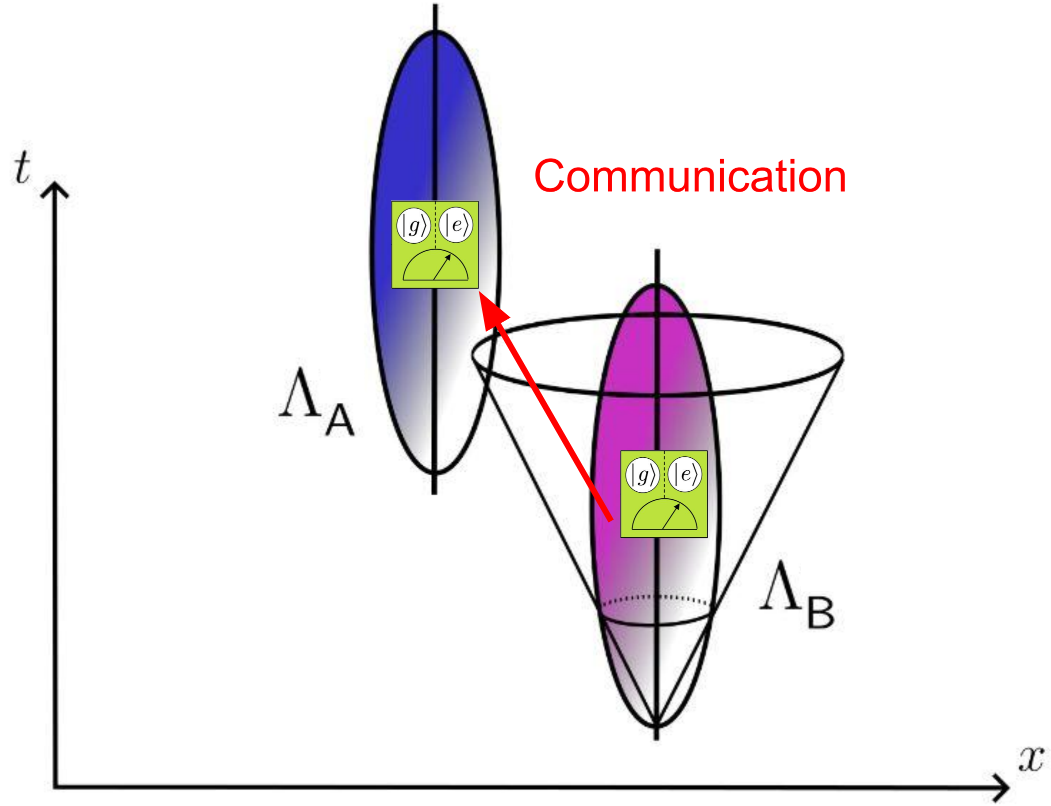

$\longrightarrow$ How to make the difference?

How to quantify signaling?

How to quantify signaling?

$\longrightarrow \quad G_{\mathsf{F}}(\Lambda^{+}_{\mathsf{A}}, \Lambda^{+}_{\mathsf{B}})$

How to quantify signaling?

$$\longrightarrow \quad G_{\mathsf{F}}(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) = \frac{1}{2}\left(H(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) + \mathsf{i}\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+})\right)$$

How to quantify signaling?

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state dependence

How to quantify signaling?

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state dependence

exchange of information

How to quantify signaling?

How to quantify signaling?

$$\Delta(\Lambda^+_{\mathsf{A}}, \Lambda^+_{\mathsf{B}}) = G_{\mathsf{R}}(\Lambda^+_{\mathsf{A}}, \Lambda^+_{\mathsf{B}}) + G_{\mathsf{R}}(\Lambda^+_{\mathsf{B}}, \Lambda^+_{\mathsf{A}})$$

How to quantify signaling?

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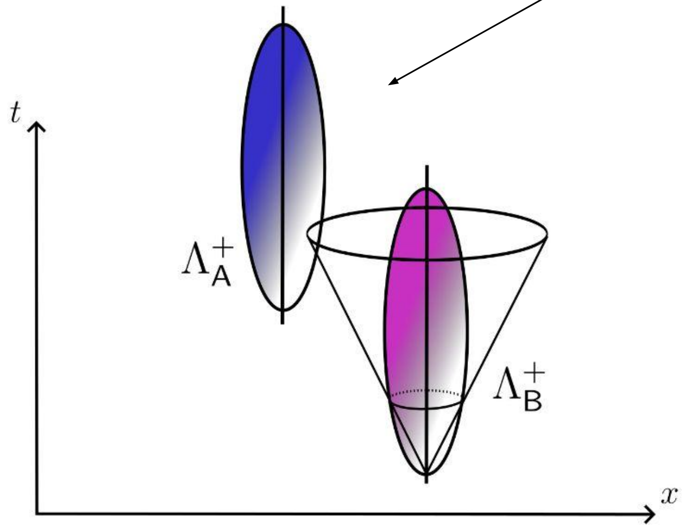

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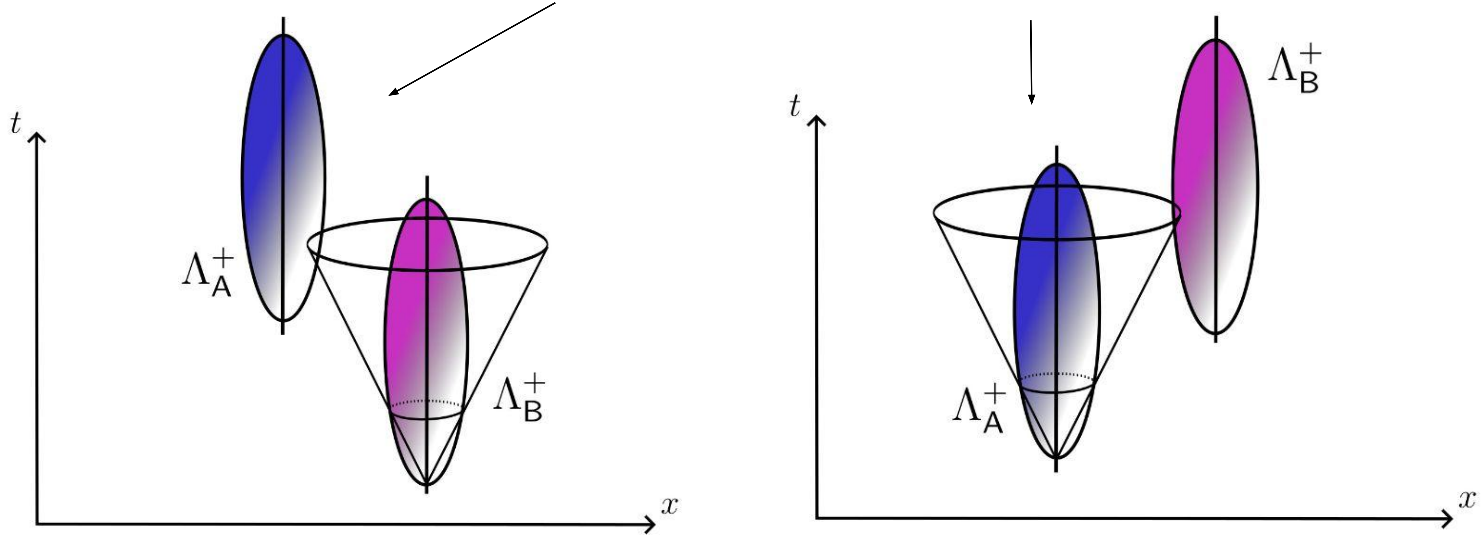

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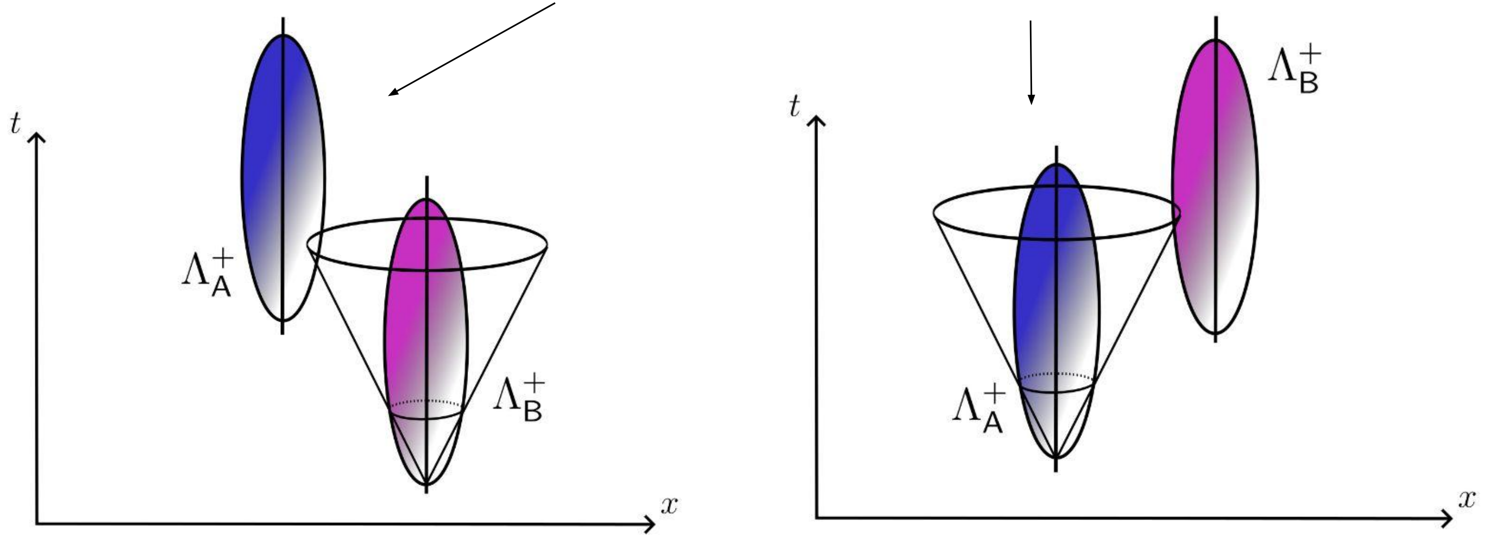

If $\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) \neq 0$, there is exchange of information

How to quantify signaling?

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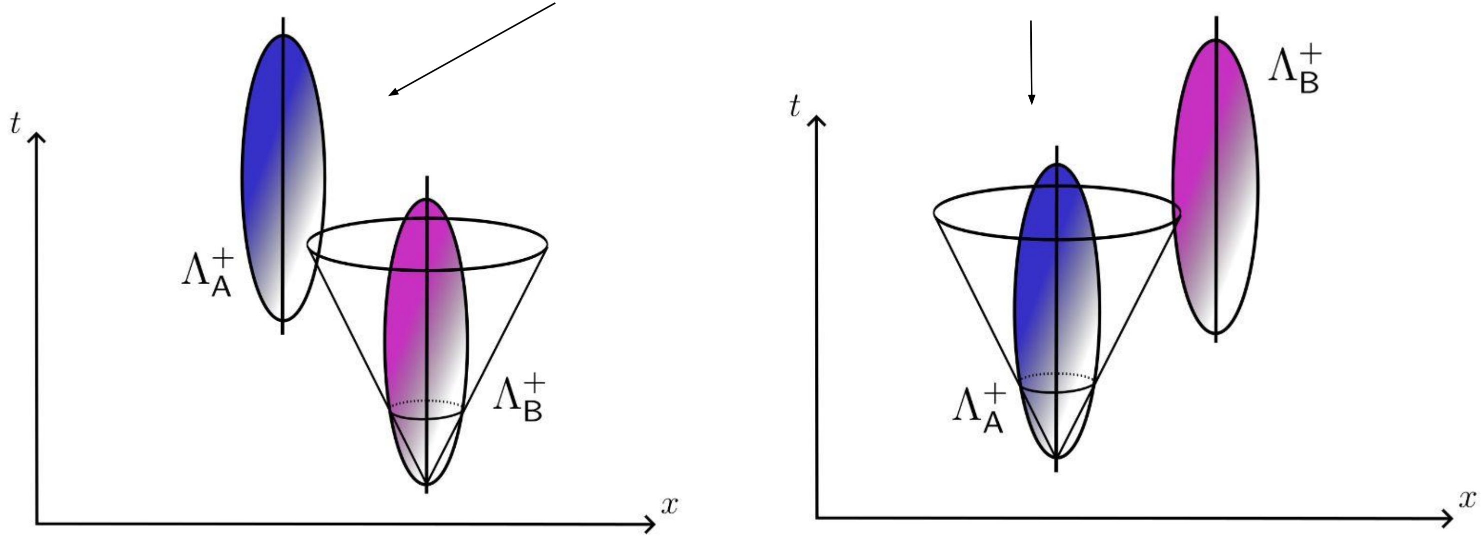

$\longrightarrow$ If $\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) \neq 0$, there is exchange of information $\longrightarrow$ "bad" entanglement

How to quantify signaling?

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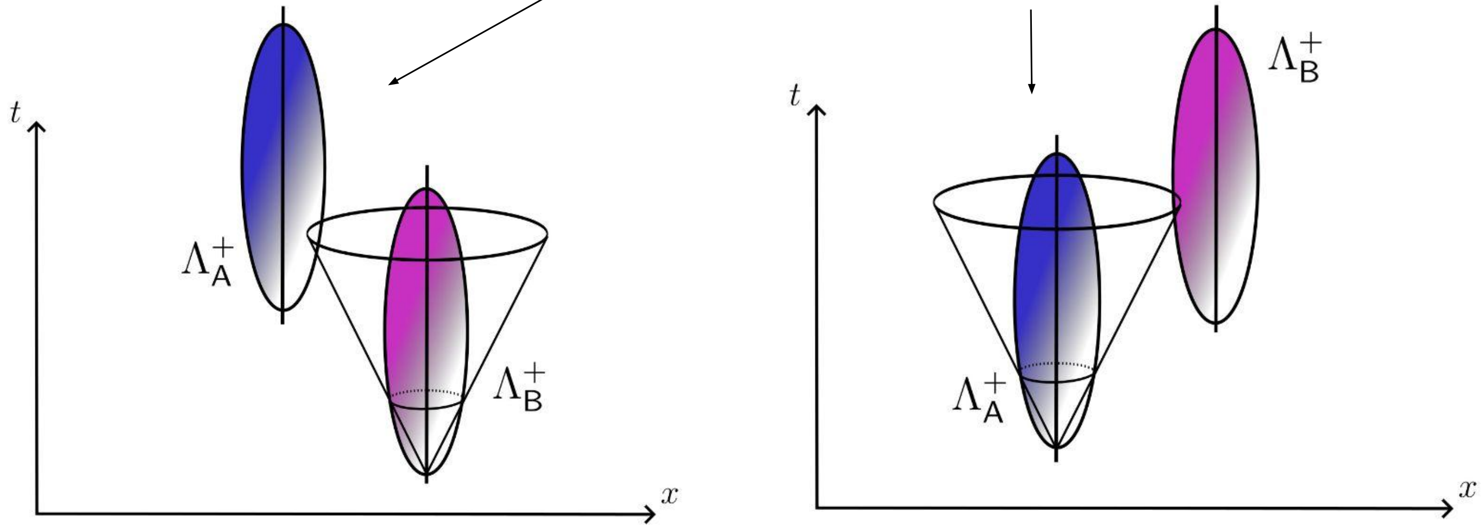
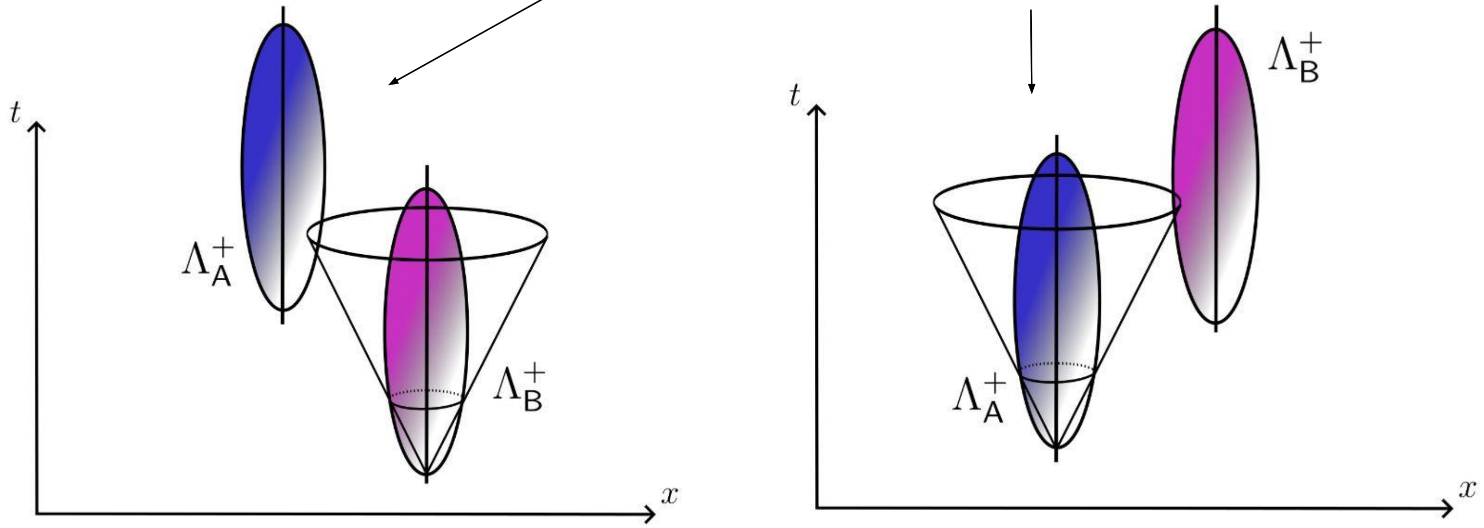


→ If $\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) \neq 0$, there is exchange of information → "bad" entanglement

→ If $\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) = 0$, there is **no** exchange of information

How to quantify signaling?

$$\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) = G_{\mathsf{R}}(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) + G_{\mathsf{R}}(\Lambda_{\mathsf{B}}^{+}, \Lambda_{\mathsf{A}}^{+})$$



→ If $\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) \neq 0$, there is exchange of information → “bad” entanglement

→ If $\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) = 0$, there is **no** exchange of information → genuine entanglement from vacuum

How to quantify signaling?

$$\Delta(\Lambda_{\mathsf{A}}^{+}, \Lambda_{\mathsf{B}}^{+}) = 0$$

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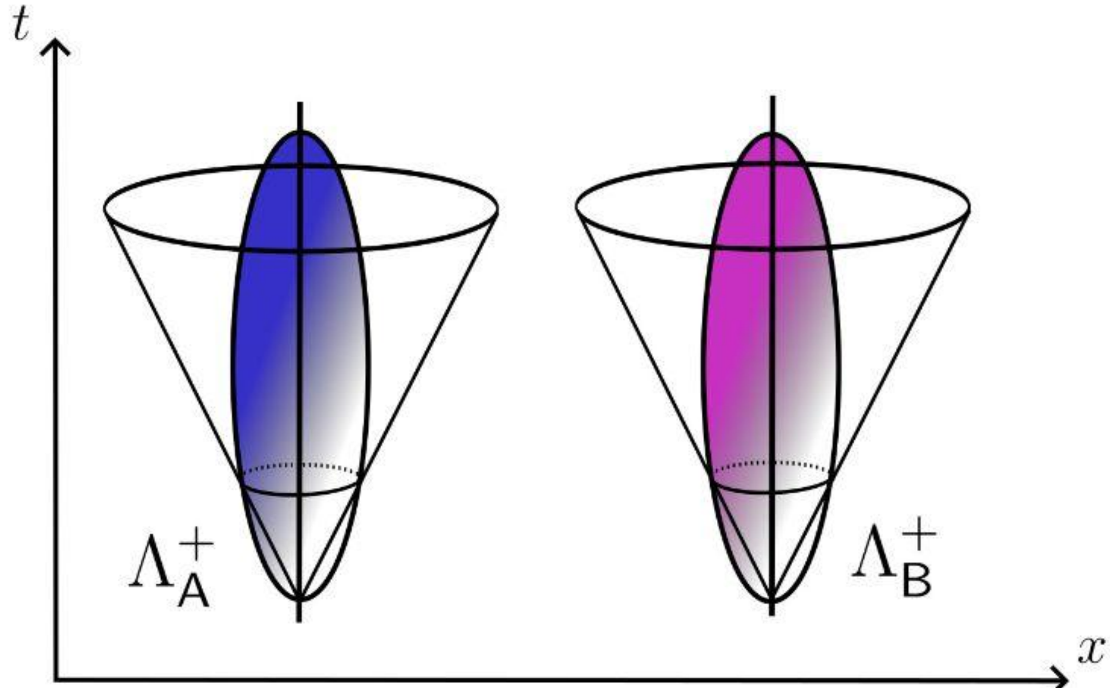

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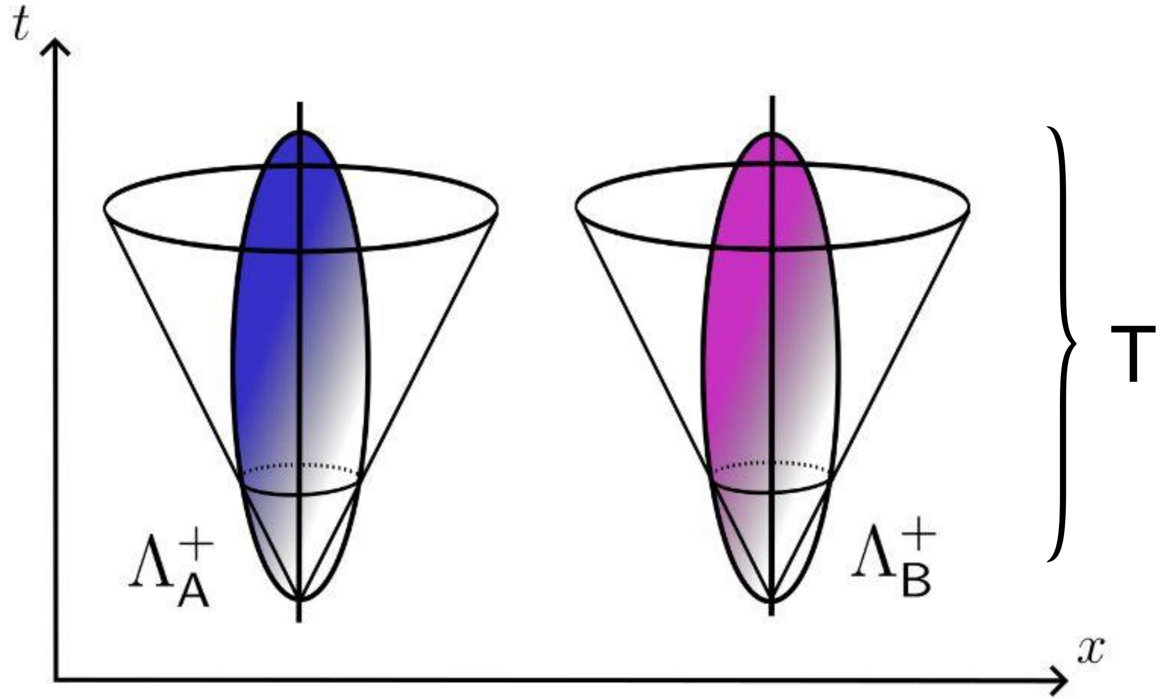

How to quantify signaling?

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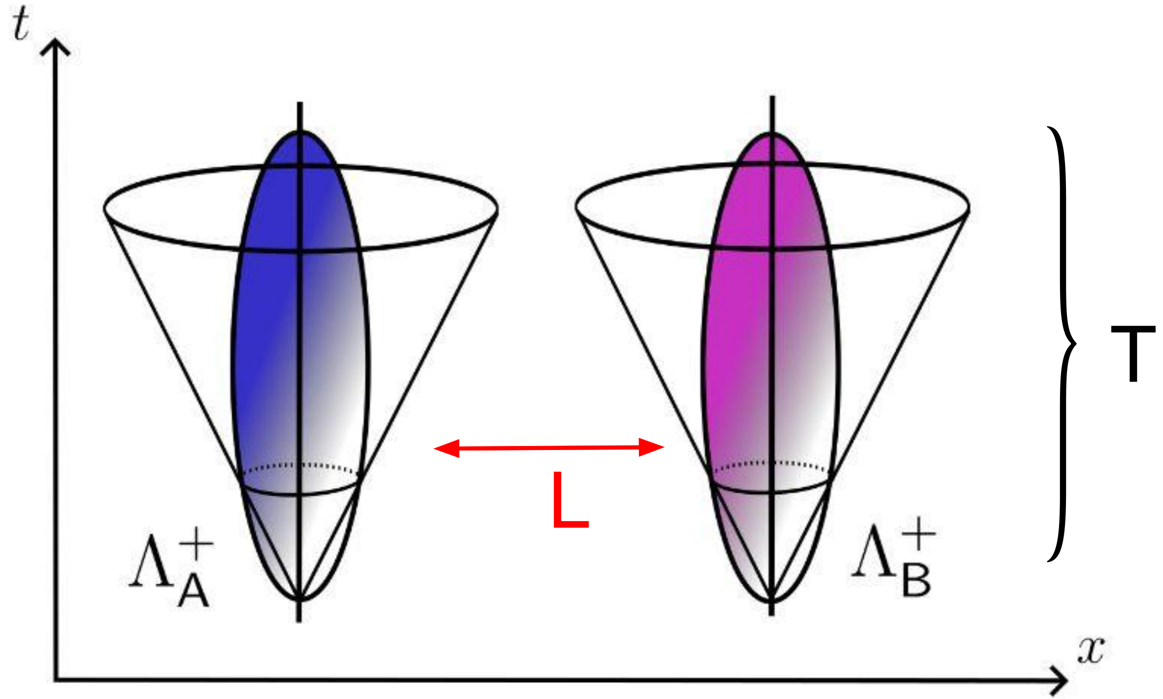
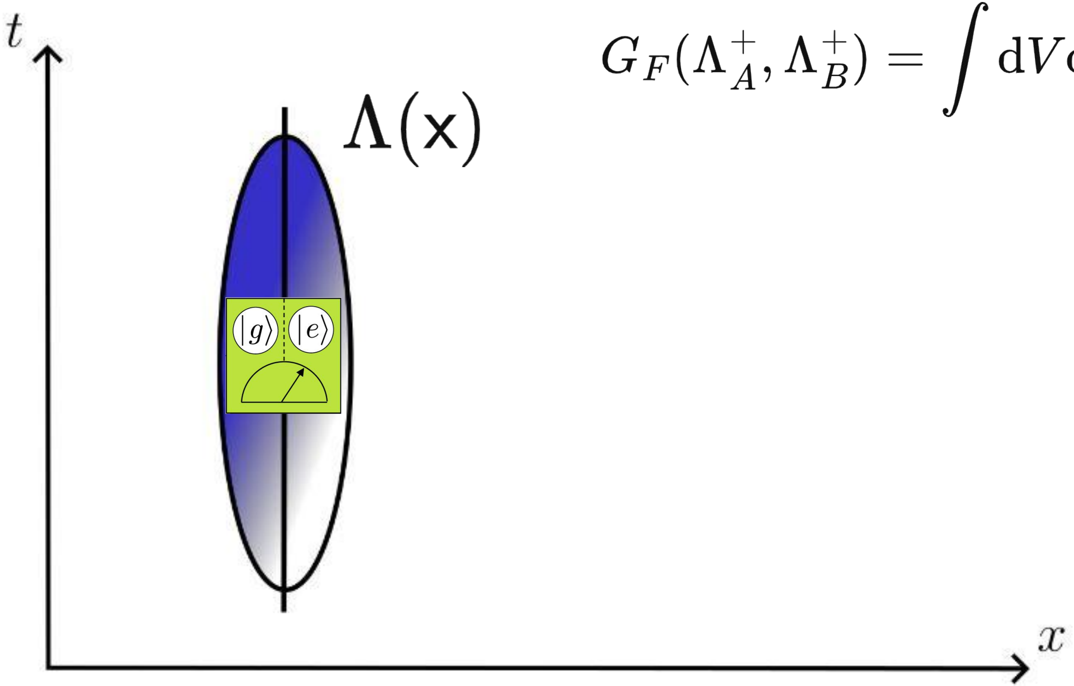


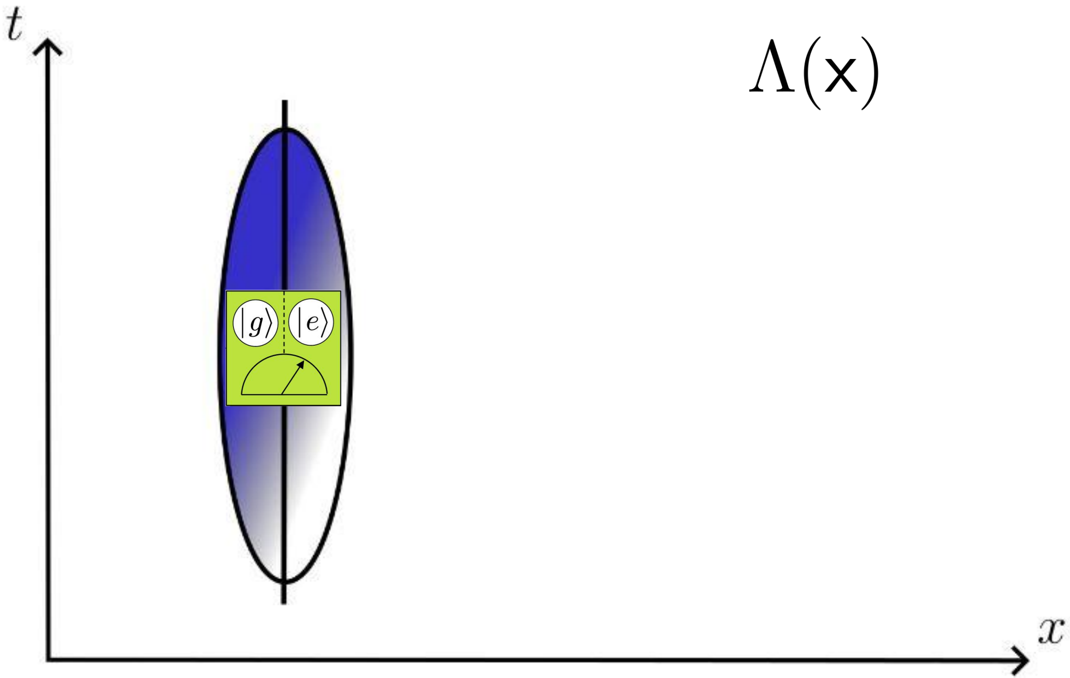
Negativity in practice

Negativity in practice



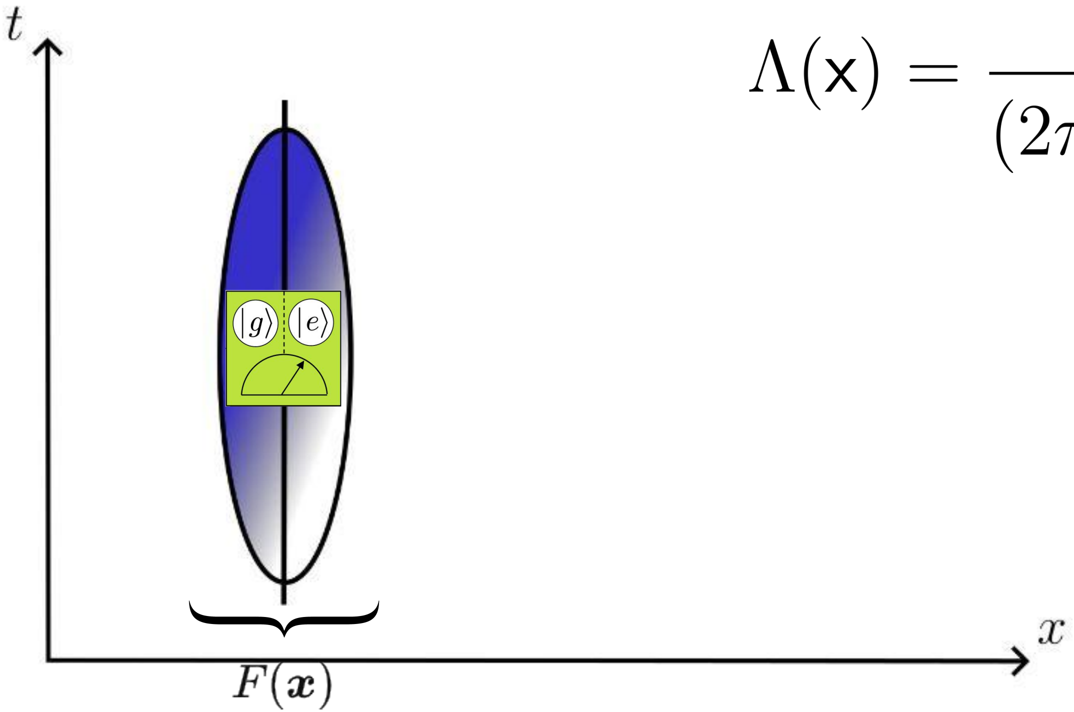
$$G_F(\Lambda_A^+, \Lambda_B^+) = \int \mathrm{d}V \mathrm{d}V' \Lambda_A^+(\mathsf{x}) \Lambda_B^+(\mathsf{x}') G_F(\mathsf{x}, \mathsf{x}')$$

Negativity in practice



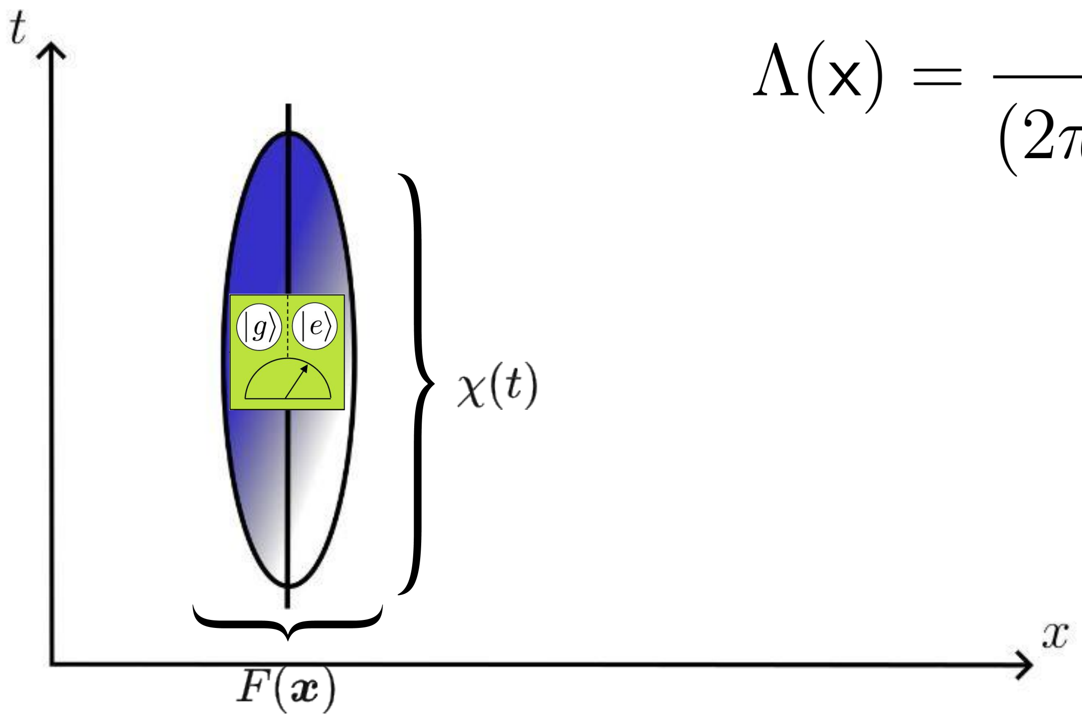
$\Lambda(x)$

Negativity in practice



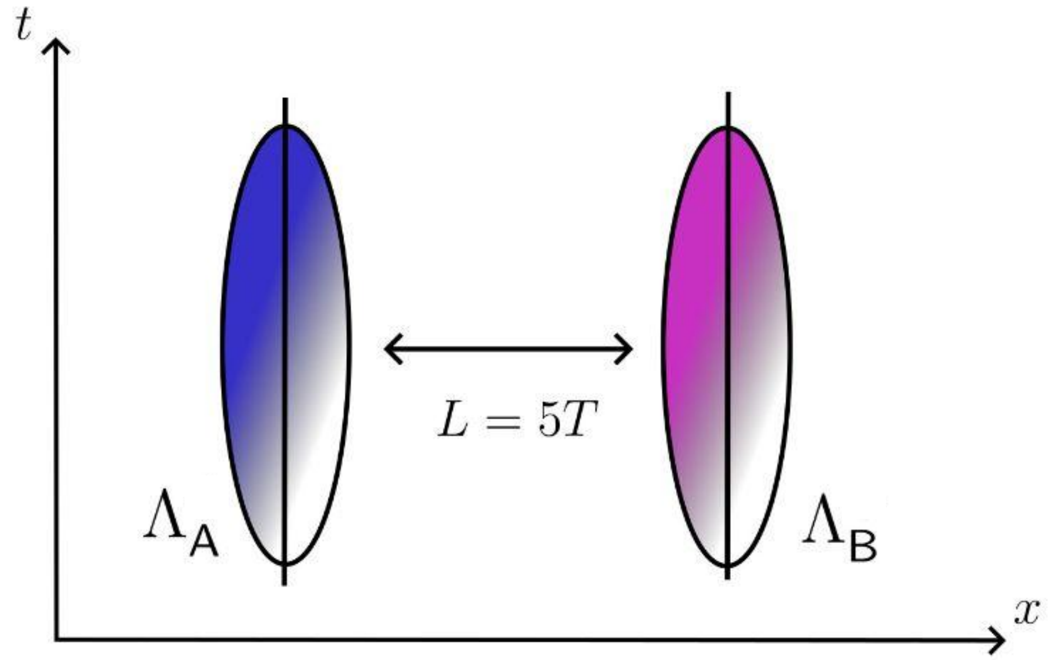
$$\Lambda(\mathrm{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}$$

Negativity in practice

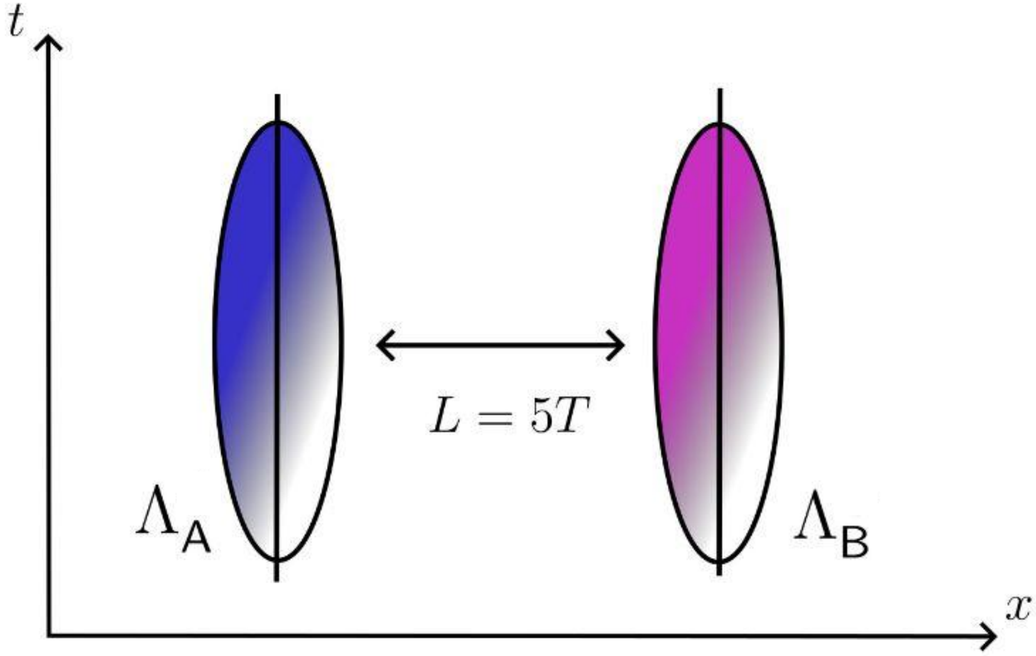


$$\Lambda(\mathsf{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}} \mathrm{e}^{-\frac{t^2}{2T^2}}$$

Negativity in practice

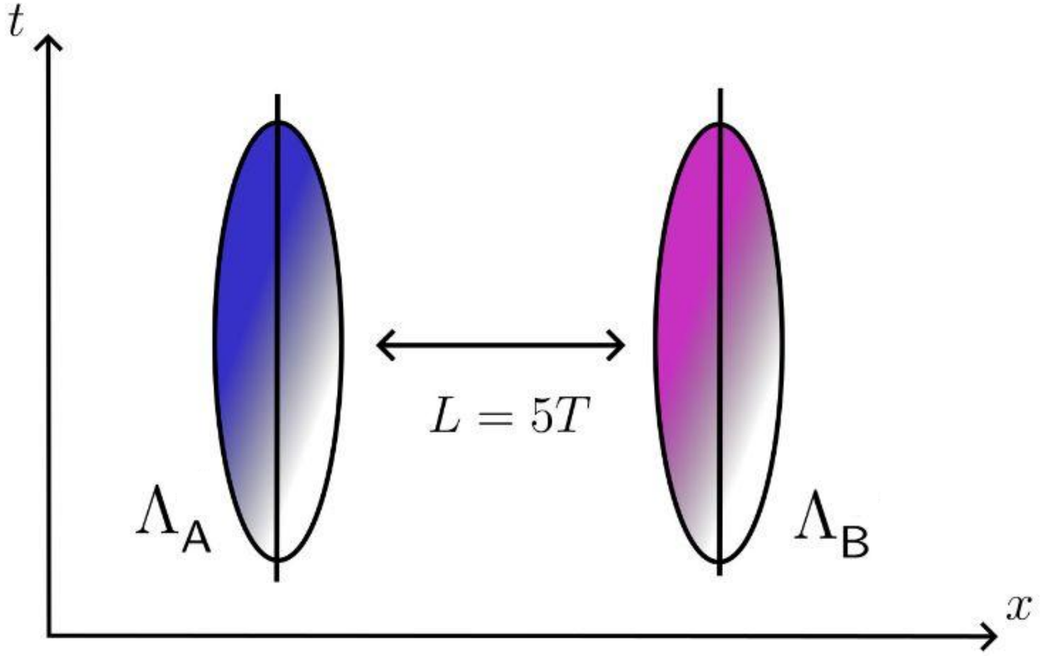


Negativity in practice



$$\Lambda_A = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{\frac{-t^2}{2T^2}} e^{\frac{-|\boldsymbol{x}|^2}{2\sigma^2}}$$

Negativity in practice



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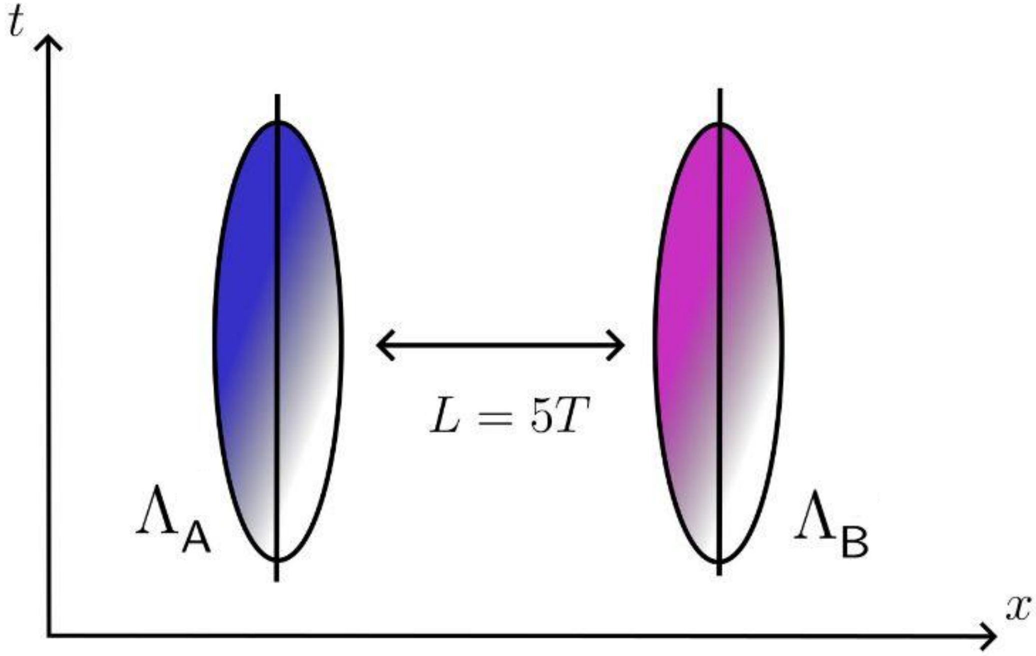
Negativity in practice



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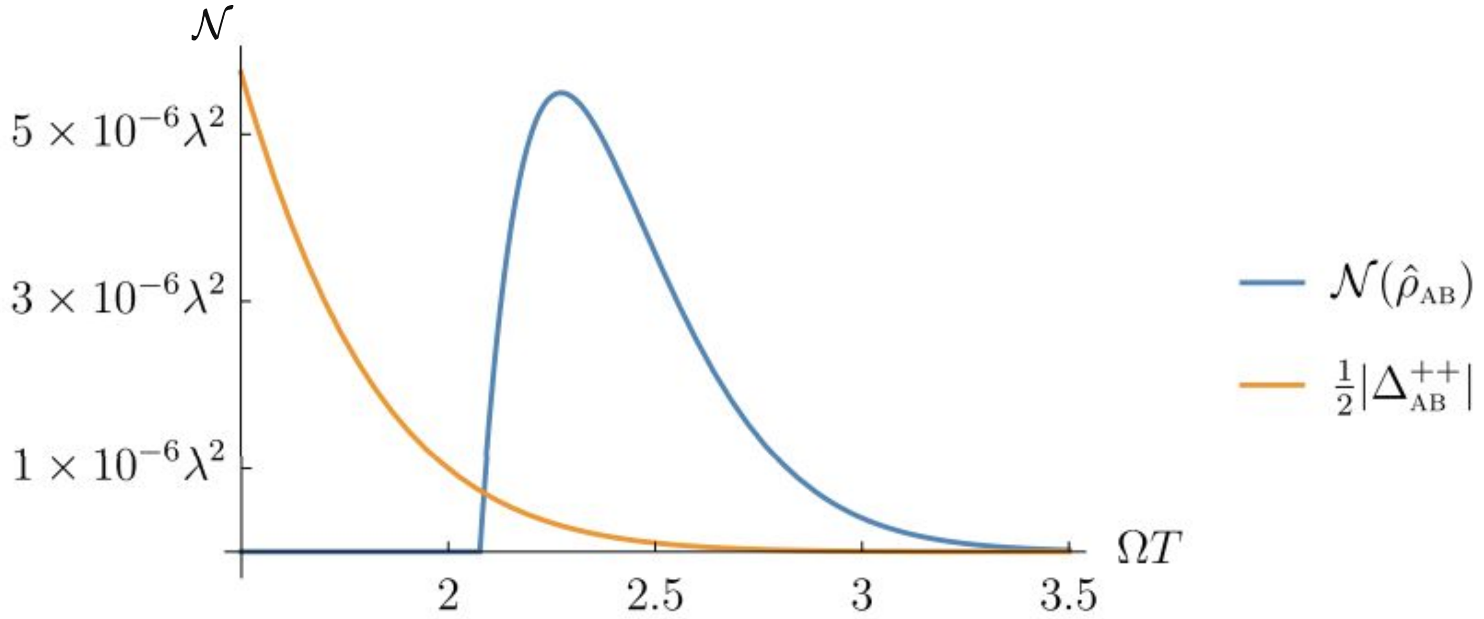
$\longrightarrow$ Closed forms!

R. T. Perche, "Closed-form expressions for smeared bi-distributions of a massless scalar field: non-perturbative and asymptotic results in relativistic quantum information", 20 Dec 2023, https://doi.org/10.48550/arXiv.2312.13343

Negativity in practice



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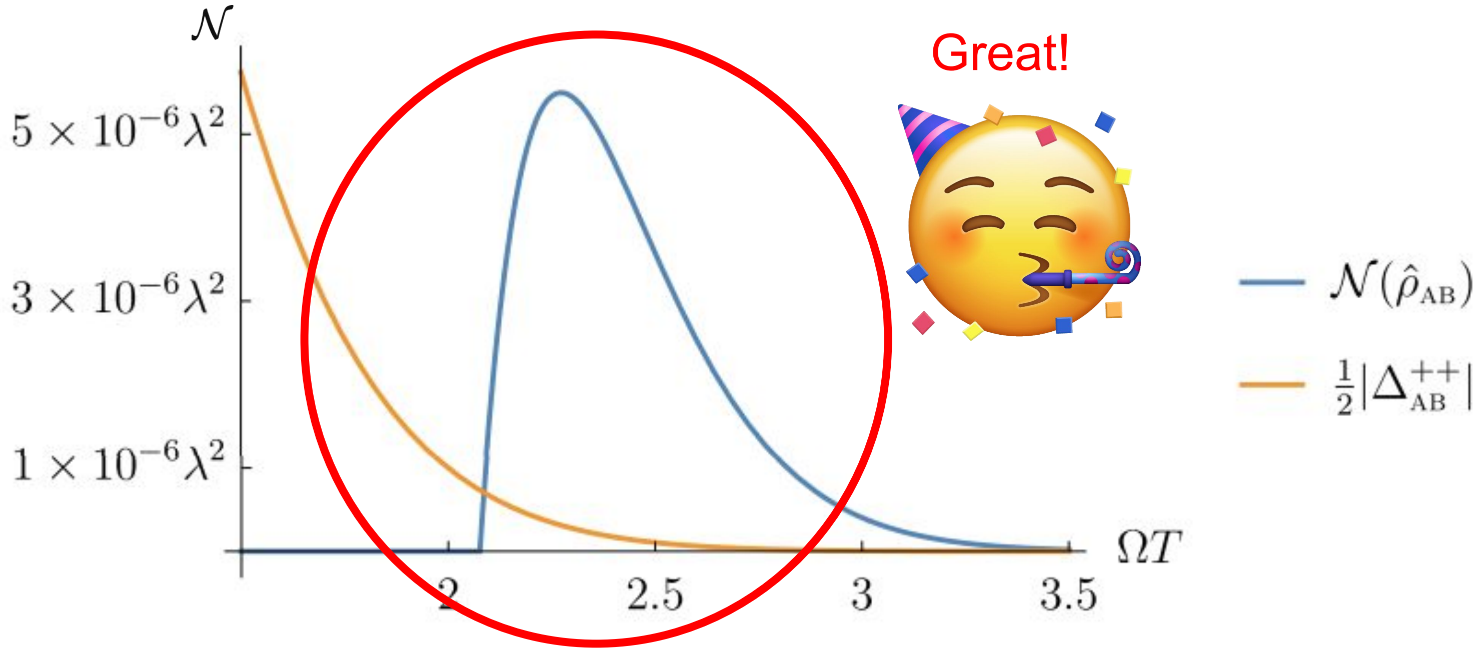

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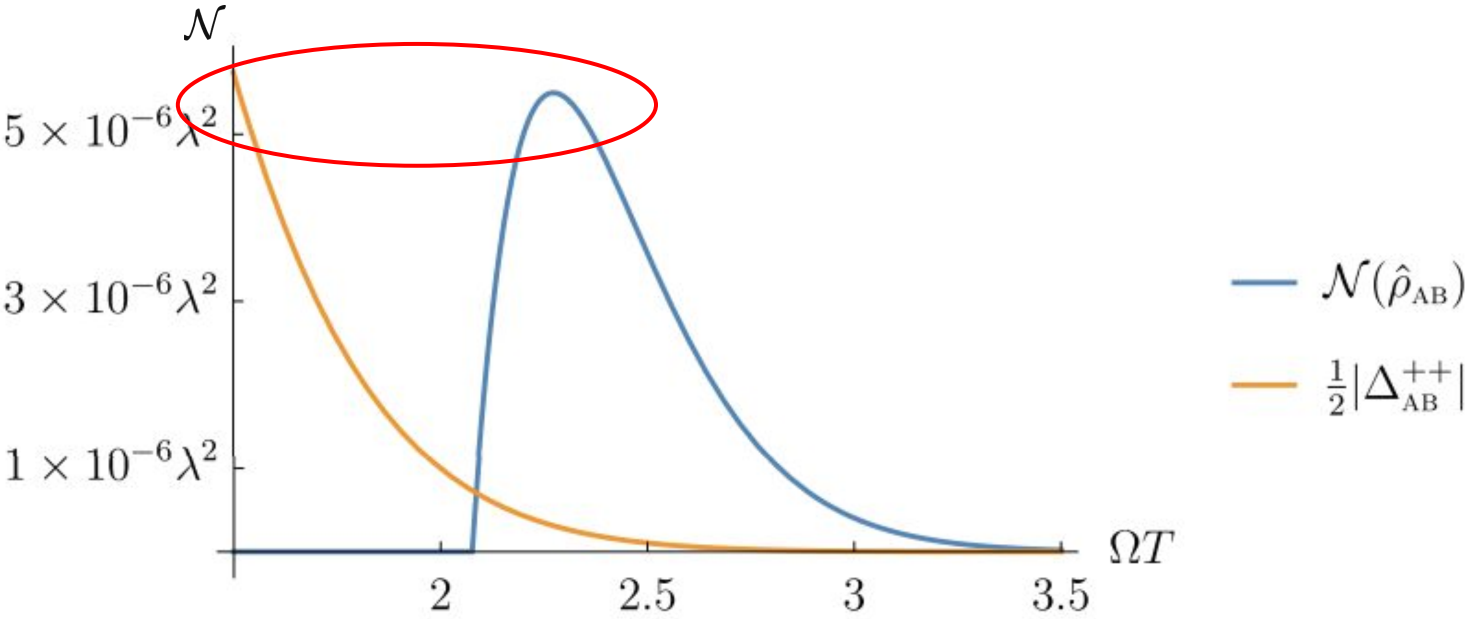

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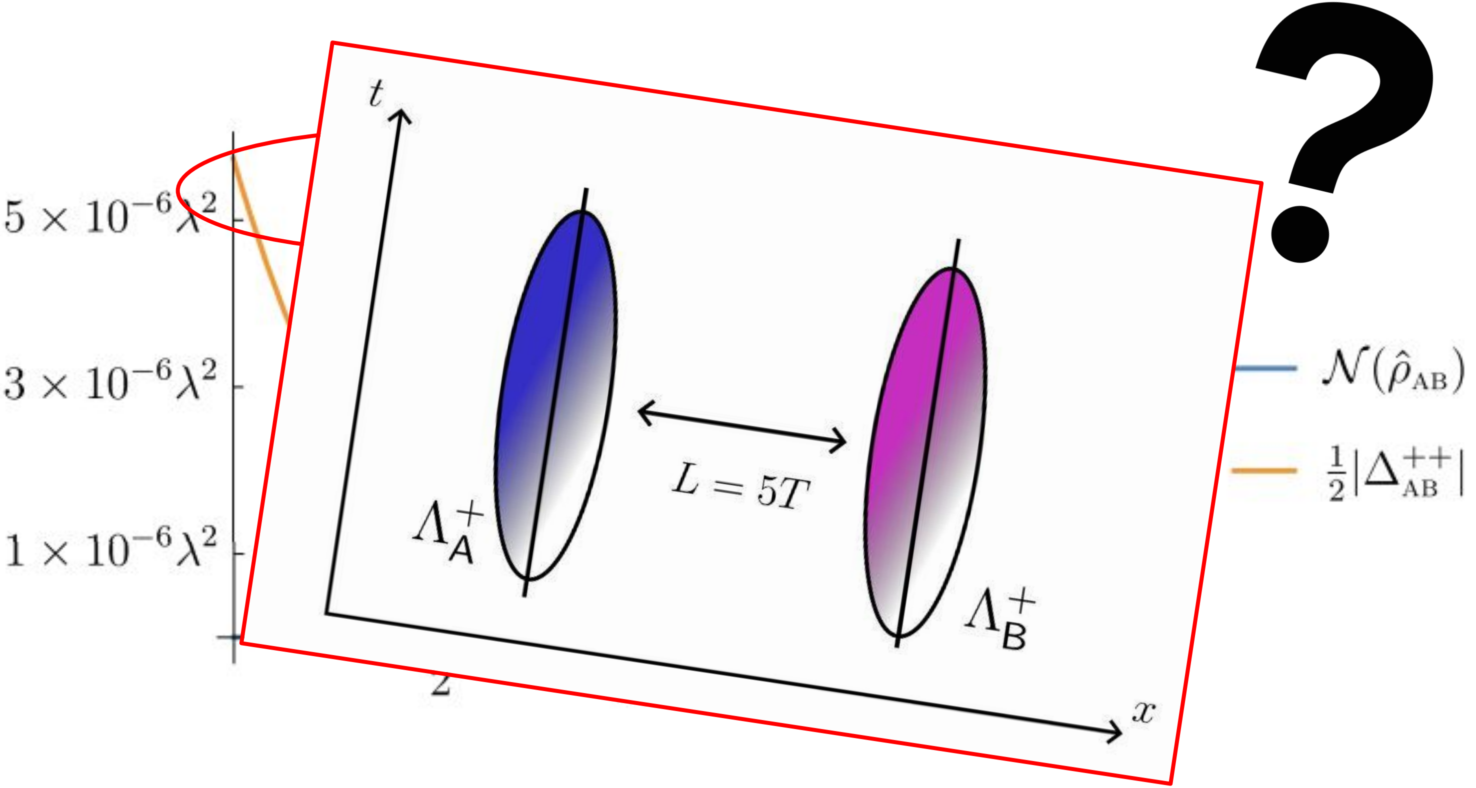

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$$\Lambda_{\mathsf{A}}^{+} = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}}\mathsf{e}^{\frac{-t^2}{2T^2}}\mathsf{e}^{\frac{-|\boldsymbol{x}|^2}{2\sigma^2}}$$

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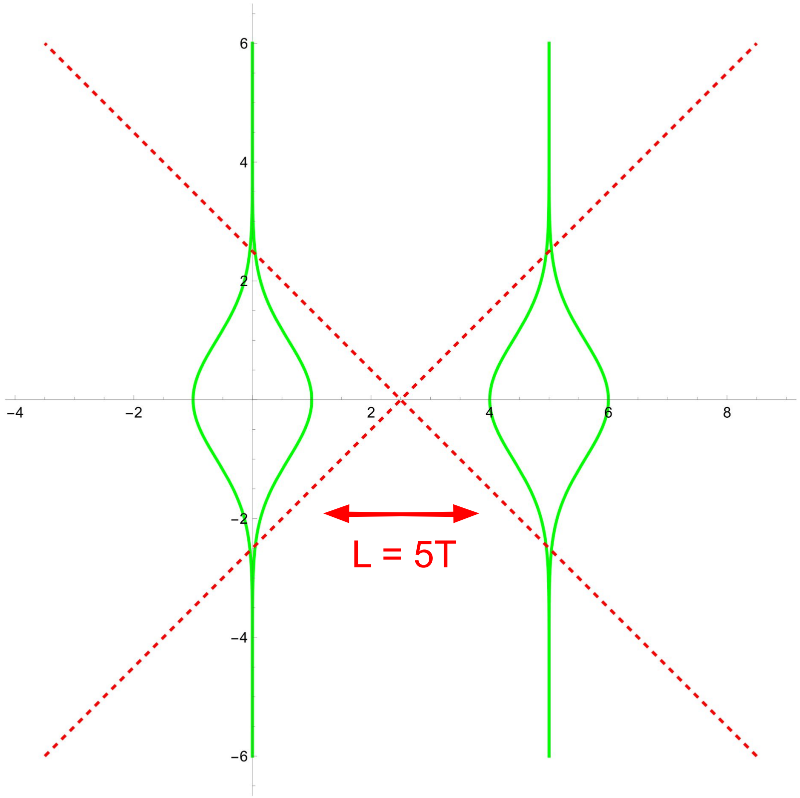

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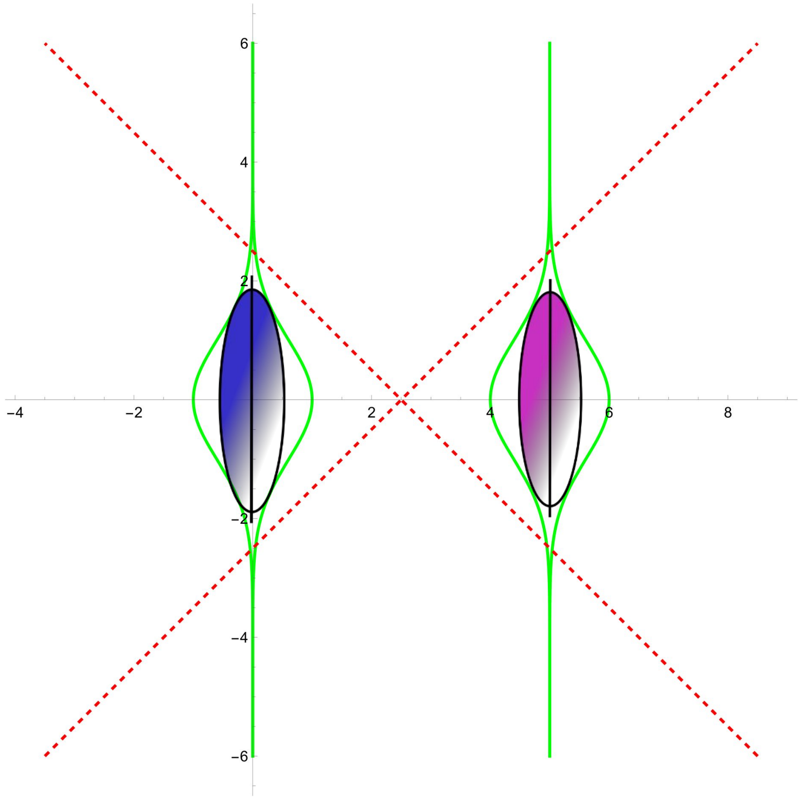

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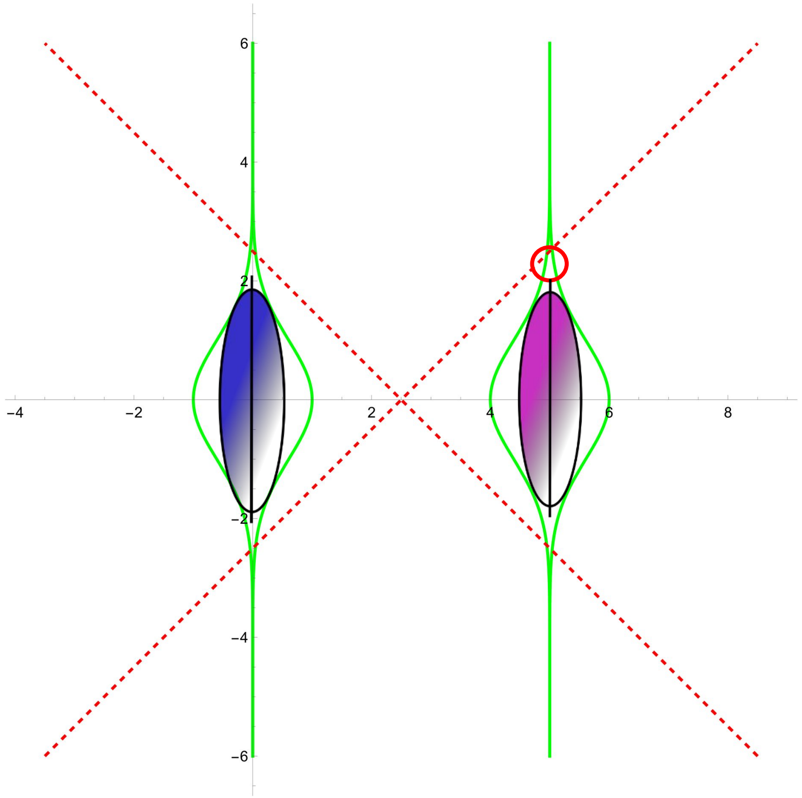

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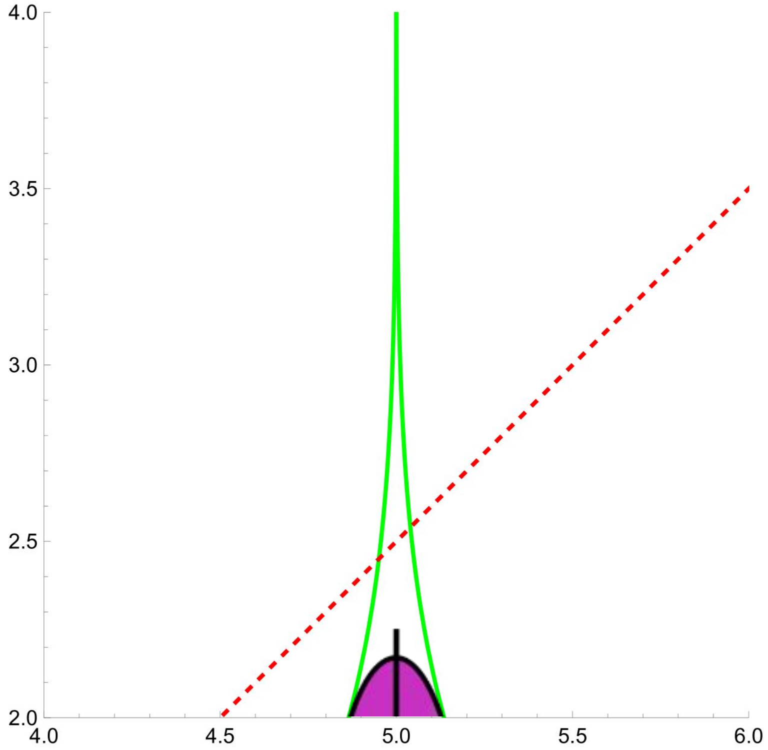

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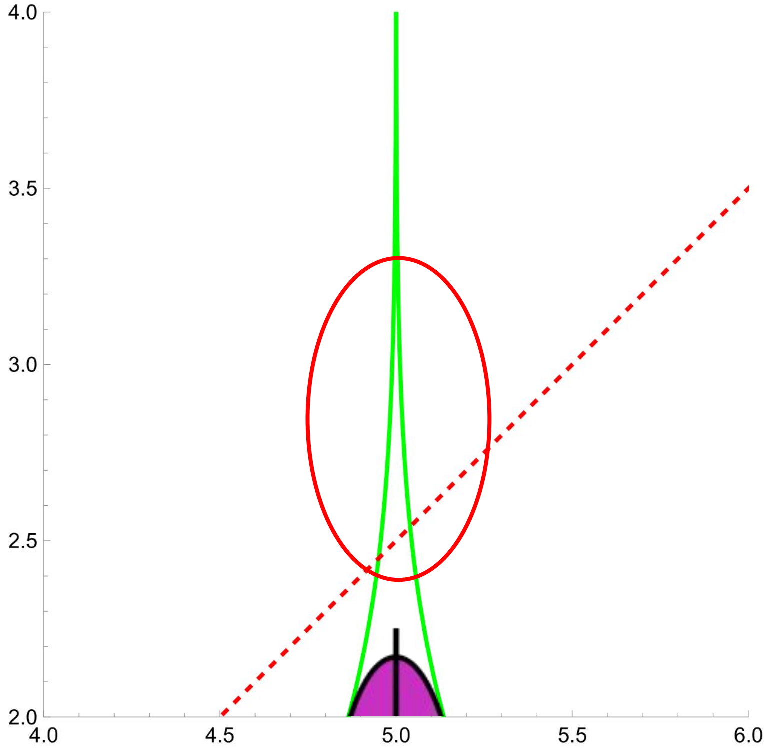

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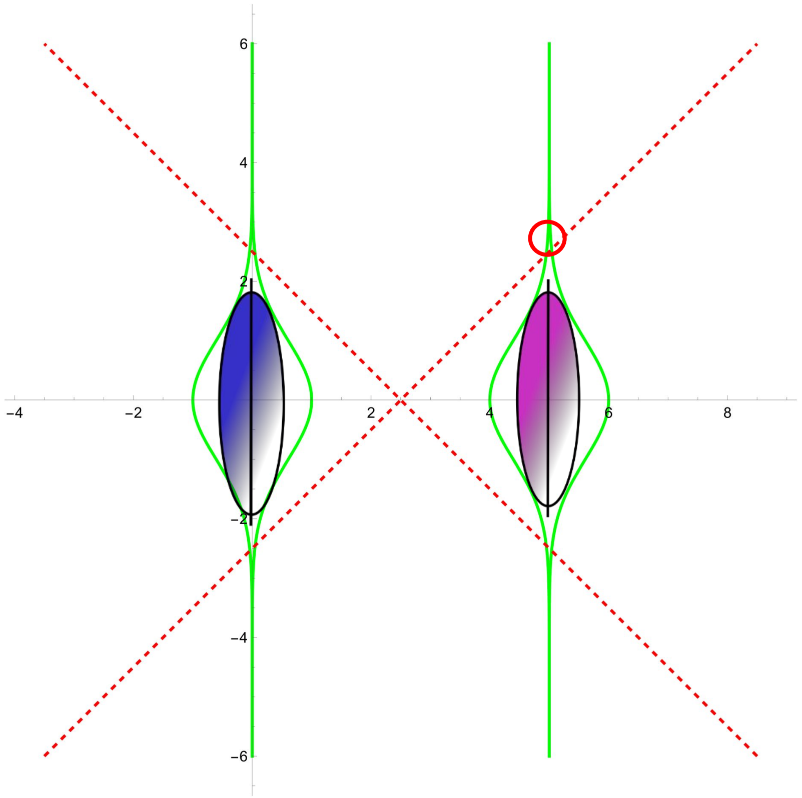

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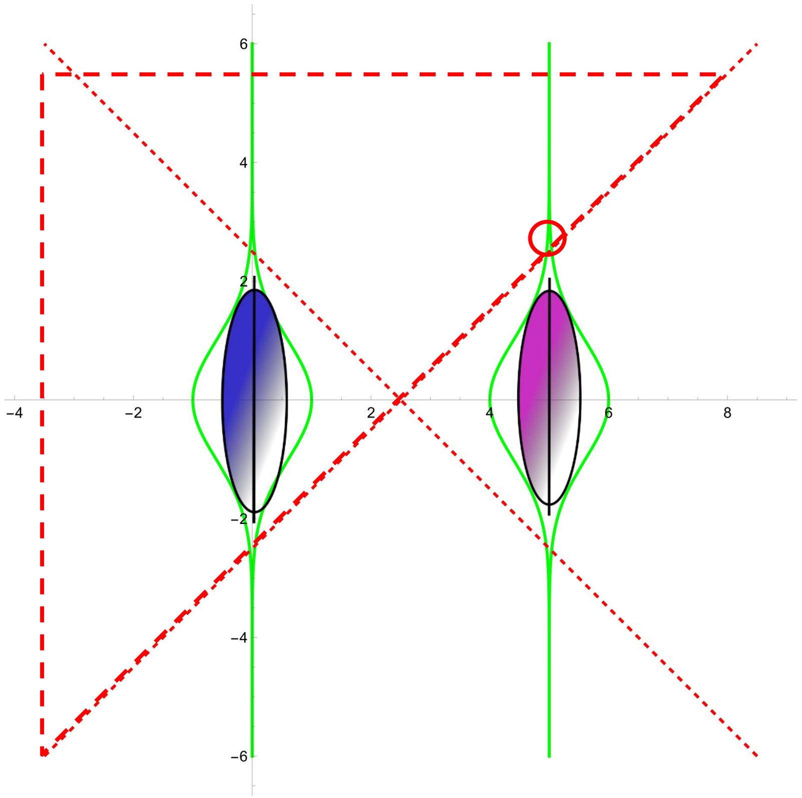

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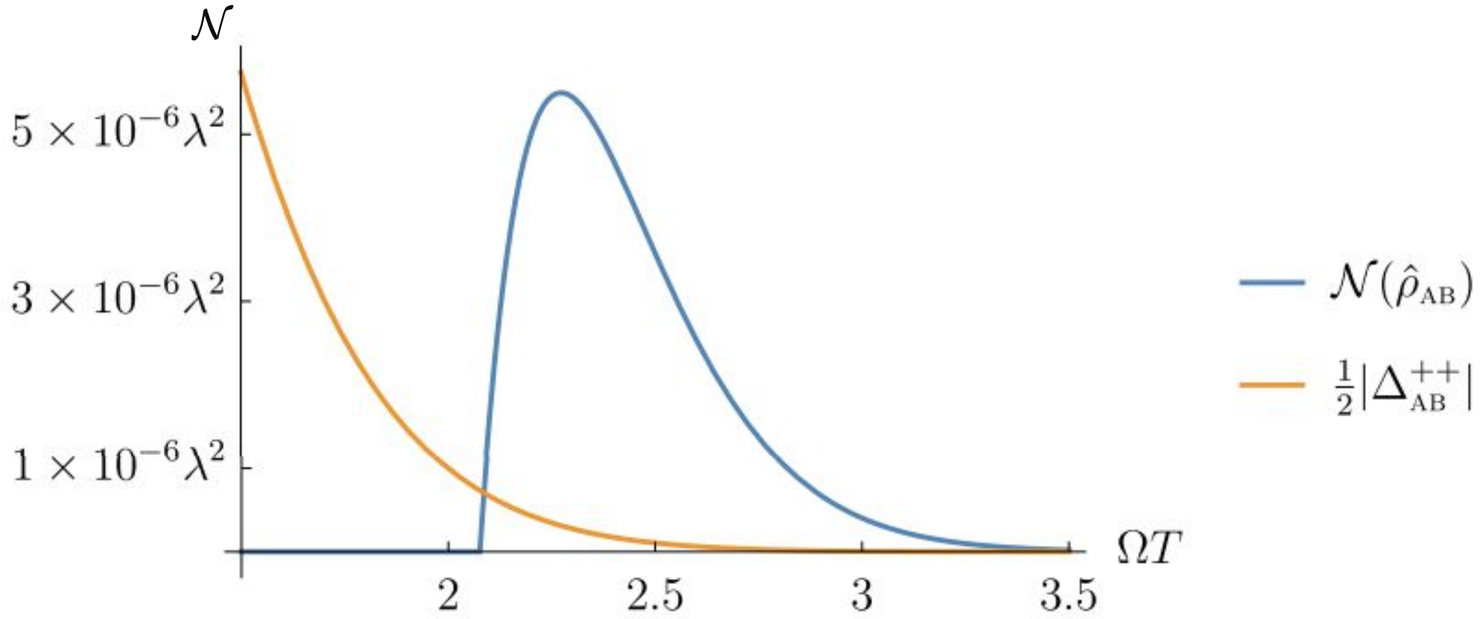

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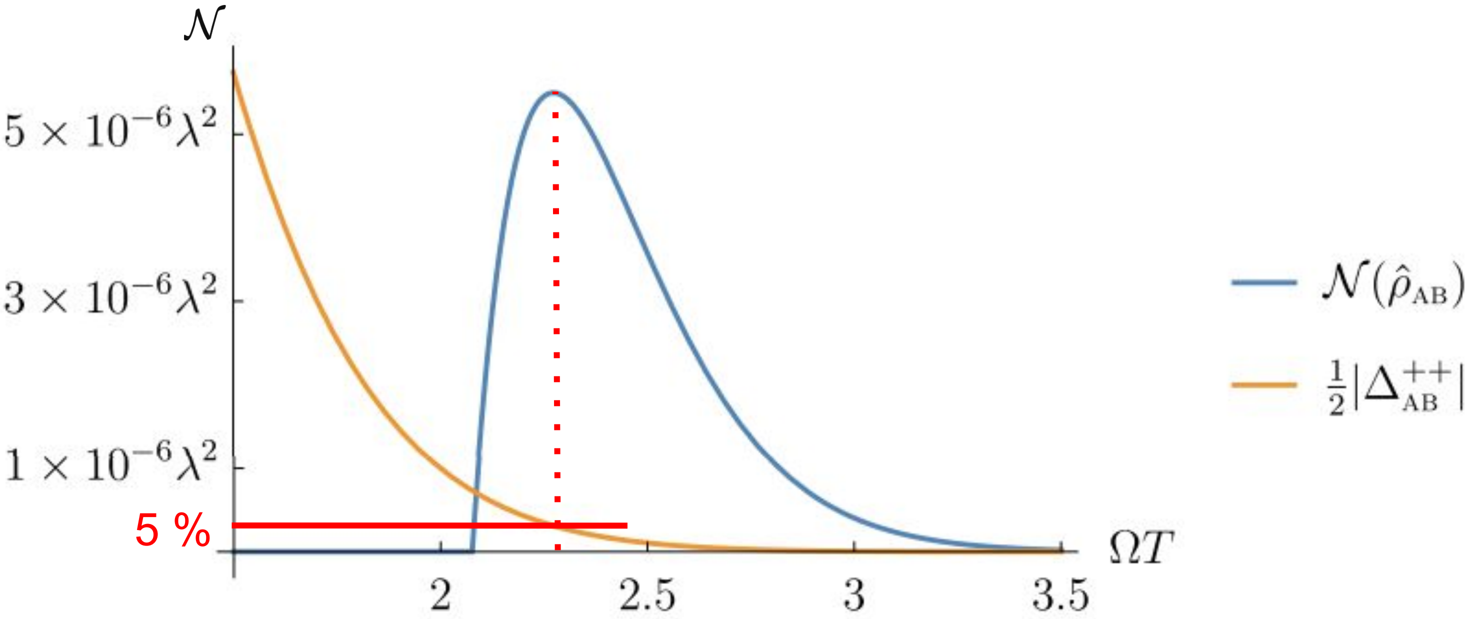

Summary of Entanglement Harvesting

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→ We place 2 UDW detectors in two regions of spacetime

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Summary of Entanglement Harvesting

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→ Compute → Feynman propagator

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  → Wightman function

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  - we consider causally disconnected regions to avoid signaling
- Smear the detectors in interaction regions
- Let the detectors interact with the field for **finite** time
- Compute
  - Feynman propagator
  - Wightman function

Negativity (entanglement)

Summary of Entanglement Harvesting

$\longrightarrow$ We place 2 UDW detectors in two regions of spacetime

  $\longrightarrow$ we consider causally disconnected regions to avoid signaling

$\longrightarrow$ Smear the detectors in interaction regions

$\longrightarrow$ Let the detectors interact with the field for **finite** time

$\longrightarrow$ Compute $\longrightarrow$ Feynman propagator

$\longrightarrow$ Wightman function

$\longrightarrow$ Symmetric propagator

$\left.\begin{array}{l}\text{Feynman propagator}\\ \text{Wightman function}\end{array}\right\} \longrightarrow$ Negativity (entanglement)

Summary of Entanglement Harvesting

- → We place 2 UDW detectors in two regions of spacetime
  - → we consider causally disconnected regions to avoid signaling
- → Smear the detectors in interaction regions
- → Let the detectors interact with the field for **finite** time
- → Compute
  - → Feynman propagator
  - → Wightman function
  - → Symmetric propagator

Feynman propagator, Wightman function → Negativity (entanglement)

Symmetric propagator → Origin of entanglement

Summary of Entanglement Harvesting

$\longrightarrow$ We place 2 UDW detectors in two regions of spacetime

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$\longrightarrow$ Let the detectors interact with the field for **finite** time

$\longrightarrow$ Compute $\longrightarrow$ Feynman propagator, Wightman function $\longrightarrow$ Negativity (entanglement)

$\longrightarrow$ Symmetric propagator $\longrightarrow$ Origin of entanglement

$\longrightarrow$ If

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$\longrightarrow$ We place 2 UDW detectors in two regions of spacetime

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Summary of Entanglement Harvesting

$\longrightarrow$ We place 2 UDW detectors in two regions of spacetime

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$\longrightarrow$ Smear the detectors in interaction regions

$\longrightarrow$ Let the detectors interact with the field for **finite** time

$\longrightarrow$ Compute $\longrightarrow$ Feynman propagator, Wightman function $\longrightarrow$ Negativity (entanglement)

$\longrightarrow$ Symmetric propagator $\longrightarrow$ Origin of entanglement

$\longrightarrow$ If $\longrightarrow$ The negativity is non zero

$\longrightarrow$ The signaling is negligible

Summary of Entanglement Harvesting

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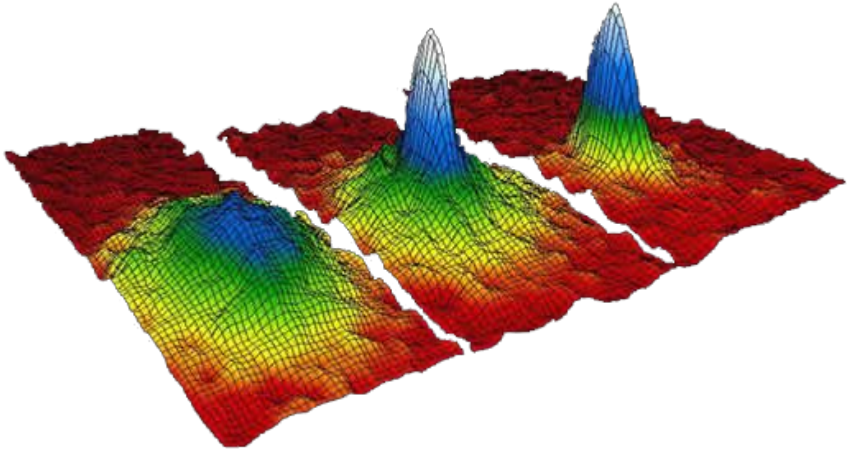
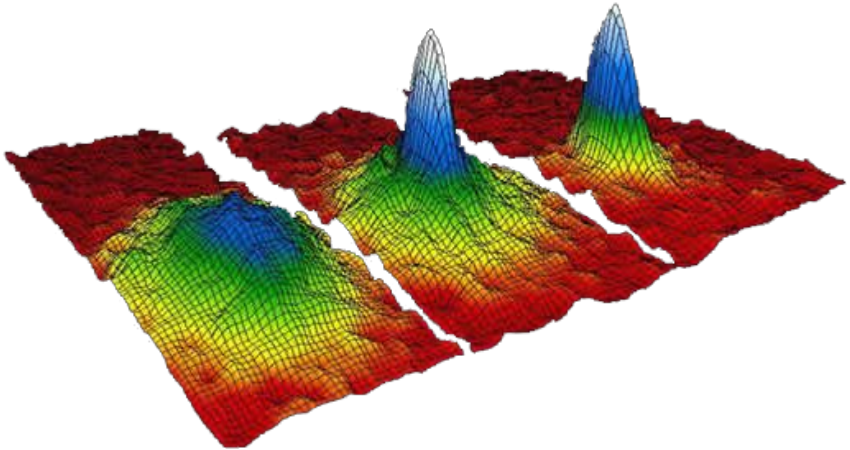
$\longrightarrow$ Symmetric propagator $\longrightarrow$ Origin of entanglement

$\longrightarrow$ If $\longrightarrow$ The negativity is non zero, $\longrightarrow$ The signaling is negligible $\Big\} \longrightarrow$ Success!

Analogous Quantum Field Theory



Analogous Quantum Field Theory



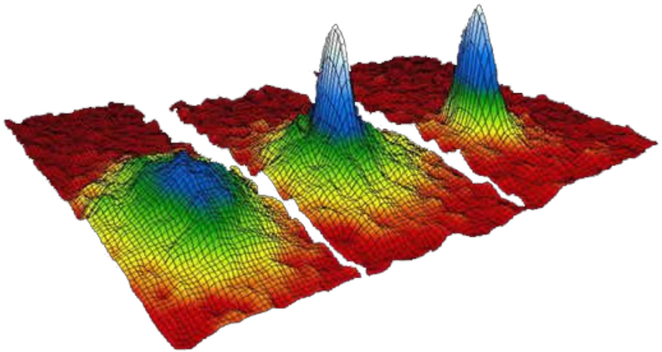
Markus Oberthaler

Francesco Gozzini

T. Rick Perche

Analogous Quantum Field Theory

→ Bose-Einstein condensate



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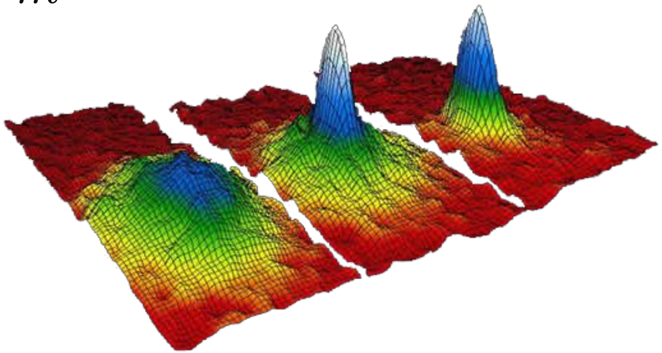

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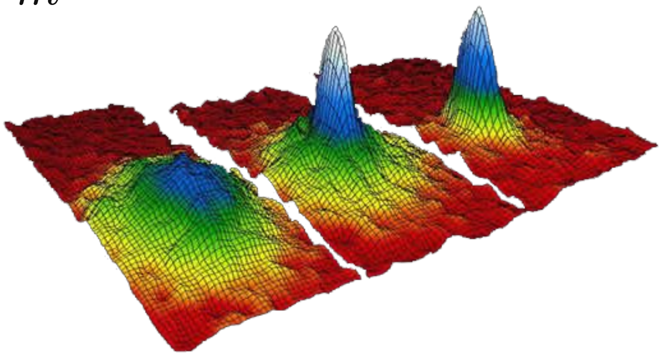

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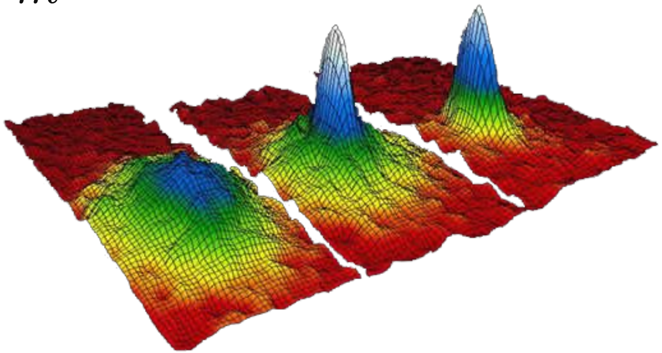

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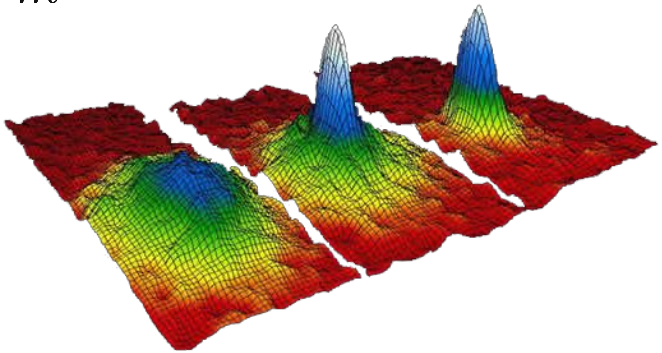

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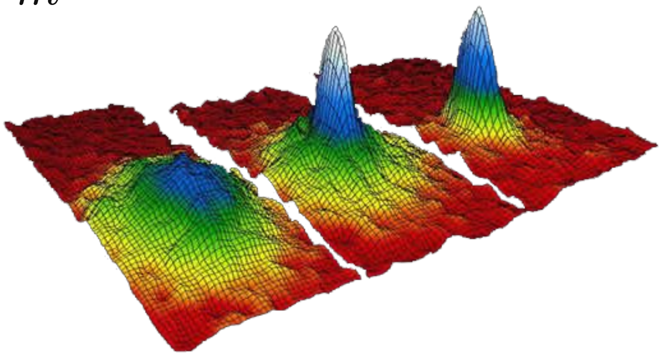

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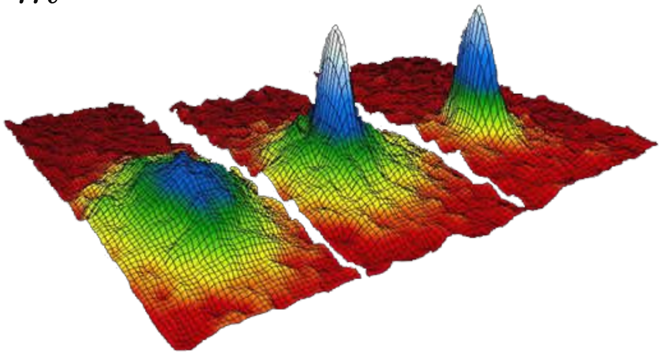

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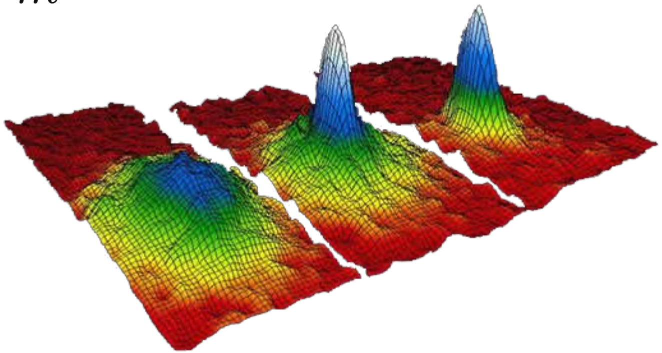

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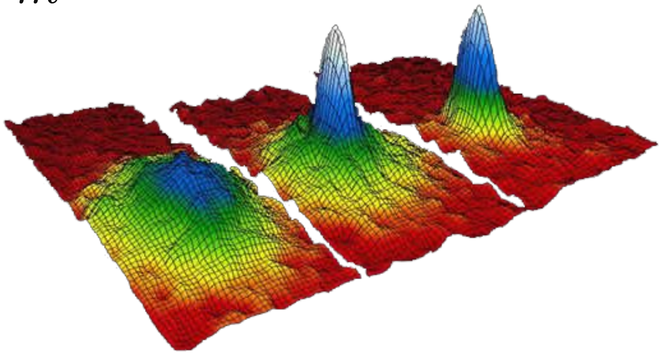

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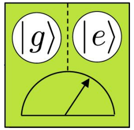

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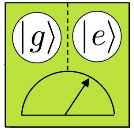

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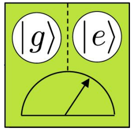
$\longrightarrow$ “the detector clicks” $\longrightarrow$ 

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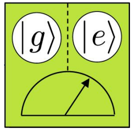

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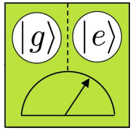

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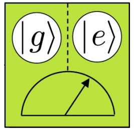

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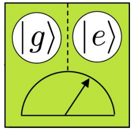

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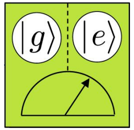

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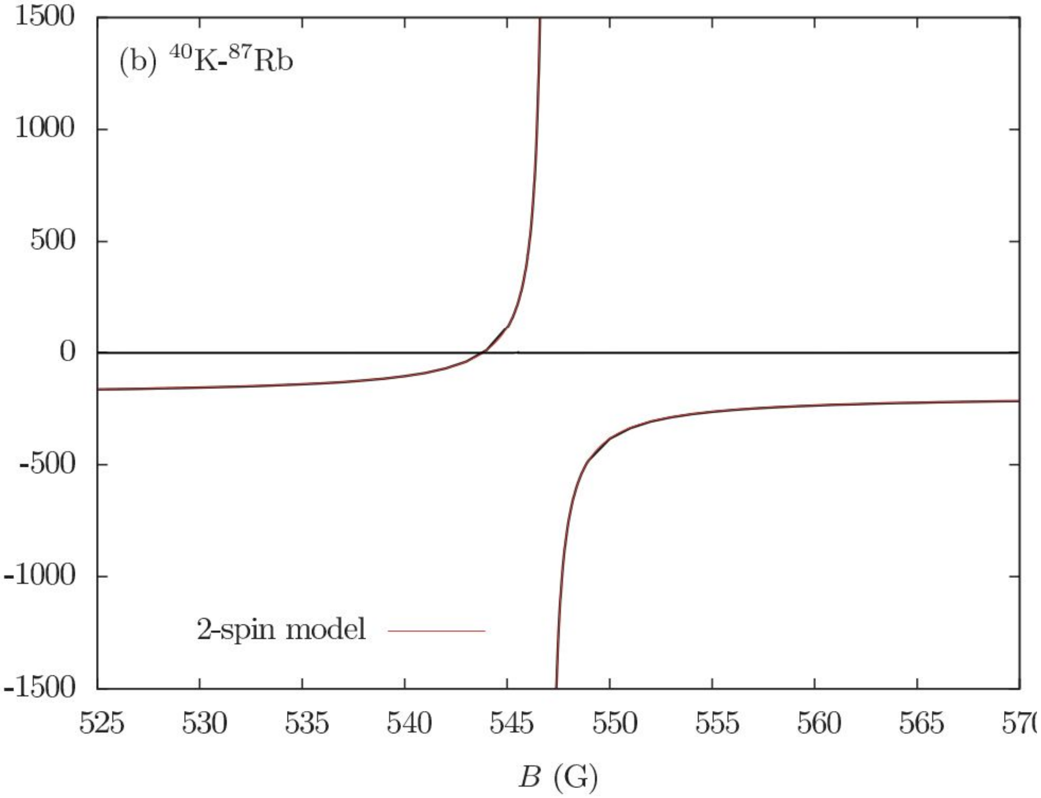

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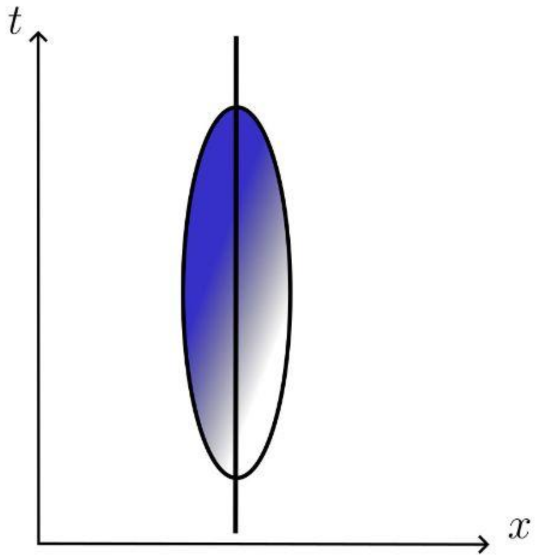

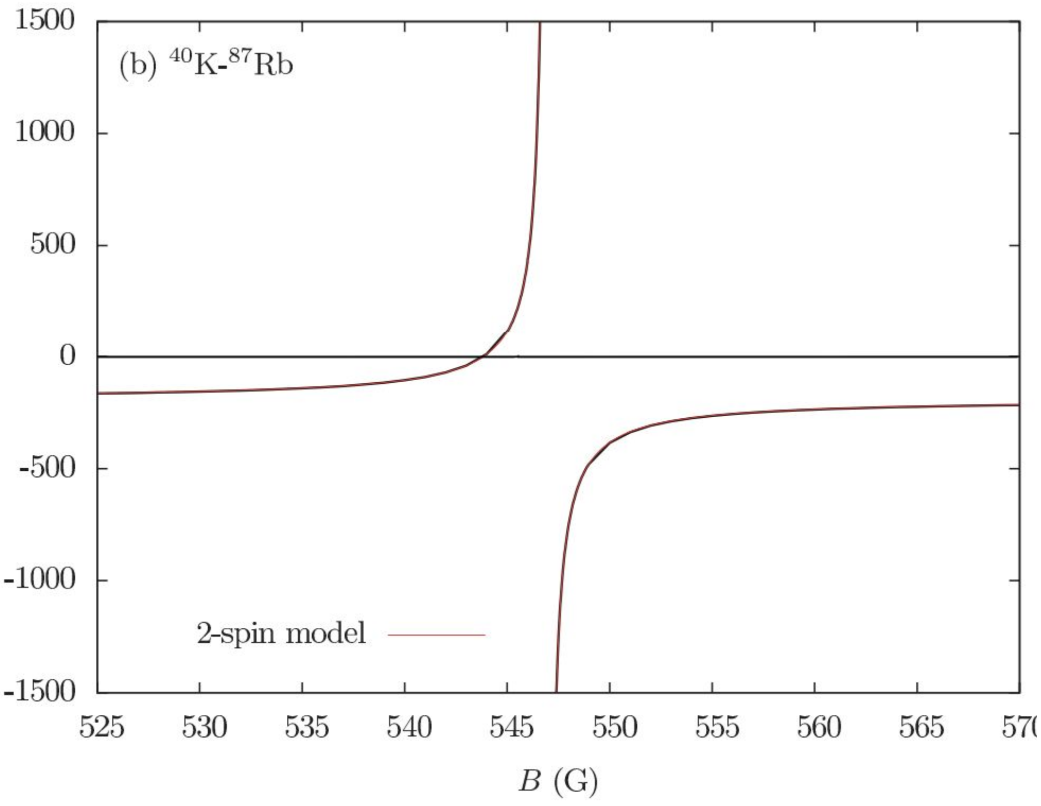

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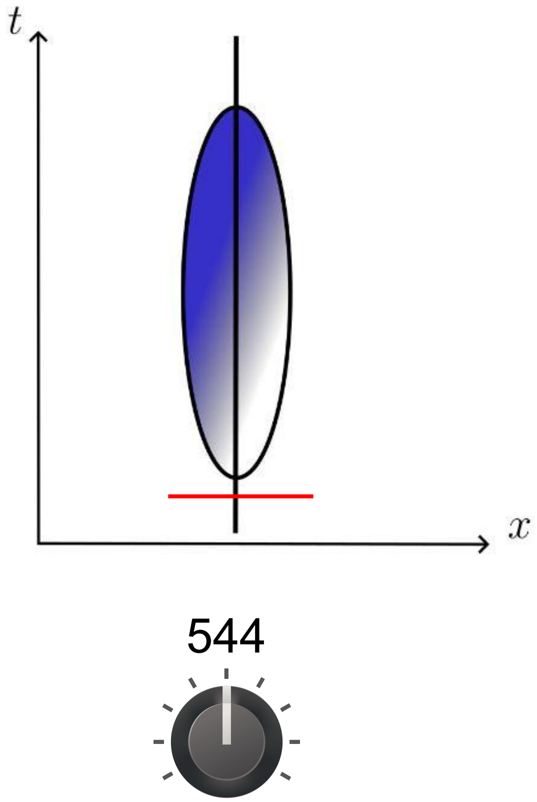

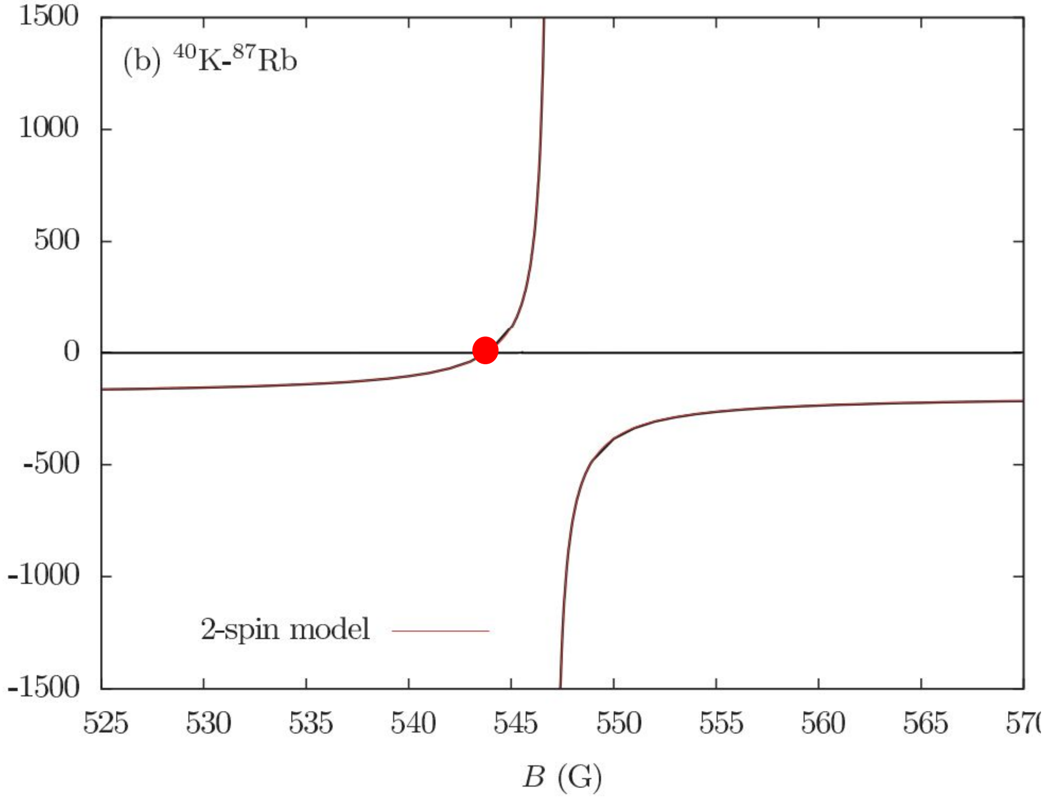

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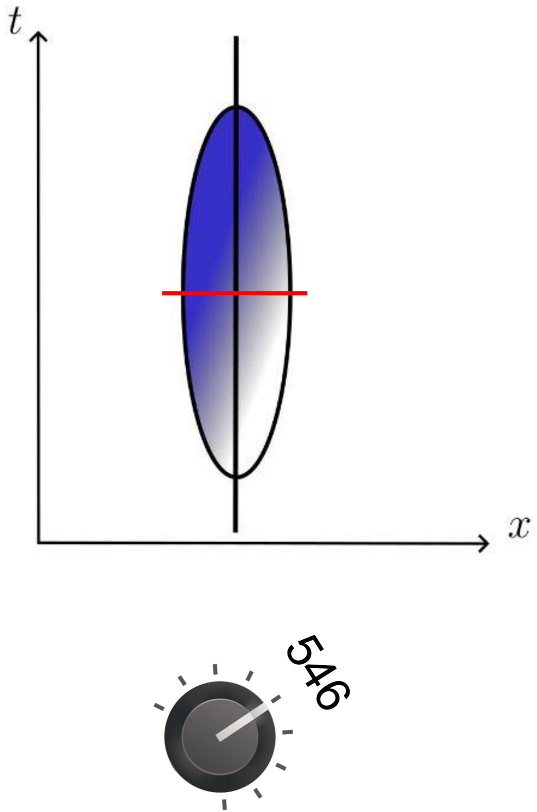

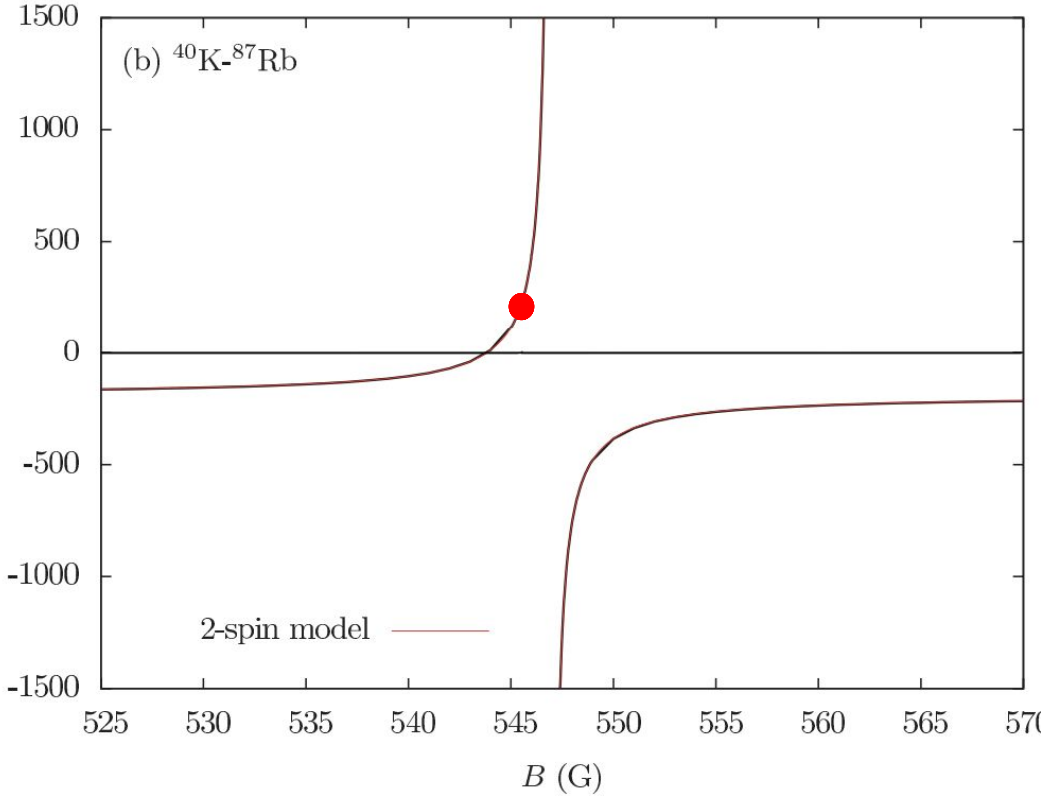

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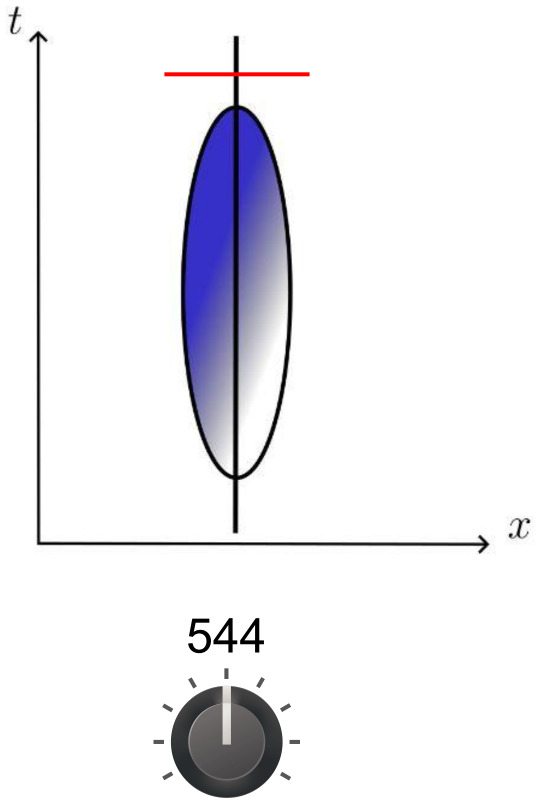

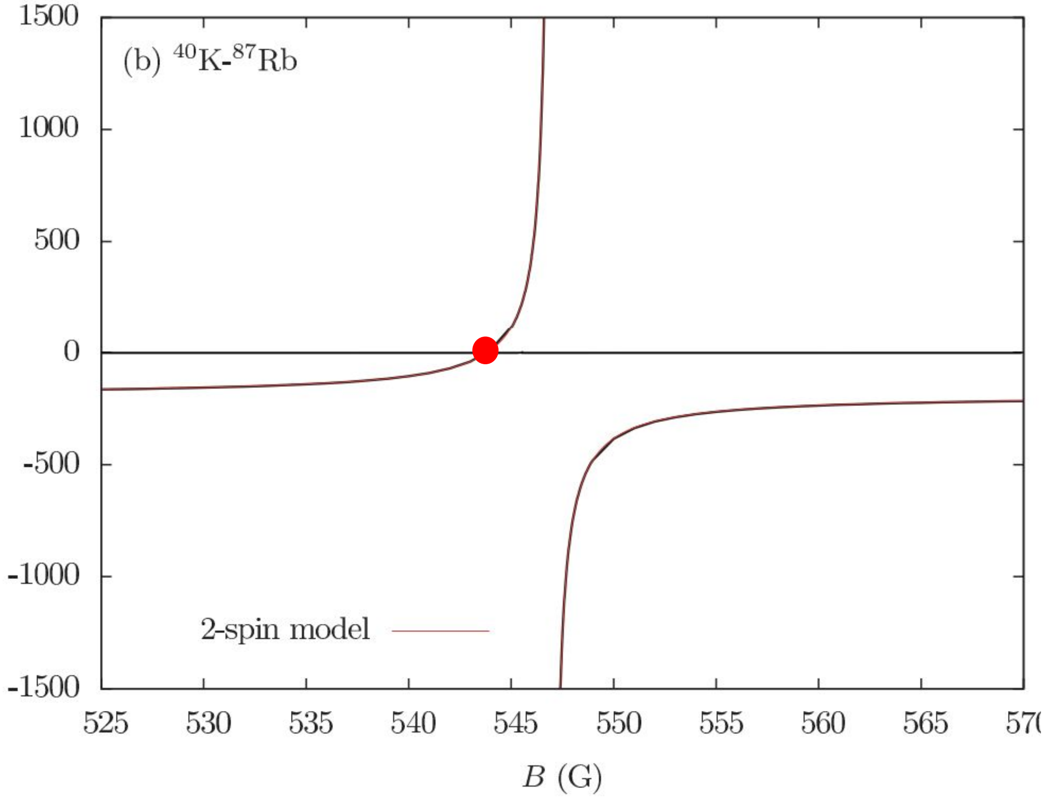

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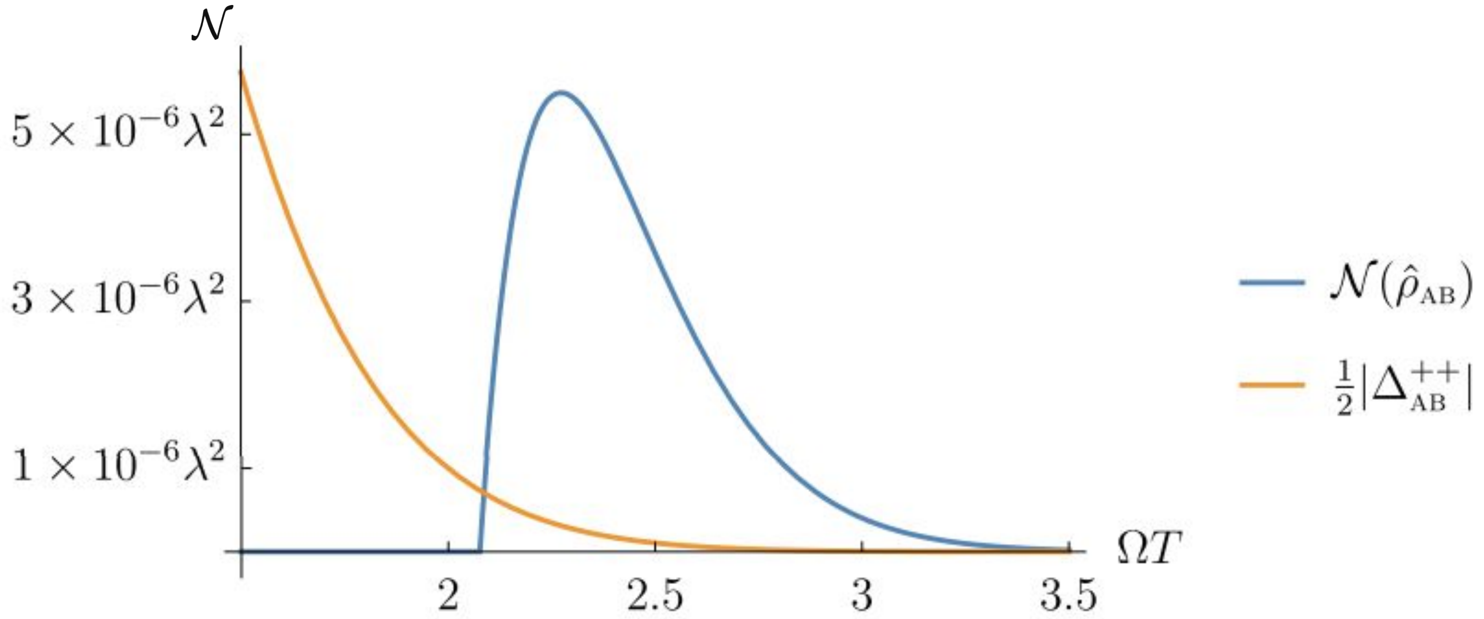
→ The time smearing (switching) is implemented by tuning the scattering length via an external magnetic field

Negativity in practice



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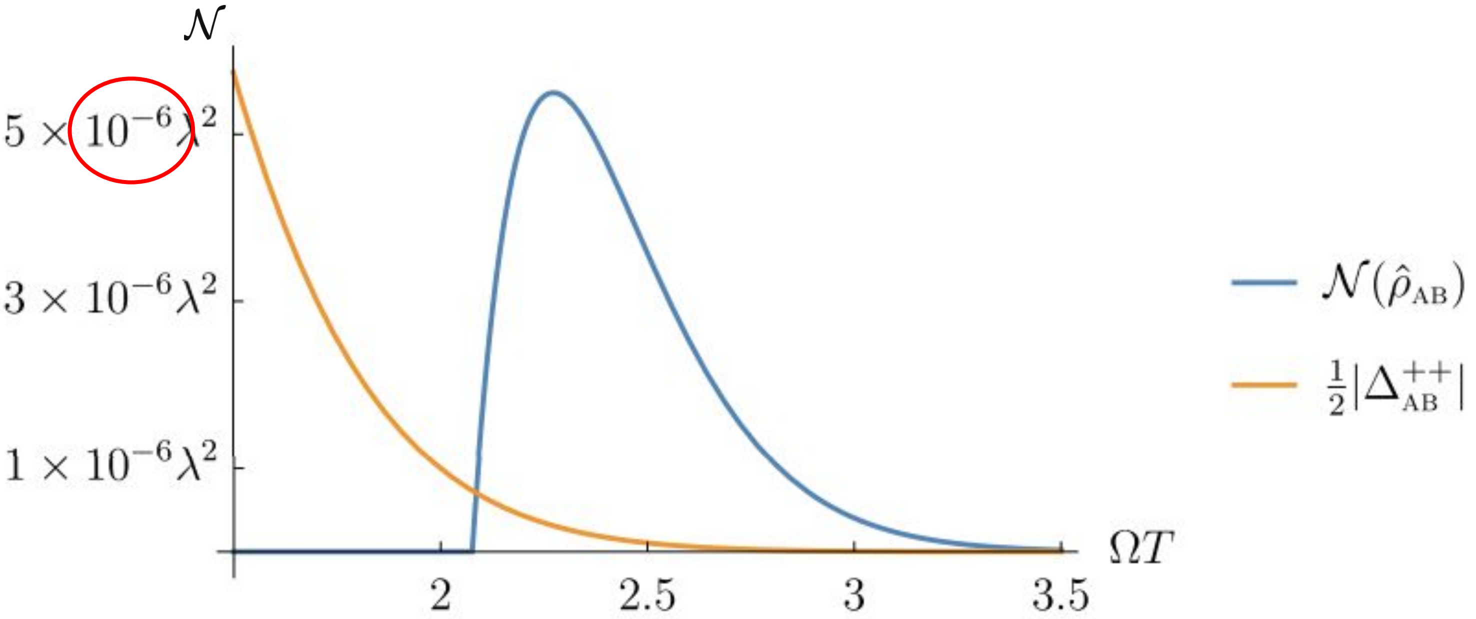

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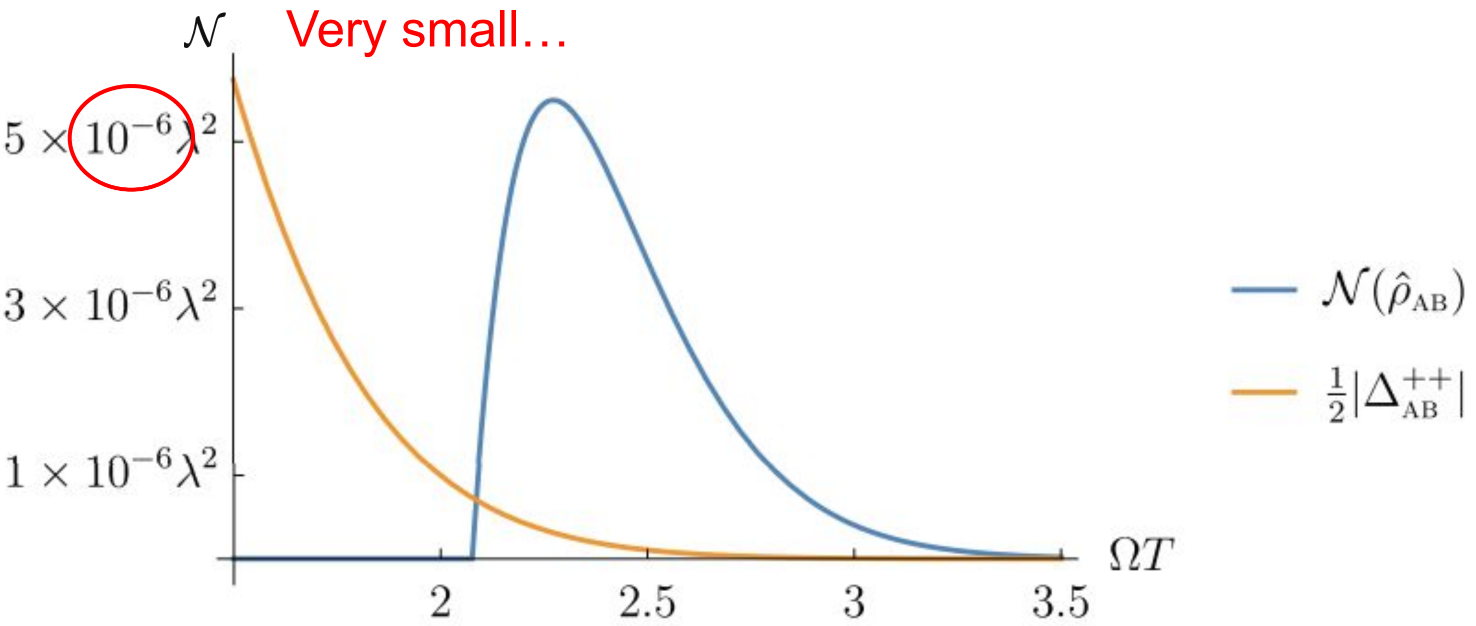

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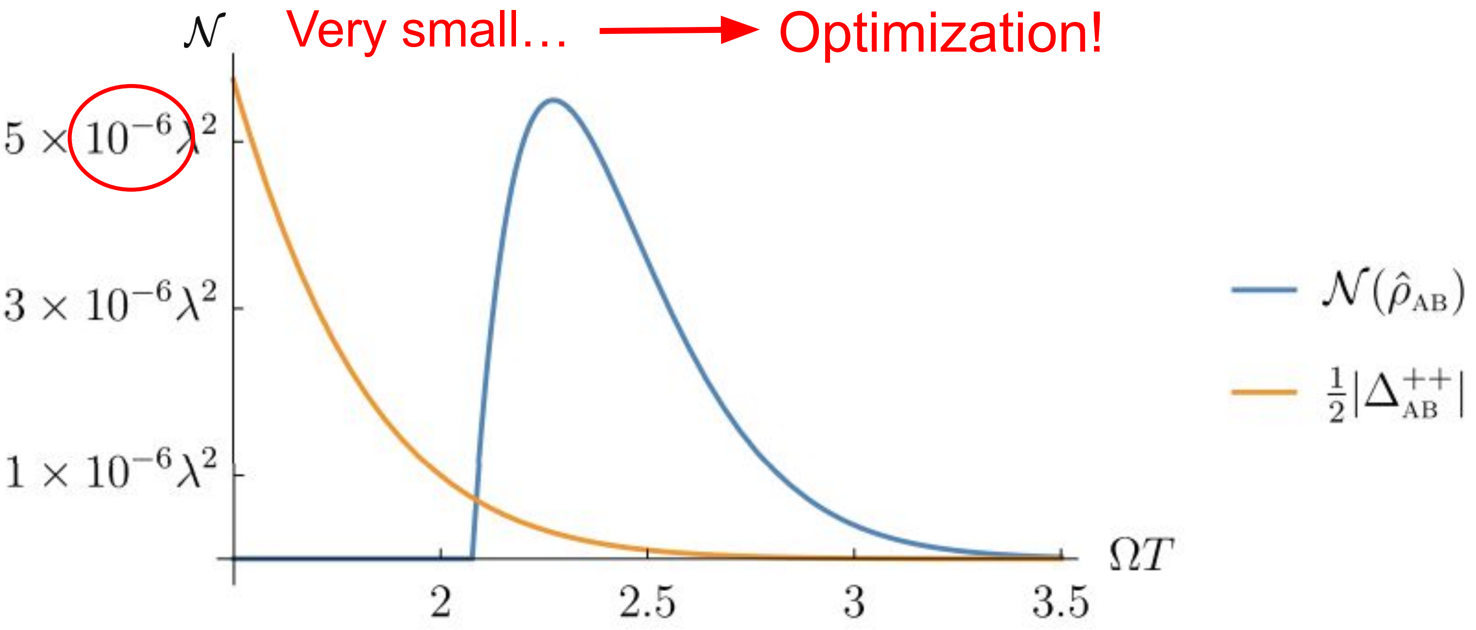

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$$W(\Lambda_{\mathsf{A}}^{-}, \Lambda_{\mathsf{A}}^{+}) = \int \mathrm{d}V \mathrm{d}V' \Lambda_{A}^{-}(\mathsf{x}) \Lambda_{A}^{+}(\mathsf{x}') W(\mathsf{x}, \mathsf{x}')$$

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$$= \frac{1}{(2\pi)^3}\frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}}\int\frac{\mathrm{d}^3\boldsymbol{k}}{2|\boldsymbol{k}|}\int\mathrm{d}t\int\mathrm{d}^3\boldsymbol{x}\int\mathrm{d}t'\int\mathrm{d}^3\boldsymbol{x}\; F(\boldsymbol{x})F(\boldsymbol{x}')\mathsf{e}^{-\mathsf{i}\Omega t}\mathsf{e}^{\mathsf{i}\Omega t'}\mathsf{e}^{\mathsf{i}|\boldsymbol{k}|(t-t')+\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')}\chi(t)\chi(t')$$

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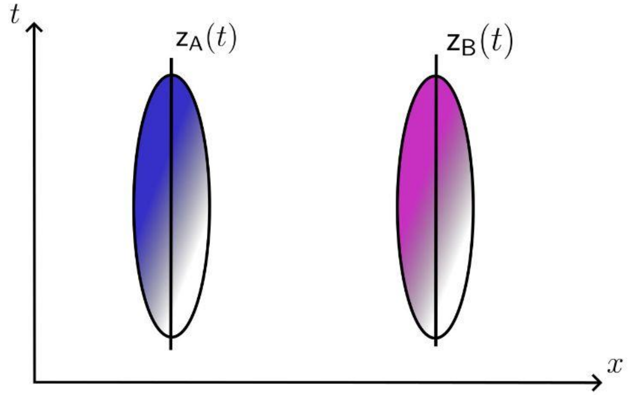

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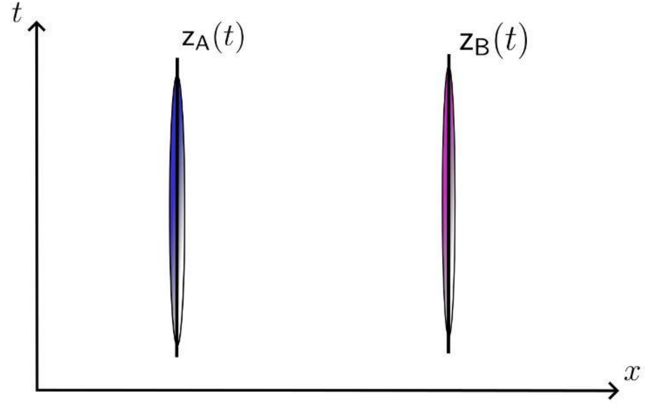

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$\longrightarrow$ closed-forms

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$W_{nm} \longrightarrow$ closed-forms

Optimization

$$\longrightarrow \chi(t) \approx \sum_{n=0}^{N} c_n \chi_n(t) \qquad \{\chi_n\} \subseteq L^2(\mathbb{R})$$

$$\longrightarrow \{H_n(t)\mathrm{e}^{-t^2/2T^2}\}$$

$$\sum_{n=0}^{N}\sum_{m=0}^{N} c_n c_m \underbrace{\int \mathrm{d}V \mathrm{d}V' H_n(t)\, H_m(t')\mathrm{e}^{-t^2/2T^2} F(\boldsymbol{x})\, \mathrm{e}^{-t'^2/2T^2} F(\boldsymbol{x}') W(\mathrm{x}, \mathrm{x}')}_{W_{nm}}$$

$$\equiv \sum_{n=0}^{N}\sum_{m=0}^{N} c_n c_m W_{nm}$$

$W_{nm} \longrightarrow$ closed-forms

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$W_{nm} \longrightarrow$ closed-forms

$$\equiv \sum_{n=0}^{N}\sum_{m=0}^{N} c_n c_m W_{nm}$$

$$\equiv c^{\mathsf{T}} W c$$

Optimization

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$$\equiv \sum_{n=0}^{N}\sum_{m=0}^{N} c_n c_m W_{nm}$$

$W_{nm} \longrightarrow$ closed-forms

Same for other propagators

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$$W_{nm} \longrightarrow \text{closed-forms}$$

$$\equiv \sum_{n=0}^{N}\sum_{m=0}^{N} c_n c_m W_{nm}$$

$$\equiv c^{\mathsf{T}} W c$$

Same for other propagators

$$G_{nm} \qquad \Delta_{nm}$$

Optimization

Optimization

Guide (old)

Optimization

Guide (old)

$\longrightarrow$ For **each** smearing function we want to test: we compute the following integral

Optimization

Guide (old)

→ For **each** smearing function we want to test: we compute the following integral

$$W(\Lambda_{\mathsf{A}}^{-},\Lambda_{\mathsf{A}}^{+}) = \frac{1}{(2\pi)^3}\frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}}\int\frac{\mathrm{d}^3\boldsymbol{k}}{2|\boldsymbol{k}|}\int\mathrm{d}t\int\mathrm{d}^3\boldsymbol{x}\int\mathrm{d}t'\int\mathrm{d}^3\boldsymbol{x}\;\mathsf{e}^{\frac{-|\boldsymbol{x}|^2}{2\sigma^2}}\,\mathsf{e}^{\frac{-|\boldsymbol{x}'|^2}{2\sigma^2}}\,\mathsf{e}^{-\mathsf{i}\Omega t}\mathsf{e}^{\mathsf{i}\Omega t'}\mathsf{e}^{\mathsf{i}|\boldsymbol{k}|(t-t')+\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')}\chi(t)\chi(t')$$

Optimization

Guide (new)

Optimization

Guide (new)

$\longrightarrow$ We take the closed-form expression of W_{nm}

Optimization

Guide (new)

→ We take the closed-form expression of W_{nm}

→ We expand in the Hermite basis the smearing function we want to test

Optimization

Guide (new)

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→ We expand in the Hermite basis the smearing function we want to test

→ We get the coefficients c_n

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→ We take the closed-form expression of W_{nm}

→ We expand in the Hermite basis the smearing function we want to test

→ We get the coefficients c_n

→ We calculate $c^\top W c$

Optimization

Guide (new)

→ We take the closed-form expression of W_{nm}

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→ We calculate $c^{\top} W c$

hard integral

Optimization

Guide (new)

→ We take the closed-form expression of W_{nm}

→ We expand in the Hermite basis the smearing function we want to test

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hard integral (not analytical)

Optimization

Guide (new)

→ We take the closed-form expression of W_{nm}

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→ We get the coefficients c_n

→ We calculate $c^{\intercal} W c$

hard integral (not analytical) → Matrix product

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→ We take the closed-form expression of W_{nm}

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hard integral (not analytical) → Matrix product

Optimization

Optimization

$\longrightarrow$ find c

Optimization

→ find c

→ maximize $$\mathcal{N}(c) = \lambda^2 \left(|c^\intercal G c| - c^\intercal W c \right)$$

Optimization

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→ minimize $$\frac{c^\intercal \Delta c}{\mathcal{N}(c)}$$

Optimization

→ find c

→ maximize $$\boxed{\mathcal{N}(c) = \lambda^2\left(|c^\intercal G c| - c^\intercal W c\right)}$$

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Maximization

Maximization

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Maximization

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\intercal G c| - c^\intercal W c \right)$$

$$\longrightarrow \chi(t) \to 10 \times \chi(t)$$

Maximization

$$\mathcal{N}(c) = \lambda^2 \left(|c^\intercal G c| - c^\intercal W c \right)$$

$$\chi(t) \to 10 \times \chi(t)$$

$$\to 100 \times \chi(t)$$

Maximization

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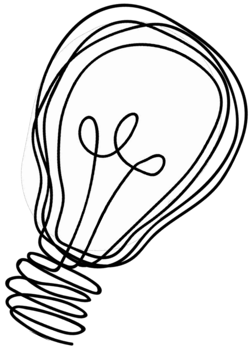
$$\longrightarrow \chi(t) \to 10 \times \chi(t)$$

$$\to 100 \times \chi(t)$$

$$\to 78440954385344538745935430 \times \chi(t)$$

Maximization

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\intercal G c| - c^\intercal W c \right)$$



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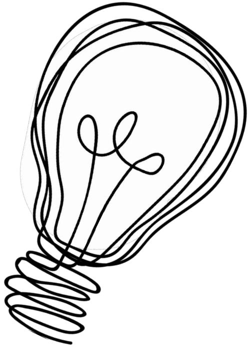


$$\longrightarrow \chi(t) \to 10 \times \chi(t)$$

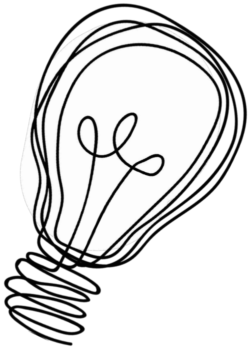
$$\to 100 \times \chi(t)$$

$$\to 78440954385344538745935430 \times \chi(t)$$

$\longrightarrow$ ill–posed

Maximization

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\intercal G c| - c^\intercal W c \right)$$



$$\longrightarrow \chi(t) \to 10 \times \chi(t)$$

$$\to 100 \times \chi(t)$$

$$\to 78440954385344538745935430 \times \chi(t)$$

$\longrightarrow$ ill–posed $\longrightarrow$ constraint

Maximization

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\intercal G c| - c^\intercal W c \right)$$

Maximization

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$$\longrightarrow ||\chi(t)||_2 = \int_{-\infty}^{\infty} |\chi(t)|^2 dt$$

Maximization

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\intercal G c| - c^\intercal W c \right)$$

$$\longrightarrow ||\chi(t)||_2 = \int_{-\infty}^{\infty} |\chi(t)|^2 dt \ \propto ||c||^2$$

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$$= ||\mathrm{e}^{-t^2/2}||_2 = \sqrt{\pi}$$

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$$= ||\mathrm{e}^{-t^2/2}||_2 = \sqrt{\pi}$$

$$\longrightarrow \frac{\sqrt{\pi}}{||c||^2}$$

Maximization

$$\longrightarrow \widetilde{\mathcal{N}}(c) = \lambda^2 \sqrt{\pi}\, \frac{|c^\intercal G c| - c^\intercal W c}{||c||^2}$$

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$$= \max_{\theta \in [0, 2\pi)} \lambda_{\max} \left(\cos(\theta) \mathrm{Re}\{G\} + \sin(\theta) \mathrm{Im}\{G\} - \mathrm{Re}\{W\} \right)$$

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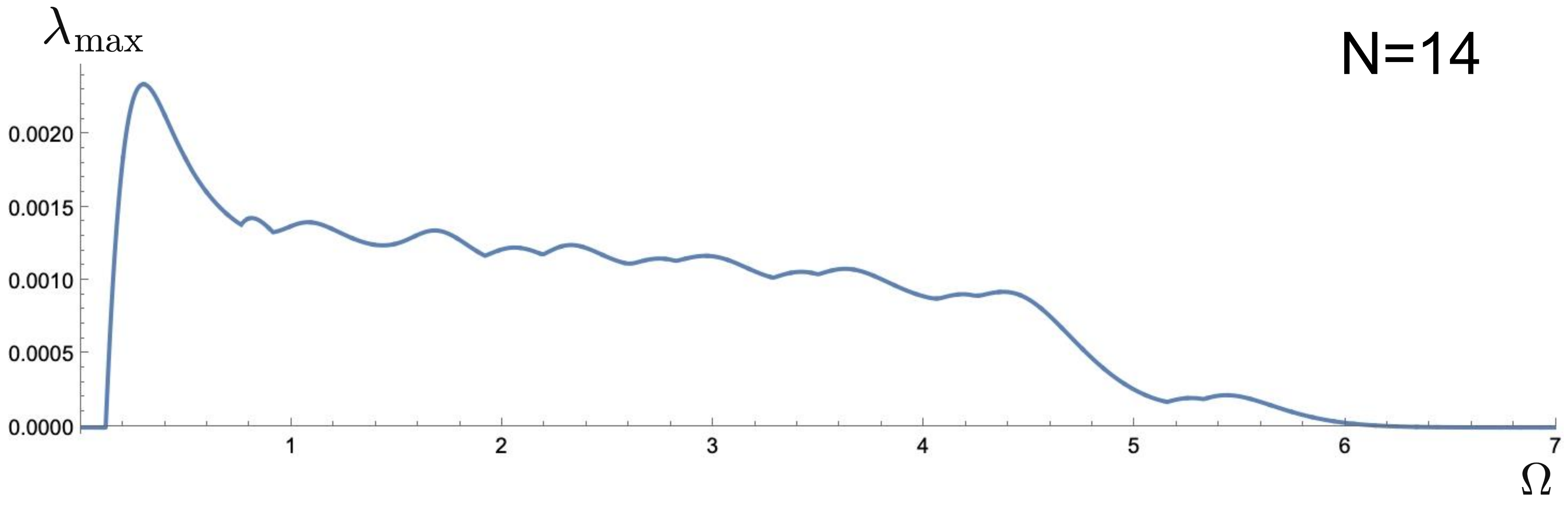

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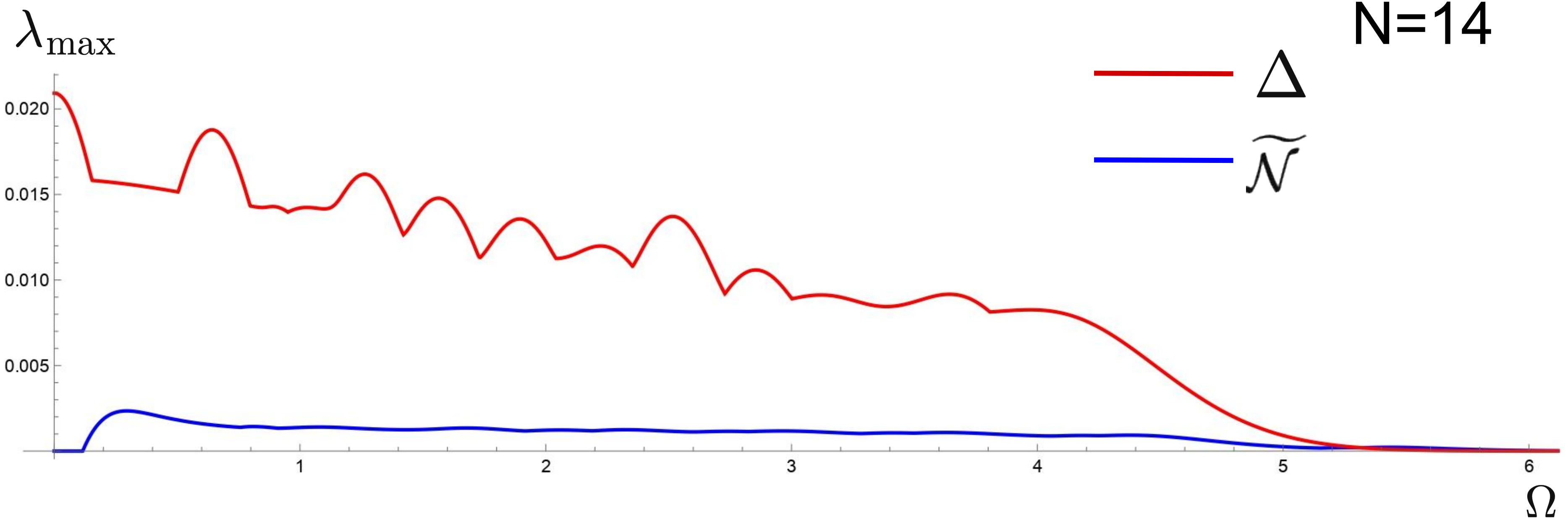

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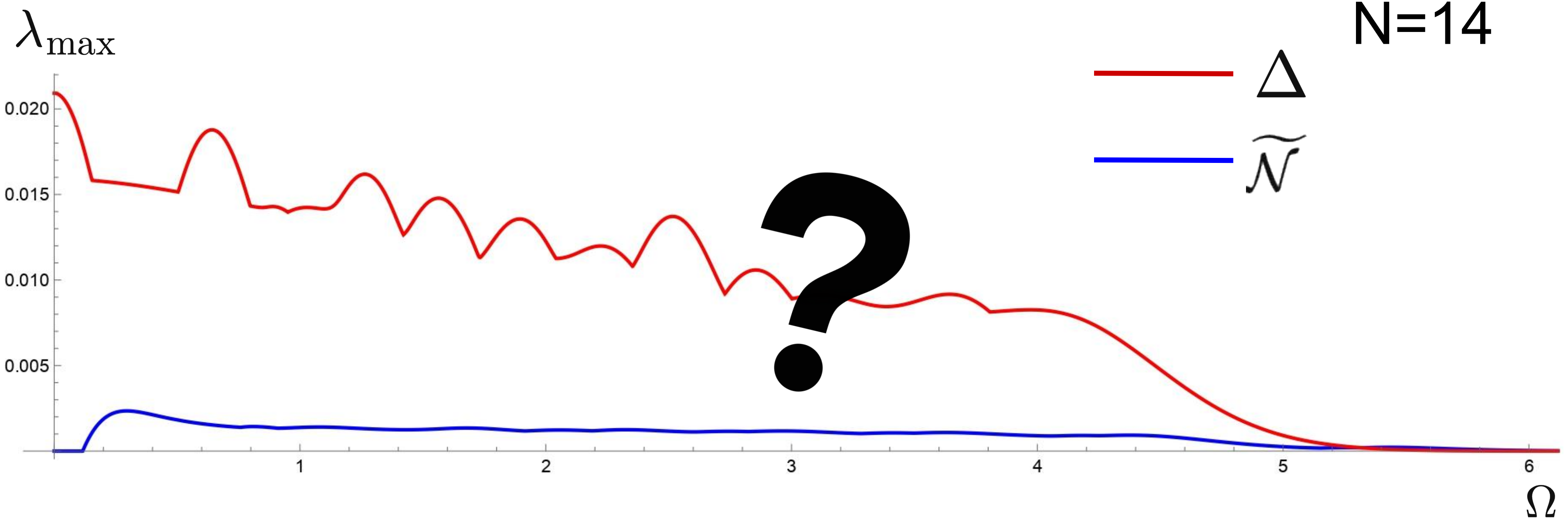

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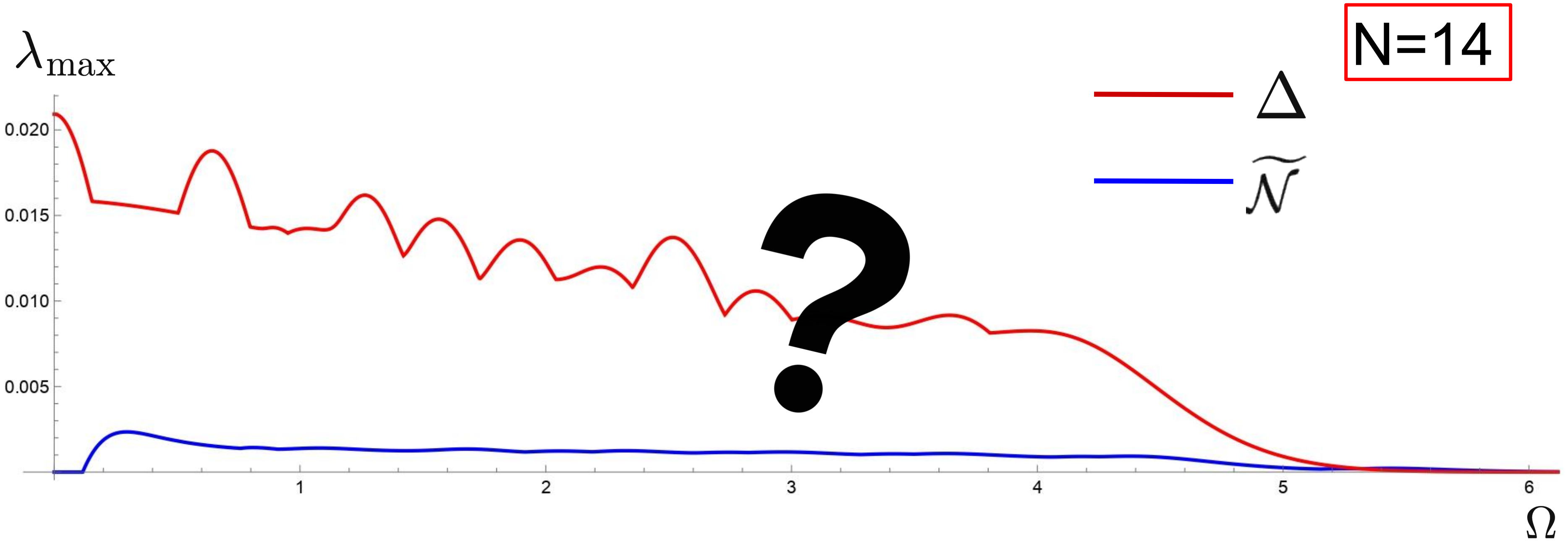

Maximization

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Maximization

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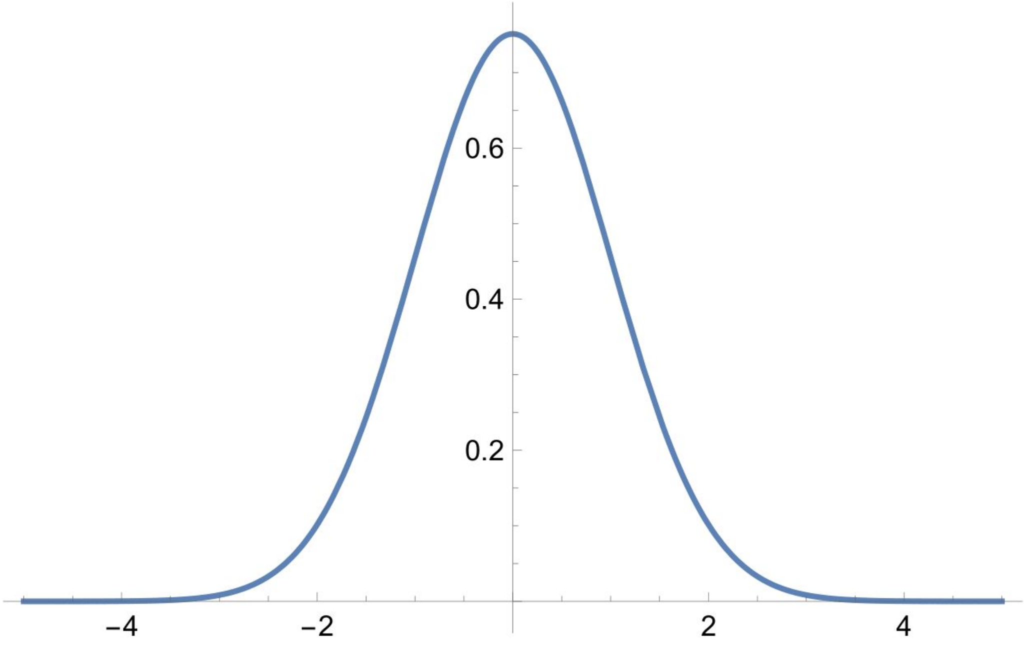

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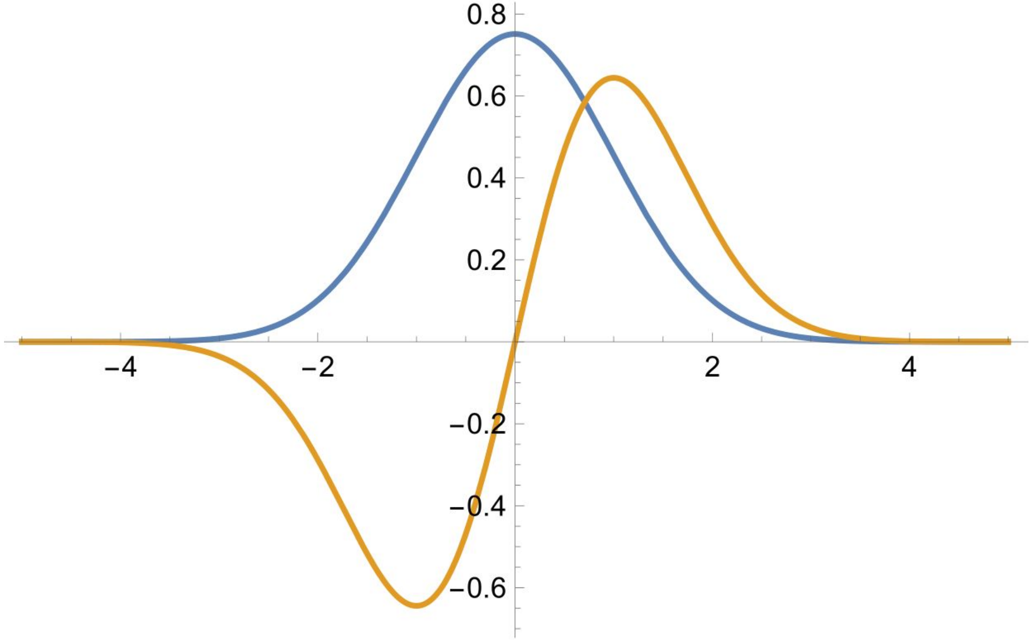

N=1

Maximization

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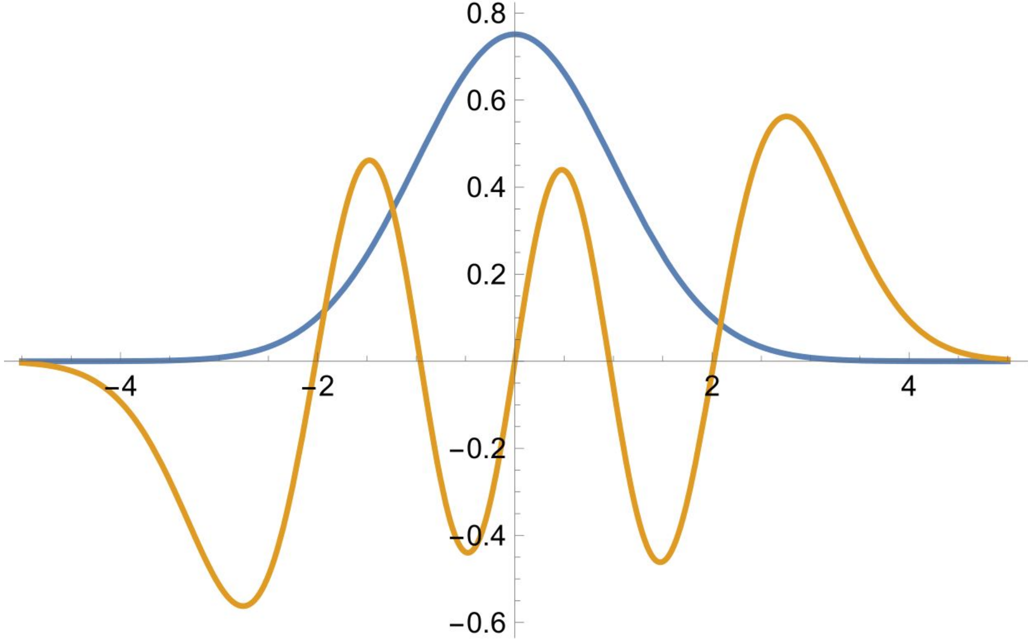
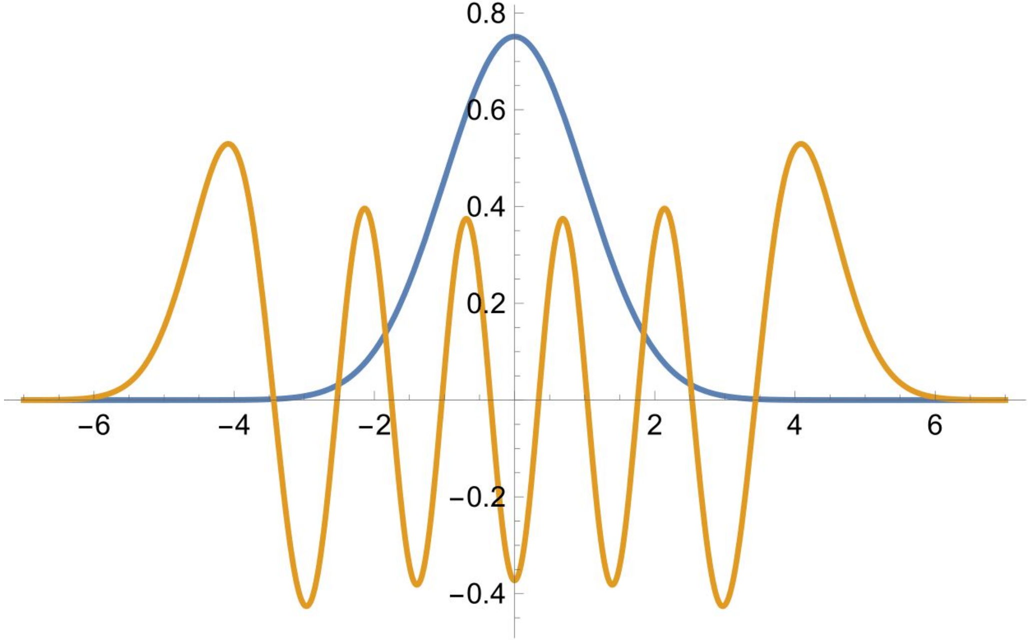


N=5

Maximization

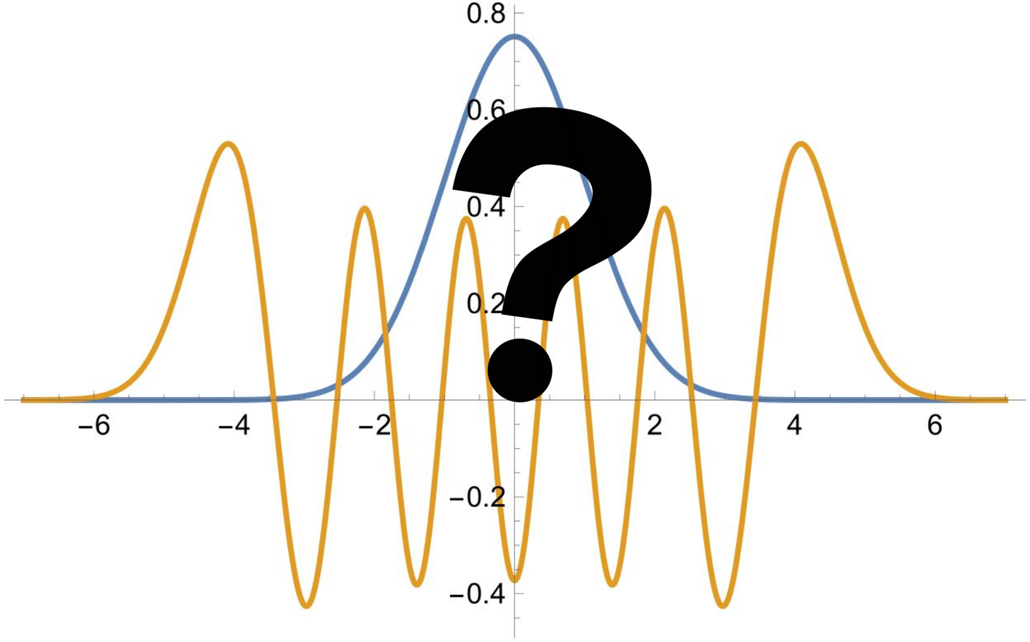
$$\longrightarrow \chi(t) \approx \sum_{n=0}^{N} c_n \chi_n(t) \equiv \sum_{n=0}^{N} c_n H_n(t) \mathrm{e}^{-t^2/2T^2}$$



N=10

Maximization

$$\longrightarrow \chi(t) \approx \sum_{n=0}^{N} c_n \chi_n(t) \equiv \sum_{n=0}^{N} c_n H_n(t) \mathrm{e}^{-t^2/2T^2}$$



N=10

Maximization

$$\longrightarrow \chi(t) \approx \sum_{n=0}^{N} c_n \chi_n(t) \equiv \sum_{n=0}^{N} c_n H_n(t) \mathrm{e}^{-t^2/2T^2}$$

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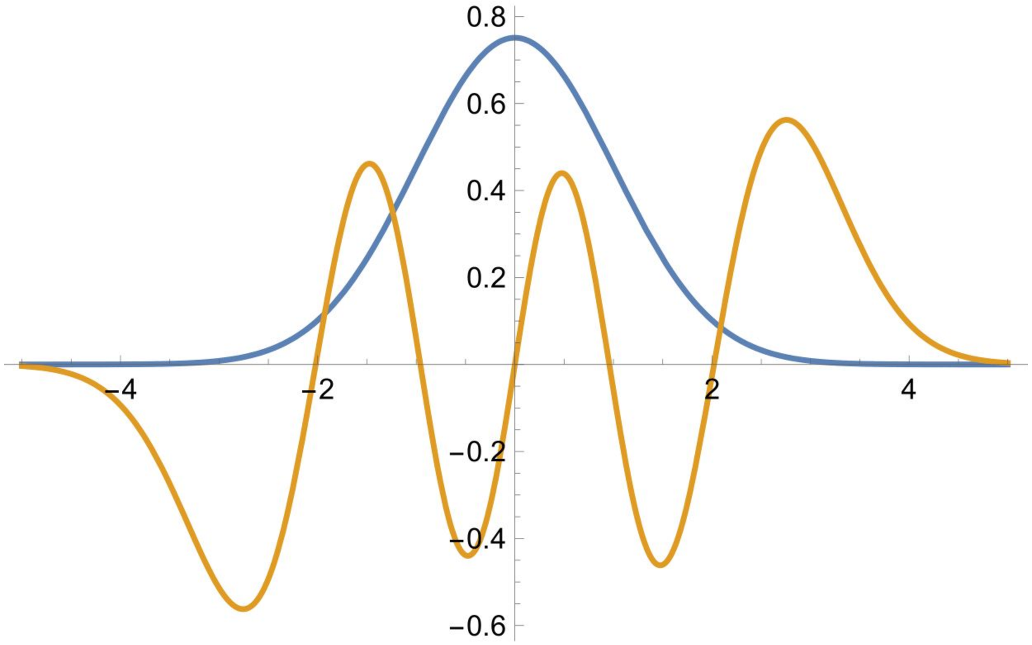
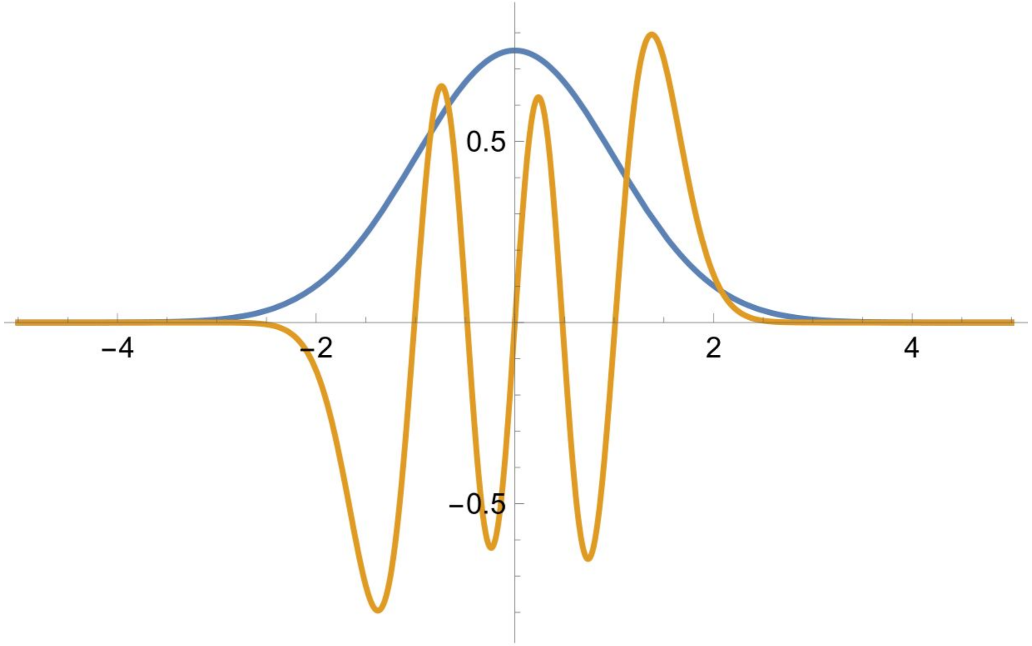


N=5
T=1*T0

Maximization

$$\chi(t) \approx \sum_{n=0}^{N} c_n \chi_n(t) \equiv \sum_{n=0}^{N} c_n H_n(t) \mathrm{e}^{-t^2/2T^2}$$

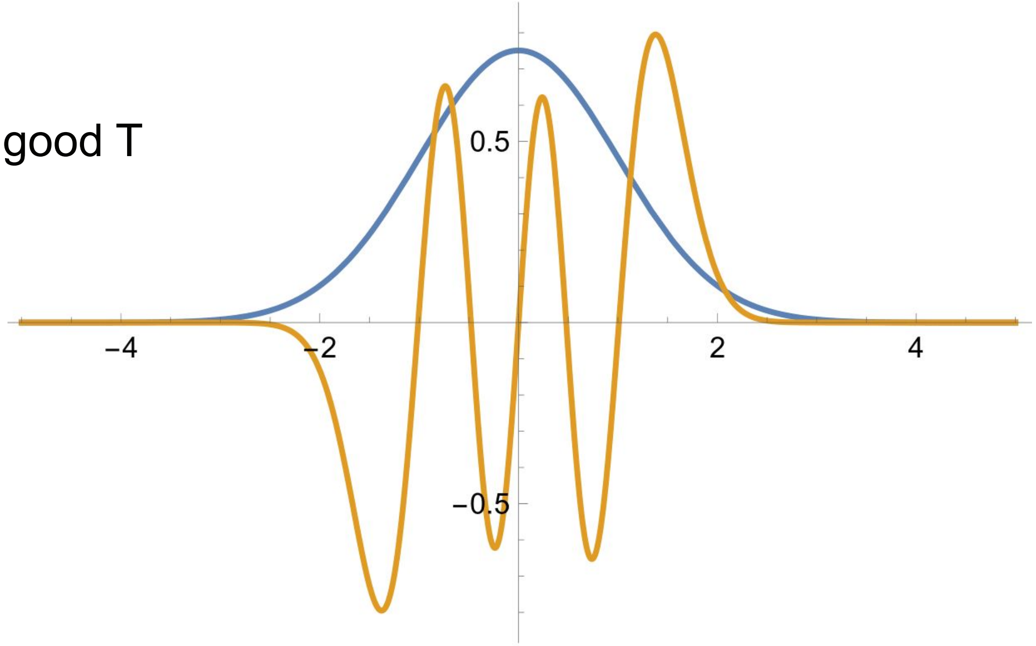


N=5
T=0.5*T0

Maximization

$$\chi(t) \approx \sum_{n=0}^{N} c_n \chi_n(t) \equiv \sum_{n=0}^{N} c_n H_n(t) \mathrm{e}^{-t^2/2T^2}$$

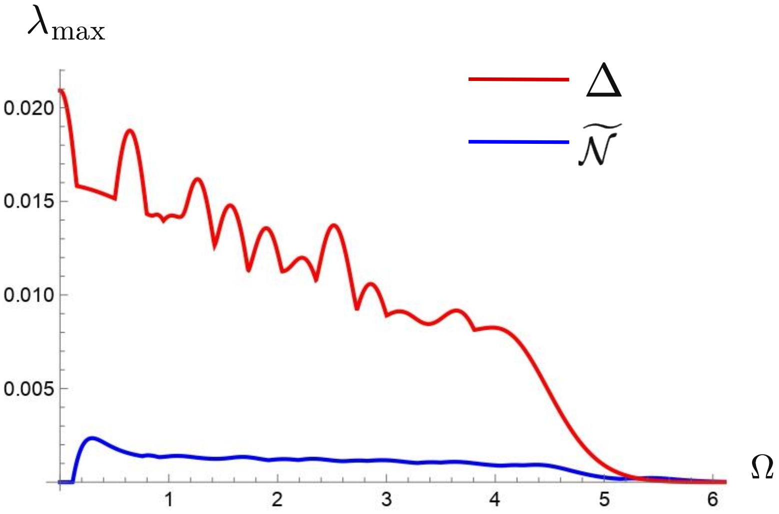
choose a good T



N=5
T=0.5*T0

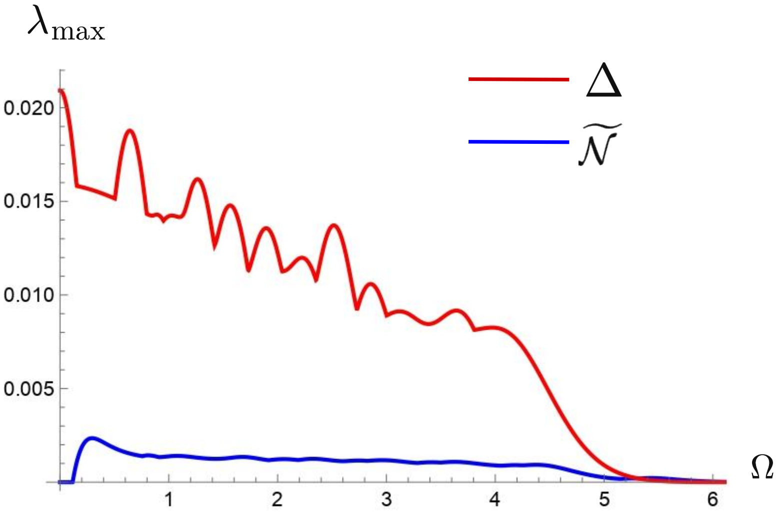
Maximization

Maximization

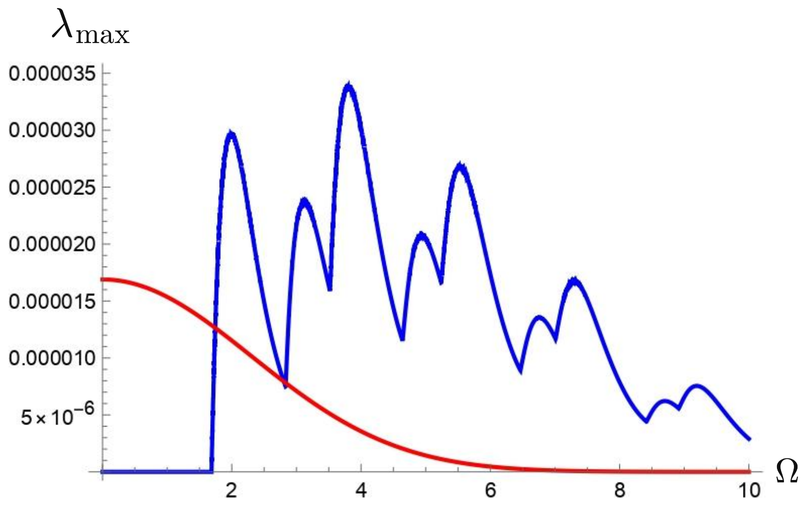


T=1*T0

Maximization

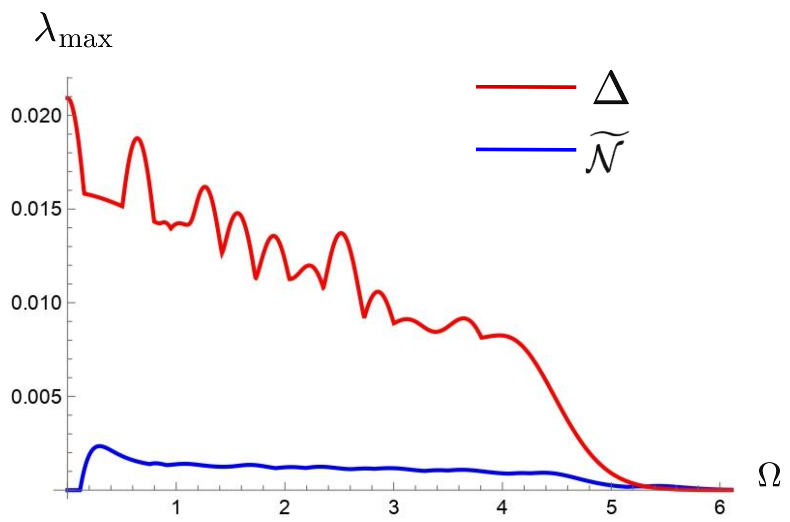


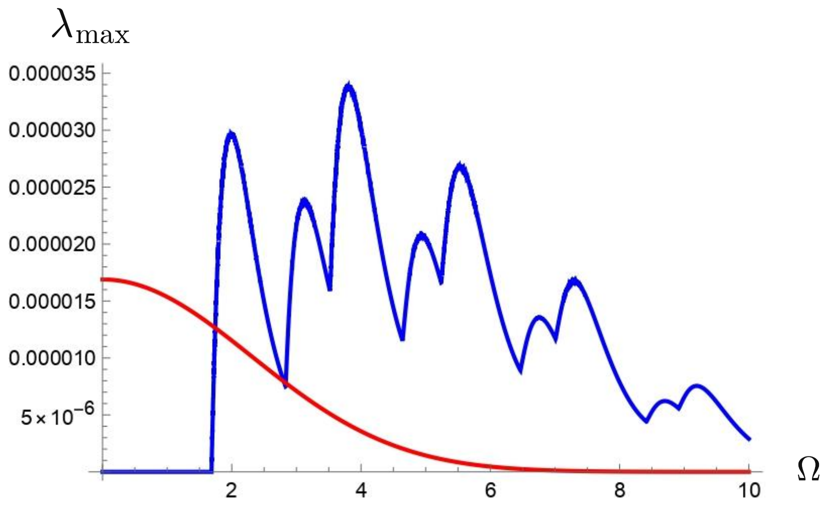
T=1*T0



T=0.4*T0

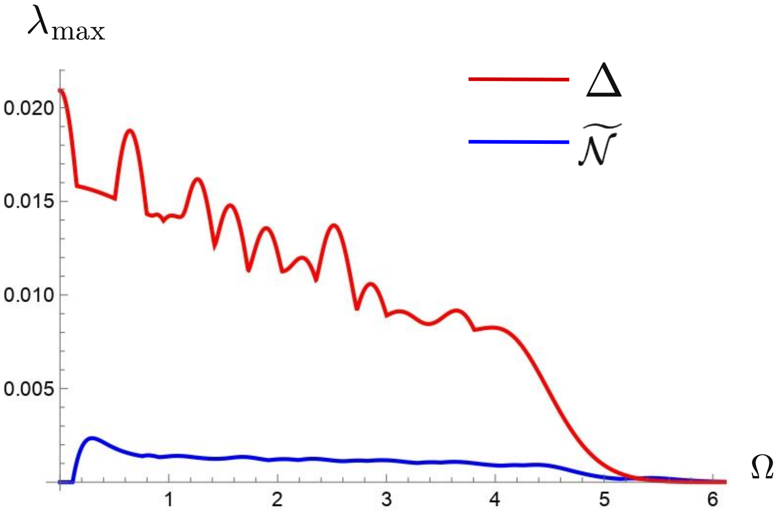
Maximization

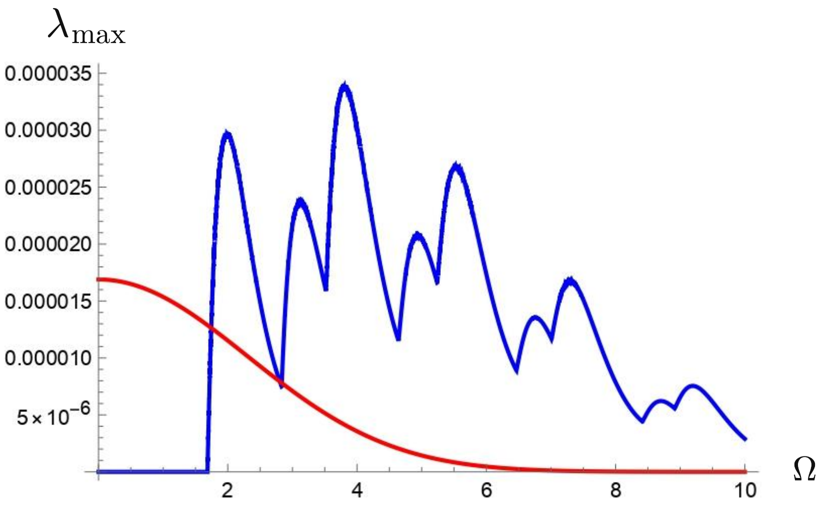




$\longrightarrow$ We virtually have no Δ

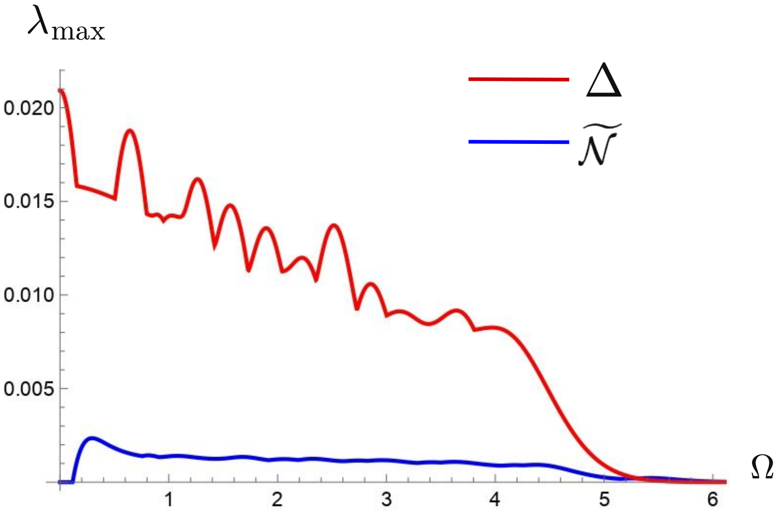
Maximization

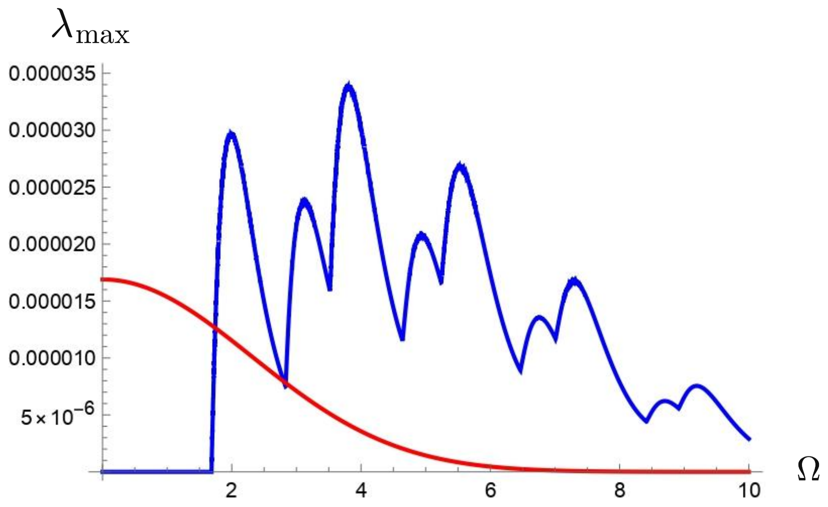




→ We virtually have no Δ , but the interaction is too short

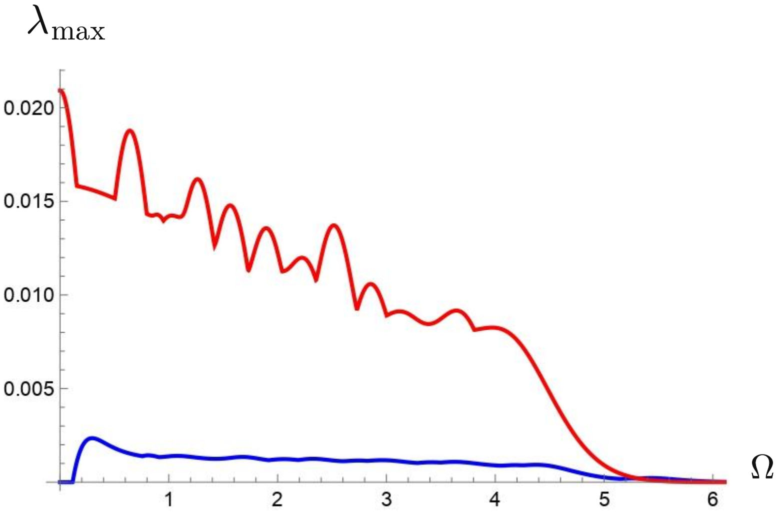
Maximization



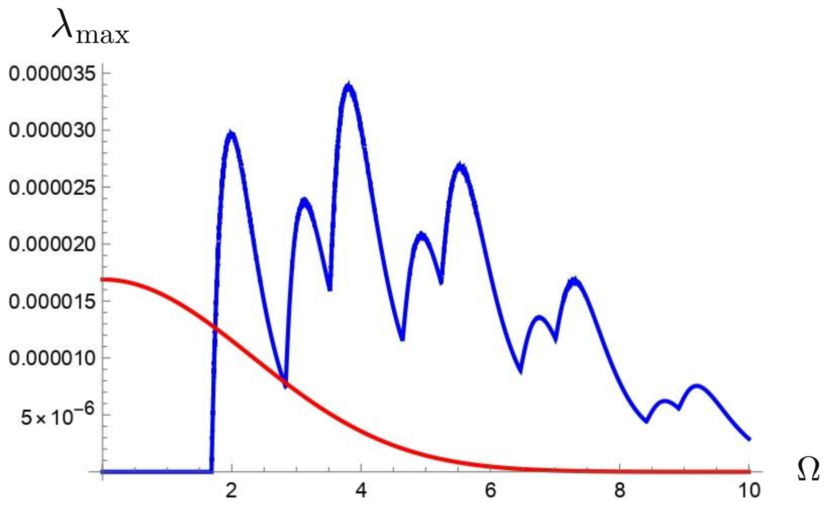


$\longrightarrow$ We virtually have no Δ , but the interaction is too short $\longrightarrow$ less entanglement

Maximization



T=?



T=1*T0

T=0.4*T0

Optimization

Optimization

$\longrightarrow$ find a good T for which we can perform the optimization

Optimization

→ find a good T for which we can perform the optimization

→ large enough to allow more entanglement

Optimization

→ find a good T for which we can perform the optimization

→ large enough to allow more entanglement

→ small enough to keep signalling at most 5 %

Optimization

→ find a good T for which we can perform the optimization

→ large enough to allow more entanglement

→ small enough to keep signalling at most 5 %

→ constrained optimization

Ongoing work

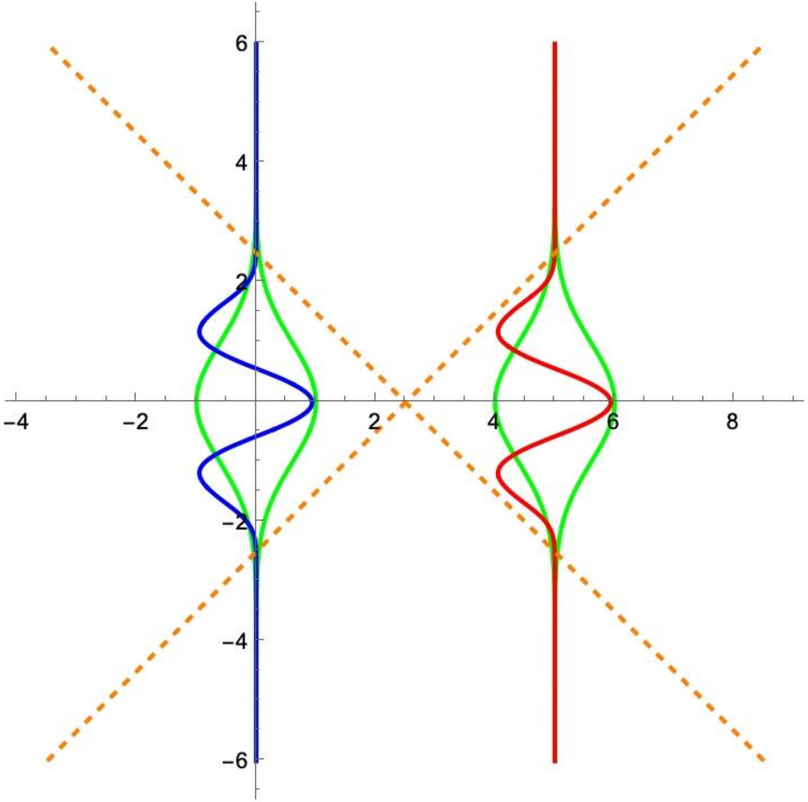
→ Point-like identical detectors, N=14

Ongoing work

→ Point-like identical detectors, N=14, T=0.53*T0

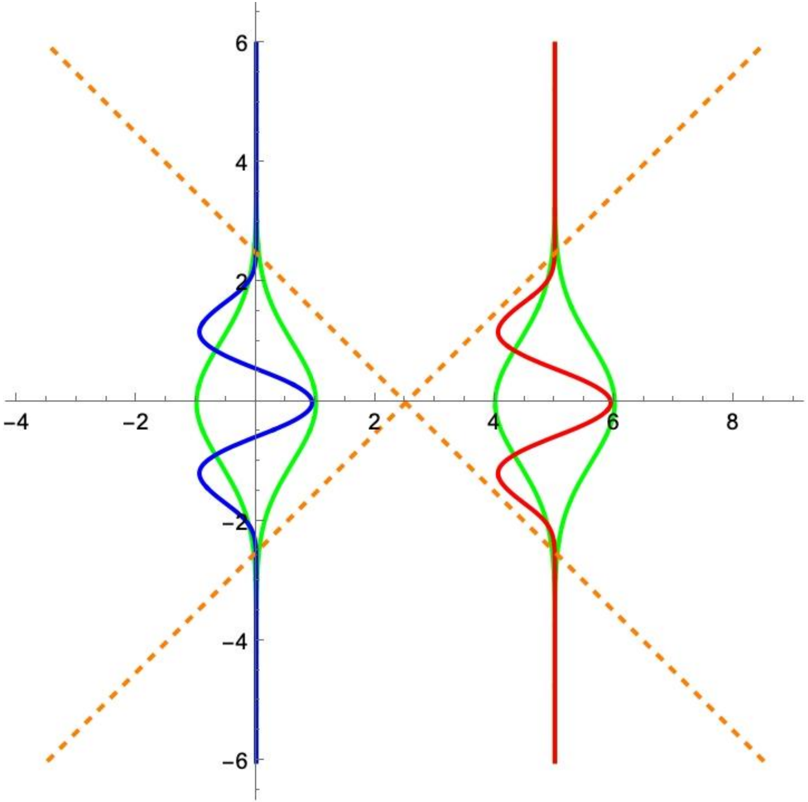
Ongoing work

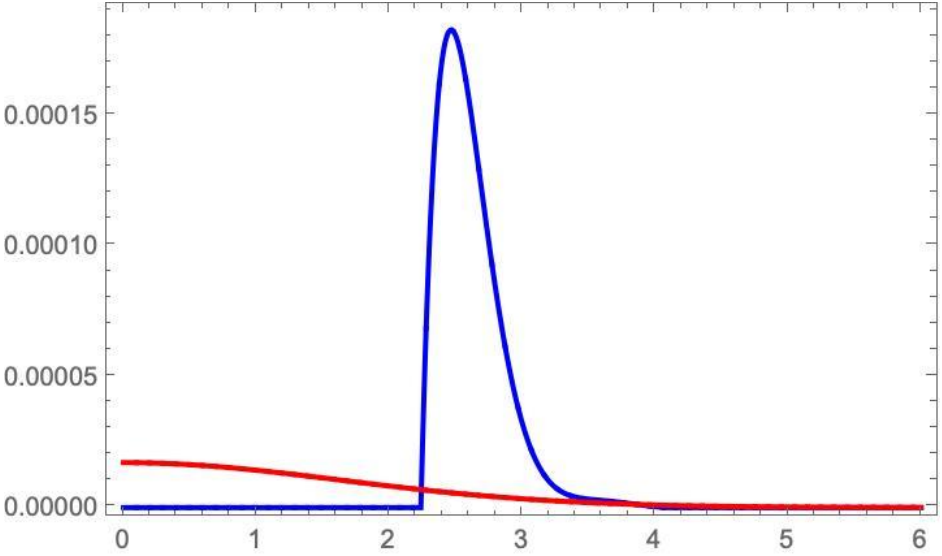
→ Point-like identical detectors, N=14, T=0.53*T0



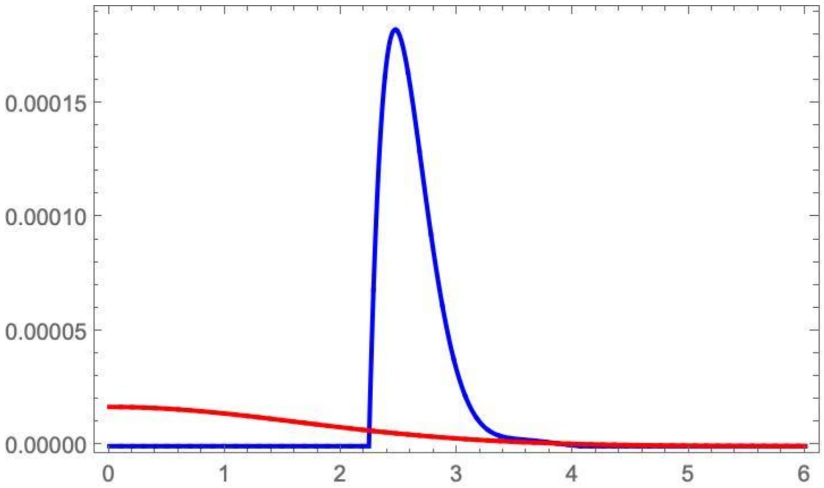
Ongoing work

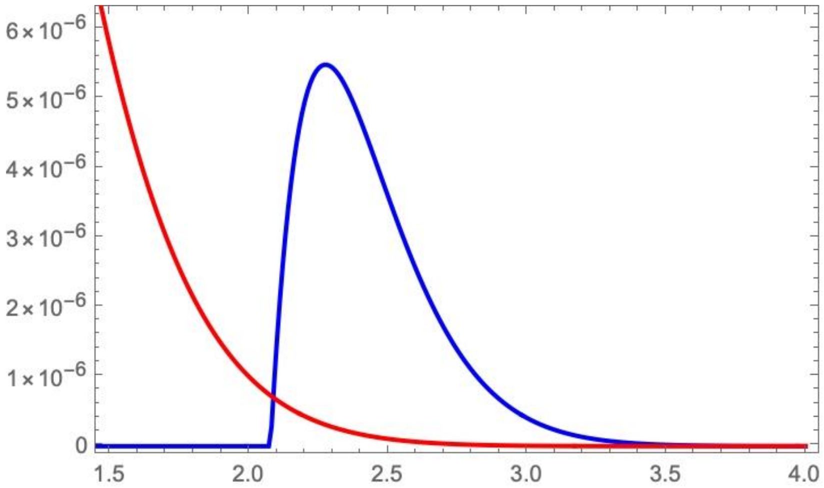
→ Point-like identical detectors, N=14, T=0.53*T0





Ongoing work





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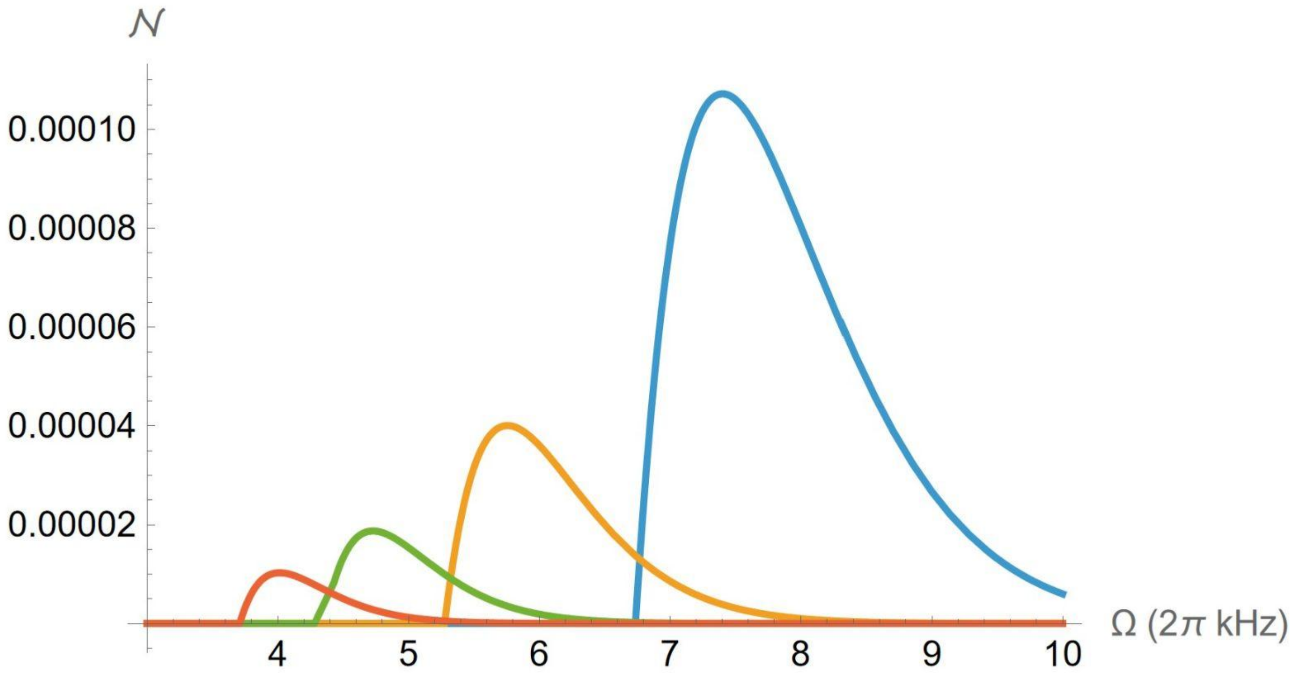
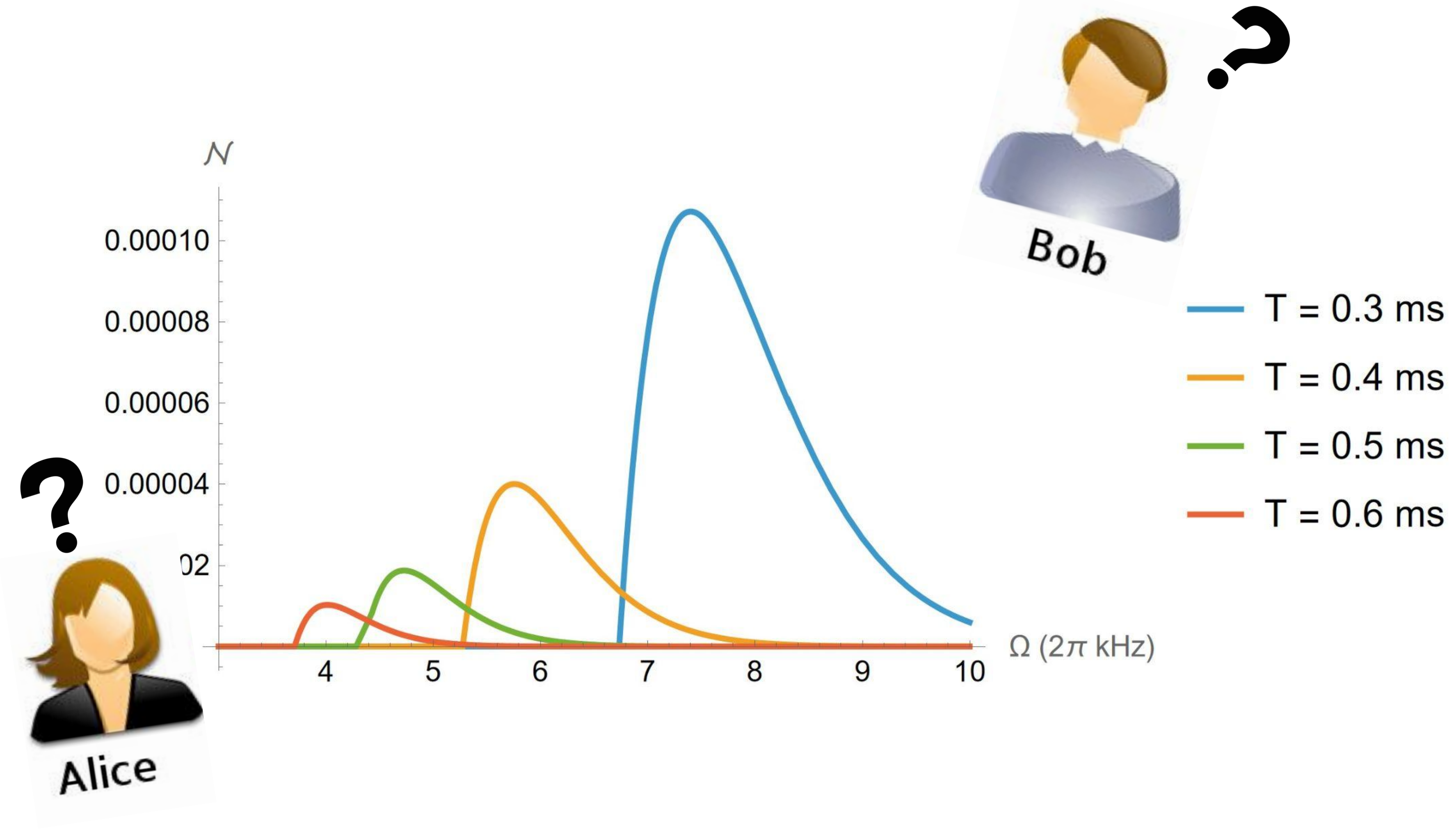


T. Rick Perche, Francesco Gozzini, and Markus Oberthaler, Bose polarons as relativistic Unruh-DeWitt detectors: Entanglement harvesting from Bose-Einstein condensates, in preparation



$\mathcal{N}$
0.00010
0.00008
0.00006
0.00004
02
4
5
6
7
8
9
10
Ω (2π kHz)
T = 0.3 ms
T = 0.4 ms
T = 0.5 ms
T = 0.6 ms
Alice
Bob
?
?

Thank you!

Optimization

$$\longrightarrow \chi(t) \approx \sum_{n=0}^{N} c_n \chi_n(t) \qquad \{\chi_n\} \subseteq L^2(\mathbb{R})$$

$$\longrightarrow \{H_n(t)\mathsf{e}^{-t^2/2T^2}\}$$

$$\sum_{n=0}^{N}\sum_{m=0}^{N} c_n c_m \int \mathrm{d}V\mathrm{d}V' H_n(t)\, H_m(t')\mathsf{e}^{-t^2/2T^2} F(\boldsymbol{x})\, \mathsf{e}^{-t'^2/2T^2} F(\boldsymbol{x}') W(\mathsf{x},\mathsf{x}')$$

$$\sum_{n=0}^{N}\sum_{m=0}^{N} c_n c_m \int \mathrm{d}V\mathrm{d}V' \qquad \mathsf{e}^{-t^2/2T^2}\mathsf{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\, \mathsf{e}^{-t'^2/2T^2}\mathsf{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathsf{x},\mathsf{x}')$$

Negativity in practice

$$\Lambda(\mathsf{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}}\mathsf{e}^{-\frac{t^2}{2T^2}}\,\mathsf{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}$$

$$G_{\text{AB}}^{++} = \frac{\lambda^2 T^2 e^{-\Omega^2 T^2} e^{\mathrm{i}\Omega t_0}}{8\sqrt{\pi}|\boldsymbol{L}|\sqrt{T^2+\sigma^2}}\Bigg(e^{-\frac{(|\boldsymbol{L}|+t_0)^2}{4(T^2+\sigma^2)}}\operatorname{erfi}\left(\frac{|\boldsymbol{L}|+t_0}{2\sqrt{T^2+\sigma^2}}\right)+e^{-\frac{(|\boldsymbol{L}|-t_0)^2}{4(T^2+\sigma^2)}}\operatorname{erfi}\left(\frac{|\boldsymbol{L}|-t_0}{2\sqrt{T^2+\sigma^2}}\right)$$
$$-\mathrm{i}\left(e^{-\frac{(|\boldsymbol{L}|+t_0)^2}{4(T^2+\sigma^2)}}\operatorname{erf}\left(\frac{|\boldsymbol{L}|T^2-t_0\sigma^2}{2T\sigma\sqrt{T^2+\sigma^2}}\right)+e^{-\frac{(|\boldsymbol{L}|-t_0)^2}{4(T^2+\sigma^2)}}\operatorname{erf}\left(\frac{|\boldsymbol{L}|T^2+t_0\sigma^2}{2T\sigma\sqrt{T^2+\sigma^2}}\right)\right)\Bigg).$$

$$\mathcal{N}(\hat{\rho}_{\text{AB}}) = \max\left(0, \frac{\lambda^2 e^{-\Omega^2T^2}}{4\pi\alpha^2}\left(\sqrt{\pi}\alpha e^{-\frac{|\boldsymbol{L}|^2}{4\alpha^2T^2}}\frac{T}{|\boldsymbol{L}|}\sqrt{\operatorname{erf}\left(\frac{|\boldsymbol{L}|}{2\alpha\sigma}\right)^2+\operatorname{erfi}\left(\frac{|\boldsymbol{L}|}{2\alpha T}\right)^2}+\frac{\sqrt{\pi}\Omega T}{\alpha}e^{\frac{\Omega^2T^2}{\alpha^2}}\operatorname{erfc}\left(\frac{\Omega T}{\alpha}\right)-1\right)\right),$$

Negativity in practice

$$\boxed{\Lambda(\mathrm{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}}\mathrm{e}^{-\frac{t^2}{2T^2}}\,\mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}}$$

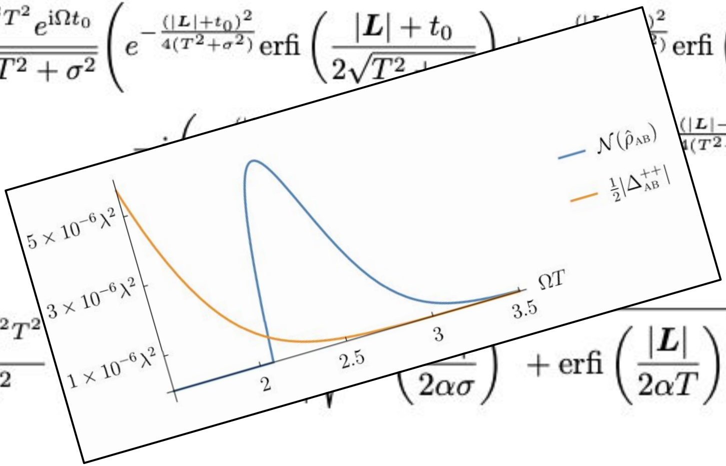
$$G^{++}_{\mathrm{AB}} = \frac{\lambda^2 T^2 e^{-\Omega^2 T^2} e^{\mathrm{i}\Omega t_0}}{8\sqrt{\pi}|\boldsymbol{L}|\sqrt{T^2+\sigma^2}}\left(e^{-\frac{(|\boldsymbol{L}|+t_0)^2}{4(T^2+\sigma^2)}}\operatorname{erfi}\left(\frac{|\boldsymbol{L}|+t_0}{2\sqrt{T^2+\sigma^2}}\right) + e^{-\frac{(|\boldsymbol{L}|-t_0)^2}{4(T^2+\sigma^2)}}\operatorname{erfi}\left(\frac{|\boldsymbol{L}|-t_0}{2\sqrt{T^2+\sigma^2}}\right)\right.$$
$$\left. - \mathrm{i}\left(e^{-\frac{(|\boldsymbol{L}|+t_0)^2}{4(T^2+\sigma^2)}}\operatorname{erf}\left(\frac{|\boldsymbol{L}|T^2 - t_0\sigma^2}{2T\sigma\sqrt{T^2+\sigma^2}}\right) + e^{-\frac{(|\boldsymbol{L}|-t_0)^2}{4(T^2+\sigma^2)}}\operatorname{erf}\left(\frac{|\boldsymbol{L}|T^2 + t_0\sigma^2}{2T\sigma\sqrt{T^2+\sigma^2}}\right)\right)\right).$$

$$\mathcal{N}(\hat{\rho}_{\mathrm{AB}}) = \max\left(0, \frac{\lambda^2 e^{-\Omega^2 T^2}}{4\pi\alpha^2}\left(\sqrt{\pi}\alpha e^{-\frac{|\boldsymbol{L}|^2}{4\alpha^2 T^2}}\frac{T}{|\boldsymbol{L}|}\sqrt{\operatorname{erf}\left(\frac{|\boldsymbol{L}|}{2\alpha\sigma}\right)^2 + \operatorname{erfi}\left(\frac{|\boldsymbol{L}|}{2\alpha T}\right)^2} + \frac{\sqrt{\pi}\Omega T}{\alpha}e^{\frac{\Omega^2 T^2}{\alpha^2}}\operatorname{erfc}\left(\frac{\Omega T}{\alpha}\right) - 1\right)\right),$$

Negativity in practice

$$\Lambda(\mathrm{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} \mathrm{e}^{-\frac{t^2}{2T^2}} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}$$

$$G^{++}_{\mathrm{AB}} = \frac{\lambda^2 T^2 e^{-\Omega^2 T^2} e^{\mathrm{i}\Omega t_0}}{8\sqrt{\pi}|\boldsymbol{L}|\sqrt{T^2+\sigma^2}} \Bigg(e^{-\frac{(|\boldsymbol{L}|+t_0)^2}{4(T^2+\sigma^2)}} \operatorname{erfi}\left(\frac{|\boldsymbol{L}|+t_0}{2\sqrt{T^2 \cdots}}\right) \cdots \operatorname{erfi}\left(\frac{|\boldsymbol{L}|-t_0}{2\sqrt{T^2+\sigma^2}}\right) \cdots e^{-\frac{(|\boldsymbol{L}|-t_0)^2}{4(T^2+\sigma^2)}} \operatorname{erf}\left(\frac{|\boldsymbol{L}|T^2+t_0\sigma^2}{2T\sigma\sqrt{T^2+\sigma^2}}\right)\Bigg)\Bigg)\Bigg).$$



$$\mathcal{N}(\hat{\rho}_{\mathrm{AB}}) = \max\left(0, \frac{\lambda^2 e^{-\Omega^2 T^2}}{4\pi\alpha^2} \cdots \left(\frac{\cdots}{2\alpha\sigma}\right) + \operatorname{erfi}\left(\frac{|\boldsymbol{L}|}{2\alpha T}\right)^2 + \frac{\sqrt{\pi}\Omega T}{\alpha} e^{\frac{\Omega^2 T^2}{\alpha^2}} \operatorname{erfc}\left(\frac{\Omega T}{\alpha}\right) - 1\right)\Bigg),$$

Optimization

$$\int \mathrm{d}V \mathrm{d}V' H_n(t)\, H_m(t') \mathrm{e}^{-t^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\; \mathrm{e}^{-t'^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathsf{x}, \mathsf{x}')$$

Optimization

$$\int \mathrm{d}V \mathrm{d}V' H_n(t)\, H_m(t') \mathrm{e}^{-t^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\, \mathrm{e}^{-t'^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathrm{x}, \mathrm{x}')$$

$$\Downarrow$$

$$\int \mathrm{d}V \mathrm{d}V' \qquad \mathrm{e}^{-t^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\, \mathrm{e}^{-t'^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathrm{x}, \mathrm{x}')$$

Optimization

$\longrightarrow H_n(t)$?

Optimization

$\longrightarrow H_n(t)$?

$$
\begin{aligned}
H_0(x) &= 1, \\
H_1(x) &= 2x, \\
H_2(x) &= 4x^2 - 2, \\
H_3(x) &= 8x^3 - 12x, \\
H_4(x) &= 16x^4 - 48x^2 + 12, \\
H_5(x) &= 32x^5 - 160x^3 + 120x, \\
H_6(x) &= 64x^6 - 480x^4 + 720x^2 - 120, \\
H_7(x) &= 128x^7 - 1344x^5 + 3360x^3 - 1680x, \\
H_8(x) &= 256x^8 - 3584x^6 + 13440x^4 - 13440x^2 + 1680, \\
H_9(x) &= 512x^9 - 9216x^7 + 48384x^5 - 80640x^3 + 30240x, \\
H_{10}(x) &= 1024x^{10} - 23040x^8 + 161280x^6 - 403200x^4 + 302400x^2 - 30240.
\end{aligned}
$$

Optimization

$\longrightarrow H_n(t)$?

$\longrightarrow t, t^2, t^3, \ldots$

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$\longrightarrow t, t^2, t^3, \ldots$

?

Optimization

$\longrightarrow H_n(t)$?

$\longrightarrow t, t^2, t^3, \ldots$

?

Optimization

$\longrightarrow H_n(t)$?

$e^{\alpha t}$

Optimization

$\longrightarrow H_n(t)$?

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\mathrm{e}^{\alpha t}$$

Optimization

$\longrightarrow H_n(t)$?

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\mathsf{e}^{\alpha t} \longrightarrow t\,\mathsf{e}^{\alpha t}$$

Optimization

$\longrightarrow H_n(t)$?

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\mathrm{e}^{\alpha t}\Big|_{\alpha=0} \longrightarrow t\,\mathrm{e}^{\alpha t}$$

Optimization

$\longrightarrow H_n(t)$?

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\mathrm{e}^{\alpha t}\Big|_{\alpha=0} \longrightarrow t$$

Optimization

$\longrightarrow H_n(t)$?

$$\frac{\mathrm{d}}{\mathrm{d}\alpha} \mathrm{e}^{\alpha t}\Big|_{\alpha=0} \longrightarrow t$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\alpha^2} \mathrm{e}^{\alpha t}\Big|_{\alpha=0}$$

Optimization

$\longrightarrow H_n(t)$?

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\mathrm{e}^{\alpha t}\Big|_{\alpha=0} \longrightarrow t$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\alpha^2}\mathrm{e}^{\alpha t}\Big|_{\alpha=0} \longrightarrow t^2$$

Optimization

$$\longrightarrow H_n(t)\ ?$$

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\mathrm{e}^{\alpha t}\Big|_{\alpha=0} \longrightarrow t$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\alpha^2}\mathrm{e}^{\alpha t}\Big|_{\alpha=0} \longrightarrow t^2$$

$$\vdots$$

Optimization

$\longrightarrow H_n(t)$

Optimization

$$\longrightarrow H_n(t) = H_n\left(\frac{\mathrm{d}}{\mathrm{d}\alpha}\right)\mathrm{e}^{\alpha t}$$

Optimization

$$\longrightarrow H_n(t) = H_n\left(\frac{\mathrm{d}}{\mathrm{d}\alpha}\right)\mathrm{e}^{\alpha t} \qquad (\alpha = 0)$$

Optimization

$$\longrightarrow H_n(t) = \underbrace{H_n\left(\frac{\mathrm{d}}{\mathrm{d}\alpha}\right)}_{\text{no time dependence}}\mathrm{e}^{\alpha t} \qquad (\alpha = 0)$$

Optimization

$$\int \mathrm{d}V \mathrm{d}V' H_n(t)\, H_m(t') \mathsf{e}^{-t^2/2T^2} \mathsf{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\, \mathsf{e}^{-t'^2/2T^2} \mathsf{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathsf{x}, \mathsf{x}')$$

Optimization

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$$H_n\left(\frac{\mathrm{d}}{\mathrm{d}\alpha}\right) H_m\left(\frac{\mathrm{d}}{\mathrm{d}\beta}\right) \int \mathrm{d}V \mathrm{d}V' \mathrm{e}^{\alpha t} \mathrm{e}^{\beta t'} \mathrm{e}^{-t^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\, \mathrm{e}^{-t'^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathrm{x}, \mathrm{x}')$$

Optimization

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$$H_n\left(\frac{\mathrm{d}}{\mathrm{d}\alpha}\right) H_m\left(\frac{\mathrm{d}}{\mathrm{d}\beta}\right)\int \mathrm{d}V\mathrm{d}V' \boxed{\mathrm{e}^{\alpha t}\mathrm{e}^{\beta t'}}\mathrm{e}^{-t^2/2T^2}\mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\,\mathrm{e}^{-t'^2/2T^2}\mathrm{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}}W(\mathrm{x},\mathrm{x}')$$

Optimization

$$\int \mathrm{d}V \mathrm{d}V' H_n(t)\, H_m(t') \mathrm{e}^{-t^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}} \mathrm{e}^{-t'^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathrm{x}, \mathrm{x}')$$

$$\Downarrow$$

$$H_n\left(\frac{\mathrm{d}}{\mathrm{d}\alpha}\right) H_m\left(\frac{\mathrm{d}}{\mathrm{d}\beta}\right) \boxed{\int \mathrm{d}V \mathrm{d}V' \mathrm{e}^{\alpha t} \mathrm{e}^{\beta t'} \mathrm{e}^{-t^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}} \mathrm{e}^{-t'^2/2T^2} \mathrm{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathrm{x}, \mathrm{x}')}$$

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$$= W(\mathsf{e}^{\alpha t}\Lambda_A^-, \mathsf{e}^{\beta t}\Lambda_A^+)$$

Optimization

$$\int \mathrm{d}V\mathrm{d}V' H_n(t)\, H_m(t')\mathsf{e}^{-t^2/2T^2}\mathsf{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\,\mathsf{e}^{-t'^2/2T^2}\mathsf{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathsf{x},\mathsf{x}')$$

$$\Downarrow$$

$$H_n\left(\frac{\mathrm{d}}{\mathrm{d}\alpha}\right) H_m\left(\frac{\mathrm{d}}{\mathrm{d}\beta}\right) \boxed{\int \mathrm{d}V\mathrm{d}V' \mathsf{e}^{\alpha t}\mathsf{e}^{\beta t'}\mathsf{e}^{-t^2/2T^2}\mathsf{e}^{-\frac{|\boldsymbol{x}|^2}{2\sigma^2}}\,\mathsf{e}^{-t'^2/2T^2}\mathsf{e}^{-\frac{|\boldsymbol{x}'|^2}{2\sigma^2}} W(\mathsf{x},\mathsf{x}')}$$

$$= W(\mathsf{e}^{\alpha t}\Lambda^-_A, \mathsf{e}^{\beta t}\Lambda^+_A) \longrightarrow \text{Closed forms!}$$